Top 75 CAT Table with Missing values Questions With Video Solutions

Questions

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.

The Dean's office recently scanned student results into the central computer system. When their character reading software cannot read something, it leaves the space blank. The scanner output read as follows:

Name	Finance	Marketing	Statistics	Strategy	Operations	GPA
Aparna		B	F			1.4
Bikas	D	D	F	F		
Chandra		D	A	F	F	2.4
Deepak	A	B		D	D	3.2
Fazal	D	F		B	D	2.4
Gowri	C	C	A		B	3.8
Hari		B	A		D	2.8
Ismet			B		A	
Jagdeep	A	A	B		C	3.8
Kunal	F		A	F	F	1.8
Leena	B	A		B	F	3.2
Manab			A	B	B	
Nisha	A	D	B	A	F	3.6
Osman	C		B	B	A	4.6
Preeti	F	D		D		3.2
Rahul	A	C	A		F	4.2
Sameer		C	F	B		
Tara	B					2.4
Utkarsh			F	C	A	3.0
Vipul	A		C	C	F	2.4

In the grading system, A, B, C, D, and F grades fetch 6, 4, 3, 2, and 0 grade points respectively. The Grade Point Average (GPA) is the arithmetic mean of the grade points obtained in the five subjects. For example Nisha's GPA is (6 + 2 + 4 + 6 + 0) / 5 = 3.6. Some additional facts are also known about the students' grades. These are

(i) Vipul obtained the same grade in Marketing as Aparna obtained in Finance and Strategy.

(ii) Fazal obtained the same grade in Strategy as Utkarsh did in Marketing.

(iii) Tara received the same grade in exactly three courses.

Question 1

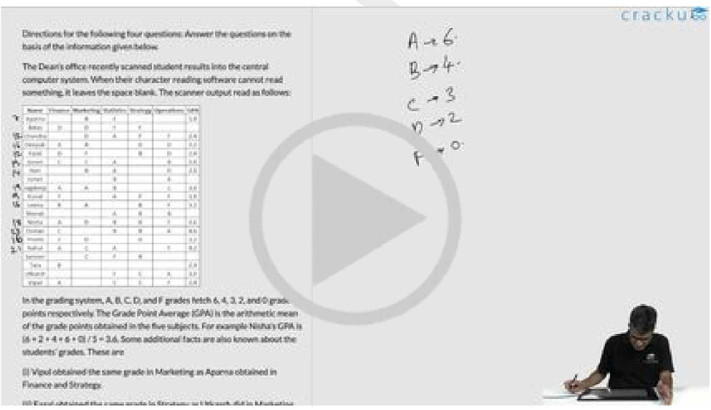
What grade did Preeti obtain in Statistics?

A A

B B

C C

D D



VIDEO SOLUTION

Instructions

Funky Pizzeria was required to supply pizzas to three different parties. The total number of pizzas it had to deliver was 800, 70% of which were to be delivered to Party 3 and the rest equally divided between Party 1 and Party 2.

Pizzas could be of Thin Crust (T) or Deep Dish (D) variety and come in either Normal Cheese (NC) or Extra Cheese (EC) versions. Hence, there are four types of pizzas: T-NC, T-EC, D-NC and D-EC. Partial information about proportions of T and NC pizzas ordered by the three parties is given below:

	Thin crust (T)	Normal cheese (NC)
Party 1	0.6	
Party 2	0.55	0.3
Party 3		0.65
Total	0.375	0.52

Question 2

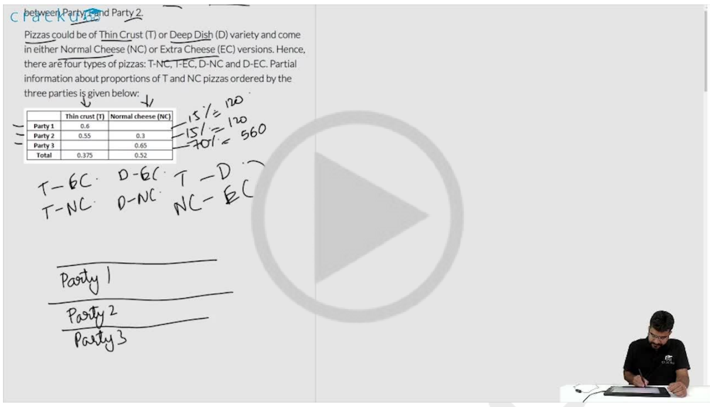
For Party 2, if 50% of the Normal Cheese pizzas were of Thin Crust variety, what was the difference between the numbers of T-EC and D-EC pizzas to be delivered to Party 2?

A 18

B 12

C 30

D 24



VIDEO SOLUTION

Instructions

A company administers a written test comprising of three sections of 20 marks each - Data Interpretation (DI), Written English (WE) and General Awareness (GA), for recruitment. A composite score for a candidate (out of 80) is calculated by doubling her marks in DI and adding it to the sum of her marks in the other two sections. Candidates who score less than 70% marks in two or more sections are disqualified. From among the rest, the four with the highest composite scores are recruited. If four or less candidates qualify, all who qualify are recruited.

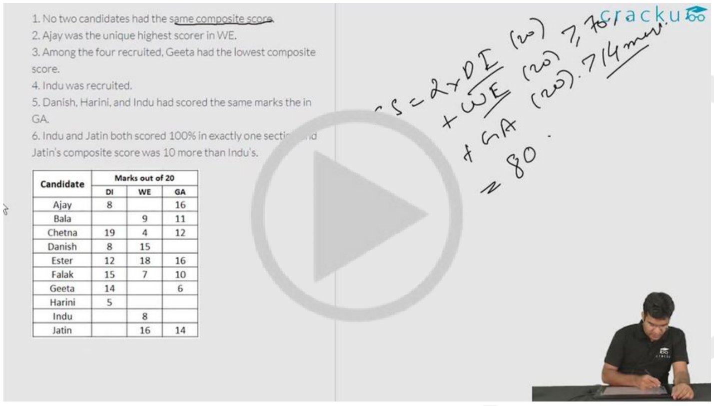
Ten candidates appeared for the written test. Their marks in the test are given in the table below. Some marks in the table are missing, but the following facts are known:

1. No two candidates had the same composite score.
2. Ajay was the unique highest scorer in WE.
3. Among the four recruited, Geeta had the lowest composite score.
4. Indu was recruited.
5. Danish, Harini, and Indu had scored the same marks the in GA.
6. Indu and Jatin both scored 100% in exactly one section and Jatin's composite score was 10 more than Indu's.

Candidate	Marks out of 20		
	DI	WE	GA
Ajay	8		16
Bala		9	11
Chetna	19	4	12
Danish	8	15	
Ester	12	18	16
Falak	15	7	10
Geeta	14		6
Harini	5		
Indu		8	
Jatin		16	14

Question 3

If all the candidates scored different marks in WE then what is the maximum marks that Harini could have scored in WE?



VIDEO SOLUTION

Instructions

The year is 2089. Beijing, London, New York, and Paris are in contention to host the 2096 Olympics. The eventual winner is determined through several rounds of voting by members of the IOC with each member representing a different city. All the four cities in contention are also represented in IOC.

Round	Total Votes Cast	Maximum Votes Cast		Eliminated	
		City	No. of Votes	City	No. of Votes
1		London	30	New York	12
2	83	Paris	32	Beijing	21
3	75				

- In any round of voting, the city receiving the lowest number of votes in that round gets eliminated. The survivor after the last round of voting gets to host the event.

- A member is allowed to cast votes for at most two different cities in all rounds of voting combined. (Hence, a member becomes ineligible to cast a vote in a given round if both the cities (s)he voted for in earlier rounds are out of contention in that round of voting.)
- A member is also ineligible to cast a vote in a round if the city (s)he represents is in contention in that round of voting.
- As long as the member is eligible, (s)he must vote and vote for only one candidate city in any round of voting.The following incomplete table shows the information on cities that received the maximum and minimum votes in different rounds, the number of votes cast in their favour, and the total votes that were cast in those rounds.It is also known that:
- All those who voted for London and Paris in round 1, continued to vote for the samecities in subsequent rounds as long as these cities were in contention. 75% of those whovoted for Beijing in round 1, voted for Beijing in round 2 as well.
- Those who voted for New York in round 1, voted either for Beijing or Paris in round 2.
- The difference in votes cast for the two contending cities in the last round was 1.
- 50% of those who voted for Beijing in round 1, voted for Paris in round 3.

Question 4

Which of the following statements must be true?

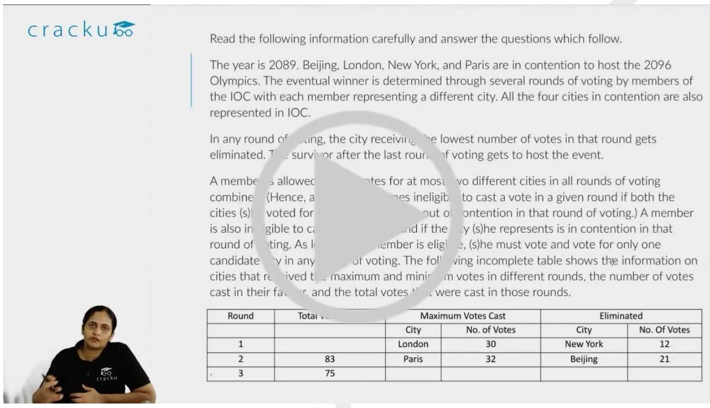
a. IOC member from New York must have voted for Paris in round 2.
b. IOC member from Beijing voted for London in round 3.

A Only a

B Only b

C Both a and b

D Neither a nor b



VIDEO SOLUTION

Instructions

Six players - Tanzi, Umeza, Wangdu, Xyla, Yonita and Zeneca competed in an archery tournament. The tournament had three compulsory rounds, Rounds 1 to 3. In each round every player shot an arrow at a target. Hitting the centre of the target (called bull's eye) fetched the highest score of 5. The only other possible scores that a player could achieve were 4, 3, 2 and 1. Every bull's eye score in the first three rounds gave a player one additional chance to shoot in the bonus rounds, Rounds 4 to 6. The possible scores in Rounds 4 to 6 were identical to the first three.

A player's total score in the tournament was the sum of his/her scores in all rounds played by him/her. The table below presents partial information on points scored by the players after completion of the tournament. In the table, NP means that the player did not participate in that round, while a hyphen means that the player participated in that round and the score information is missing.

	Round-1	Round-2	Round-3	Round-4	Round-5	Round-6
Tanzi	-	4	-	5	NP	NP
Umeza	-	-	-	1	2	NP
Wangdu	-	4	-	NP	NP	NP
Xyla	-	-	-	1	5	-
Yonita	-	-	3	5	NP	NP
Zeneca	-	-	-	5	5	NP

The following facts are also known.
1.Tanzi, Umeza and Yonita had the same total score.
2.Total scores for all players, except one, were in multiples of three.
3.The highest total score was one more than double of the lowest total score.
4.The number of players hitting bull's eye in Round 2 was double of that in Round 3.
5.Tanzi and Zeneca had the same score in Round 1 but different scores in Round 3.

Question 5

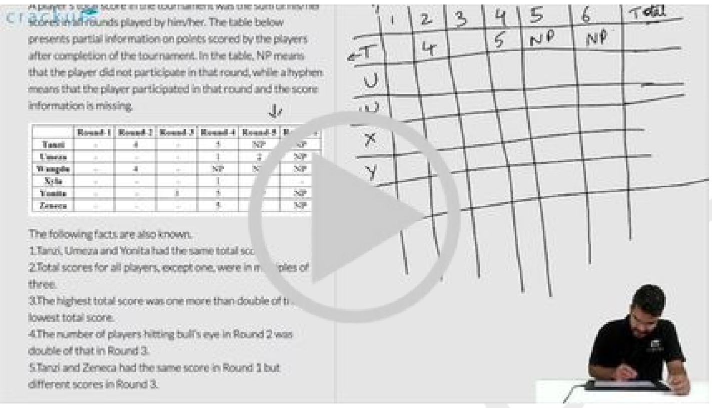
What was the highest total score?

A 25

B 21

C 24

D 23



VIDEO SOLUTION

Question 6

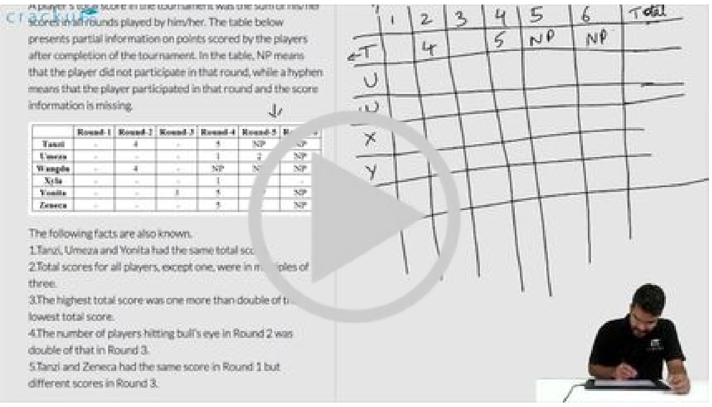
Which of the following statements is true?

A Xyla's score was 23.

B Zeneca's score was 23.

C Zeneca was the highest scorer.

D Xyla was the highest scorer.



VIDEO SOLUTION

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Question 7

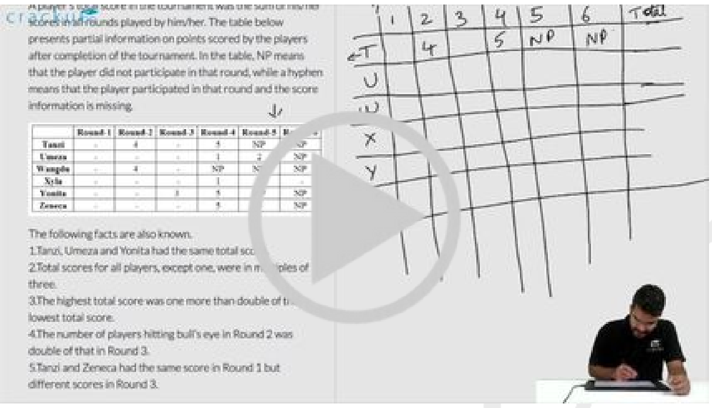
What was Tanzi's score in Round 3?

A 4

B 5

C 3

D 1



VIDEO SOLUTION

Instructions

Amudha, Bharatan, Chandran, Dhinesh, Ezhil, Fani and Gowtham are seven people in a town. Any pair of them could either be strangers, acquaintances, or friends. All relationships are mutual. For example, if Amudha is a friend of Bharatan, then Bharatan is also a friend of Amudha. Similarly, if Amudha is a stranger to Bharatan, then Bharatan is also a stranger to Amudha.

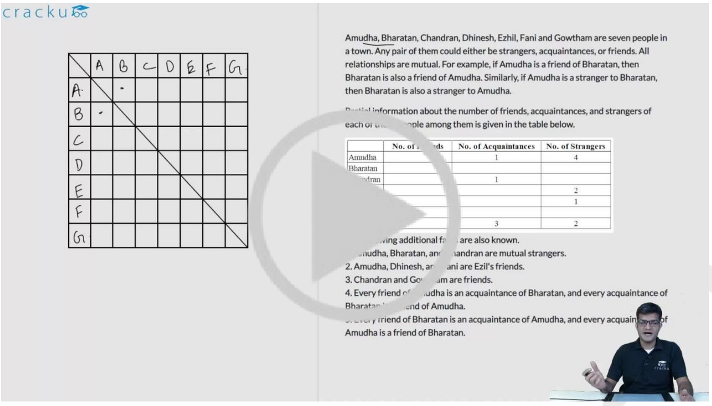
Partial information about the number of friends, acquaintances, and strangers of each of these people among them is given in the table below.

	No. of Friends	No. of Acquaintances	No. of Strangers
Amudha		1	4
Bharatan			
Chandran		1	
Dhinesh			2
Ezhil			1
Fani	1		
Gowtham		3	2

The following additional facts are also known.
1. Amudha, Bharatan, and Chandran are mutual strangers.
2. Amudha, Dhinesh, and Fani are Ezil's friends.
3. Chandran and Gowtham are friends.
4. Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance of Bharatan is a friend of Amudha.
5. Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan.

Question 8

How many people are either a friend or a friend-of-a-friend of Ezhil?



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Total	0.375	0.52

Question 9

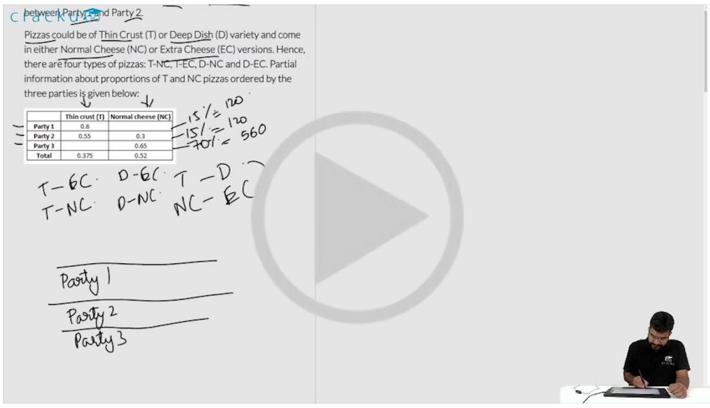
Suppose that a T-NC pizza cost as much as a D-NC pizza, but 3/5th of the price of a D-EC pizza.A D-EC pizza costs Rs. 50 more than a T-EC pizza, and the latter costs Rs. 500.
If 25% of the Normal Cheese pizzas delivered to Party 1 were of Deep Dish variety, what was the total bill for Party 1?

A Rs. 59480

B Rs. 59840

C Rs. 42520

D Rs. 45240



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Danish	8	15	
Ester	12	18	16
Falak	15	7	10
Geeta	14		6
Harini	5		
Indu		8	
Jatin		16	14

Question 10

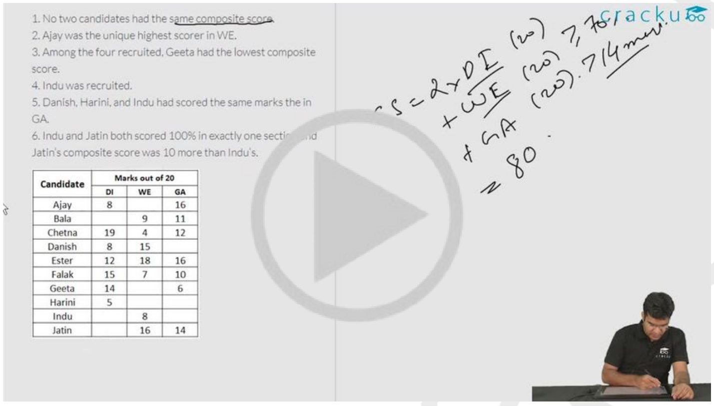
Which of the following statements MUST be true?
1. Jatin's composite score was more than that of Danish.
2. Indu scored less than Chetna in DI.
3. Jatin scored more than Indu in GA.

A Both 2 and 3

B Only 1

C Only 2

D Both 1 and 2



VIDEO SOLUTION

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Directions for the following four questions: Answer the questions on the basis of the information given below.

The Dean’s office recently scanned student results into the central computer system. When their character reading software cannot read something, it leaves the space blank. The scanner output read as follows:

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Bikas	D	D	F	F		
Chandra		D	A	F	F	2.4
Deepak	A	B		D	D	3.2
Fazal	D	F		B	D	2.4
Gowri	C	C	A		B	3.8
Hari		B	A		D	2.8
Ismet			B		A	
Jagdeep	A	A	B		C	3.8
Kunal	F		A	F	F	1.8
Leena	B	A		B	F	3.2
Manab			A	B	B	
Nisha	A	D	B	A	F	3.6
Osman	C		B	B	A	4.6
Preeti	F	D		D		3.2
Rahul	A	C	A		F	4.2
Sameer		C	F	B		
Tara	B					2.4
Utkarsh			F	C	A	3.0
Vipul	A		C	C	F	2.4

In the grading system, A, B, C, D, and F grades fetch 6, 4, 3, 2, and 0 grade points respectively. The Grade Point Average (GPA) is the arithmetic mean of the grade points obtained in the five subjects. For example Nisha's GPA is (6 + 2 + 4 + 6 + 0) / 5 = 3.6. Some additional facts are also known about the students' grades. These are

(i) Vipul obtained the same grade in Marketing as Aparna obtained in Finance and Strategy.

(ii) Fazal obtained the same grade in Strategy as Utkarsh did in Marketing.

(iii) Tara received the same grade in exactly three courses.

Question 11

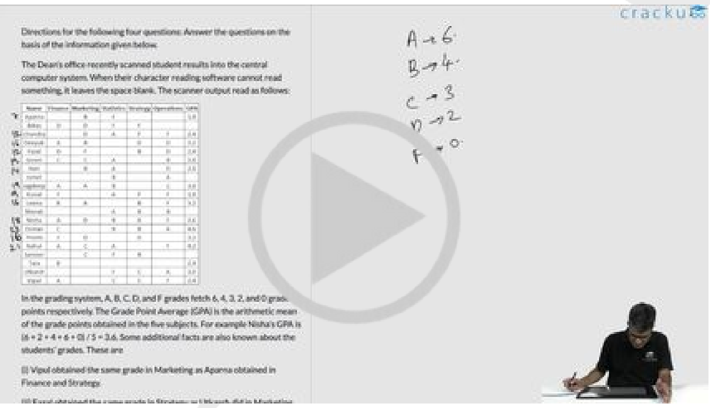
In Strategy, Gowri's grade point was higher than that obtained by

A Fazal

B Hari

C Nisha

D Rahul



VIDEO SOLUTION

Question 12

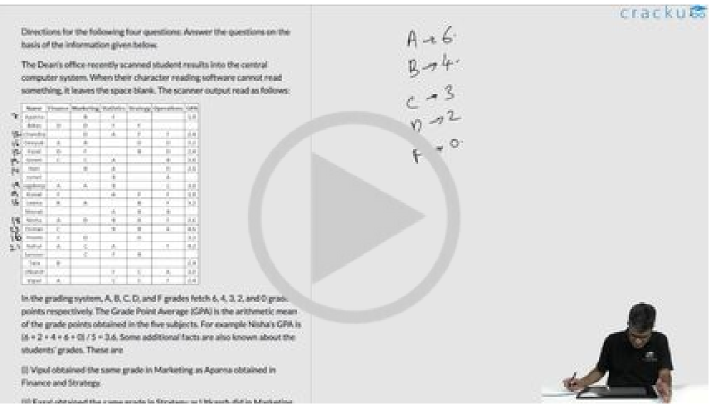
What grade did Utkarsh obtain in Finance?

A B

B C

C D

D F



VIDEO SOLUTION

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Instructions

The following table gives details regarding the total earnings of 15 employees and the number of days they have worked on complex, medium and simple operation in the month of June 2002. Even though the employees might have worked on an operation, they would be eligible for earnings only if they have minimum level of efficiency.

	Total Earning				Total Days			
Emp No.	Complex	Medium	Simple	Total	Complex	Medium	Simple	Total
2001147	82.98		636.53	719.51	3	0	23	26
2001148	51.53		461.73	513.26	3.3	1.67	16	21
2001149	171.10		79.10	250.81	5.5	4	8.5	18
2001150	100.47		497.47	597.95	6	4.67	7.33	18
2001151	594.43	159.64		754.06	9.67	13.33	0	23
2001156	83.70			89.70	8	0	1	9
2001158	472.31	109.73		582.04	1.39	9.61	0	11
2001164	402.25	735.22	213.67	1351.14	5.27	12.07	0.67	18
2001170	576.57			576.57	21	0	0	21
2001171	286.47	6.10		292.57	8.38	4.25	0.38	13
2001172	512.10	117.46		629.56	10	8.5	3.5	22
2001173	1303.88			1303.88	25.5	0	0.5	26
2001174	1017.94			1017.9	26	0	0	26
2001179	46.56	776.19		822.75	2	19	0	21
2001180	116.40	1262.79		1379.19	5	19	0	24

Question 13

Among the employees who were engaged in complex and medium operations, the number of employees whose average earning per day in complex operations is more than average earning per day in medium operations is

A 2

B 3

C 5

D 8

VIEW SOLUTION

Instructions

The year is 2089. Beijing, London, New York, and Paris are in contention to host the 2096 Olympics. The eventual winner is determined through several rounds of voting by members of the IOC with each member representing a different city. All the four cities in contention are also represented in IOC.

Round	Total Votes Cast	Maximum Votes Cast		Eliminated	
		City	No. of Votes	City	No. of Votes
1		London	30	New York	12
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3	75				

- In any round of voting, the city receiving the lowest number of votes in that round gets eliminated. The survivor after the last round of voting gets to host the event.
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- All those who voted for London and Paris in round 1, continued to vote for the samecities in subsequent rounds as long as these cities were in contention. 75% of those whovoted for Beijing in round 1, voted for Beijing in round 2 as well.
- Those who voted for New York in round 1, voted either for Beijing or Paris in round 2.
- The difference in votes cast for the two contending cities in the last round was 1.
- 50% of those who voted for Beijing in round 1, voted for Paris in round 3.

Question 14

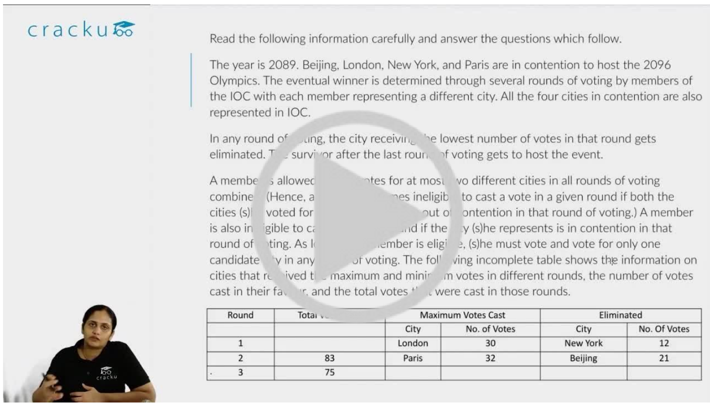
What percentage of members from among those who voted for New York in round 1, voted for Beijing in round 2?

A 31.33

B 50

C 66.67

D 75



VIDEO SOLUTION

Instructions

An agency entrusted to accredit colleges looks at four parameters: faculty quality (F), reputation (R), placement quality (P), and infrastructure (I). The four parameters are used to arrive at an overall score, which the agency uses to give an accreditation to the colleges. In each parameter, there are five possible letter grades given, each carrying certain points: A (50 points), B (40 points), C (30 points), D (20 points), and F (0 points). The overall score for a college is the weighted sum of the points scored in the four parameters. The weights of the parameters are 0.1, 0.2, 0.3 and 0.4 in some order, but the order is not disclosed. Accreditation is awarded based on the following scheme:

Range	Accreditation
Overall score ≥ 45	AAA
35 ≤ Overall score < 45	BAA
25 ≤ Overall score < 35	BBA
15 ≤ Overall score < 25	BBB
Overall score < 15	Junk

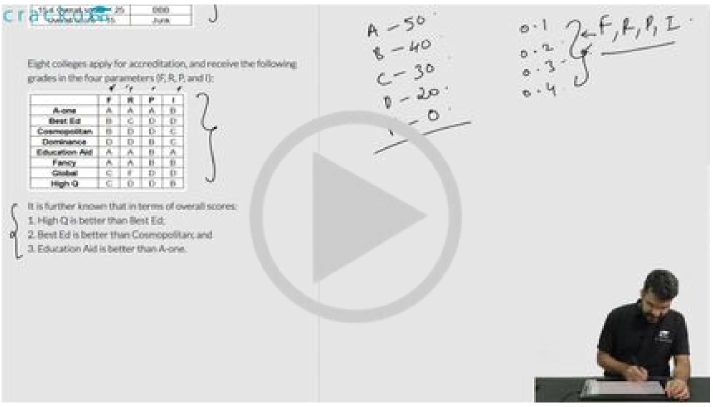
Eight colleges apply for accreditation, and receive the following grades in the four parameters (F, R, P, and I):

	F	R	P	I
A-one	A	A	A	B
Best Ed	B	C	D	D
Cosmopolitan	B	D	D	C
Dominance	D	D	B	C
Education Aid	A	A	B	A
Fancy	A	A	B	B
Global	C	F	D	D
High Q	C	D	D	B

It is further known that in terms of overall scores:
1. High Q is better than Best Ed;
2. Best Ed is better than Cosmopolitan; and
3. Education Aid is better than A-one.

Question 15

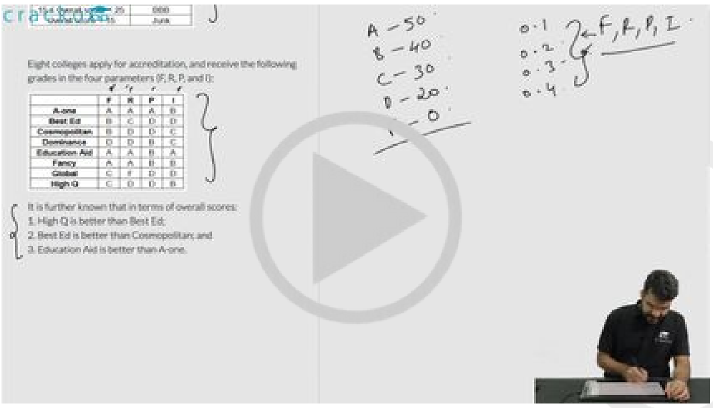
What is the highest overall score among the eight colleges ?



VIDEO SOLUTION

Question 16

How many colleges receive the accreditation of AAA?



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The following table gives details regarding the total earnings of 15 employees and the number of days they have worked on complex, medium and simple operation in the month of June 2002. Even though the employees might have worked on an operation, they would be eligible for earnings only if they have minimum level of efficiency.

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2001180	116.40	1262.79		1379.19	5	19	0	24

Question 17

The employee number of the person who has earned the maximum earnings per day in medium operation is

A 2001180

B 2001164

C 2001172

D 2001179

VIEW SOLUTION

Question 18

The number of employees who have earned more than Rs. 600 and having more than 80% attendance (there are 25 regular working days in June 2002; some might be coming on overtime too) is

A 4

B 5

C 6

D 7

VIEW SOLUTION

Question 19

The number of employees who have earned more than Rs. 50 per day in complex operations is

A 4

B 3

C 5

D 6

VIEW SOLUTION

Instructions

Amudha, Bharatan, Chandran, Dhinesh, Ezhil, Fani and Gowtham are seven people in a town. Any pair of them could either be strangers, acquaintances, or friends. All relationships are mutual. For example, if Amudha is a friend of Bharatan, then Bharatan is also a friend of Amudha. Similarly, if Amudha is a stranger to Bharatan, then Bharatan is also a stranger to Amudha.

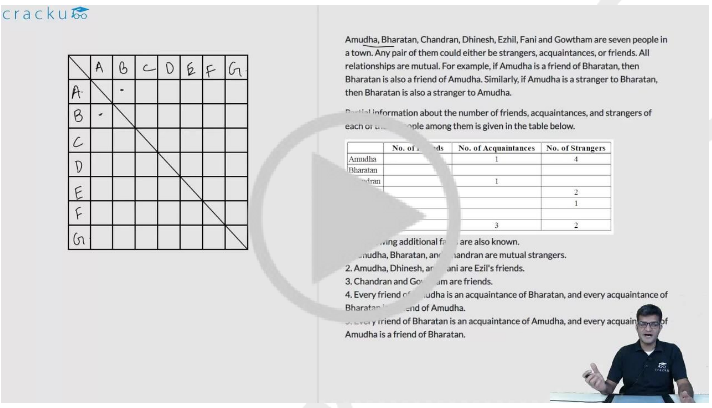
Partial information about the number of friends, acquaintances, and strangers of each of these people among them is given in the table below.

	No. of Friends	No. of Acquaintances	No. of Strangers
Amudha		1	4
Bharatan			
Chandran		1	
Dhinesh			2
Ezhil			1
Fani	1		
Gowtham		3	2

The following additional facts are also known.
1. Amudha, Bharatan, and Chandran are mutual strangers.
2. Amudha, Dhinesh, and Fani are Ezil's friends.
3. Chandran and Gowtham are friends.
4. Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance of Bharatan is a friend of Amudha.
5. Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan.

Question 20

How many friends does Ezhil have?



VIDEO SOLUTION

Instructions

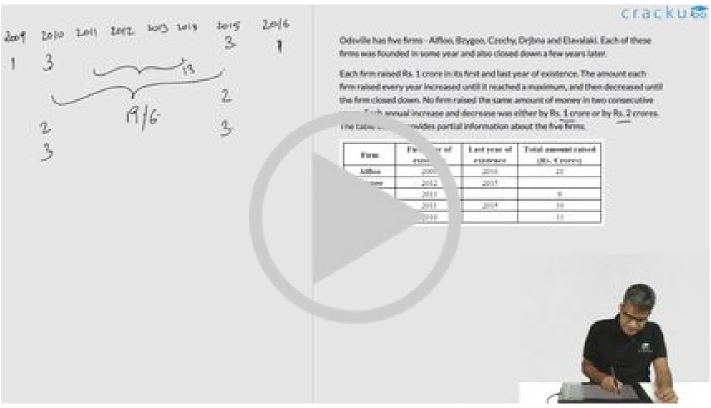
Odsville has five firms - Alfloo, Bzygoo, Czechy, Drjbna and Elavalaki. Each of these firms was founded in some year and also closed down a few years later.

Each firm raised Rs. 1 crore in its first and last year of existence. The amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down. No firm raised the same amount of money in two consecutive years. Each annual increase and decrease was either by Rs. 1 crore or by Rs. 2 crores. The table below provides partial information about the five firms.

Firm	First year of existence	Last year of existence	Total amount raised (Rs. Crores)
Alfloo	2009	2016	21
Bzygoo	2012	2015	
Czechy	2013		9
Drjbna	2011	2015	10
Elavalaki	2010		13

Question 21

What is the largest possible total amount of money (in Rs. crores) that could have been raised in 2013?



VIDEO SOLUTION

Instructions

Six players - Tanzi, Umeza, Wangdu, Xyla, Yonita and Zeneca competed in an archery tournament. The tournament had three compulsory rounds, Rounds 1 to 3. In each round every player shot an arrow at a target. Hitting the centre of the target (called bull's eye) fetched the highest score of 5. The only other possible scores that a player could achieve were 4, 3, 2 and 1. Every bull's eye score in the first three rounds gave a player one additional chance to shoot in the bonus rounds, Rounds 4 to 6. The possible scores in Rounds 4 to 6 were identical to the first three.

A player's total score in the tournament was the sum of his/her scores in all rounds played by him/her. The table below presents partial information on points scored by the players after completion of the tournament. In the table, NP means that the player did not participate in that round, while a hyphen means that the player participated in that round and the score information is missing.

	Round-1	Round-2	Round-3	Round-4	Round-5	Round-6
Tanzi	-	4	-	5	NP	NP
Umeza	-	-	-	1	2	NP
Wangdu	-	4	-	NP	NP	NP
Xyla	-	-	-	1	5	-
Yonita	-	-	3	5	NP	NP
Zeneca	-	-	-	5	5	NP

The following facts are also known.

1.Tanzi, Umeza and Yonita had the same total score.

2.Total scores for all players, except one, were in multiples of three.

3.The highest total score was one more than double of the lowest total score.
4.The number of players hitting bull's eye in Round 2 was double of that in Round 3.
5.Tanzi and Zeneca had the same score in Round 1 but different scores in Round 3.

Question 22

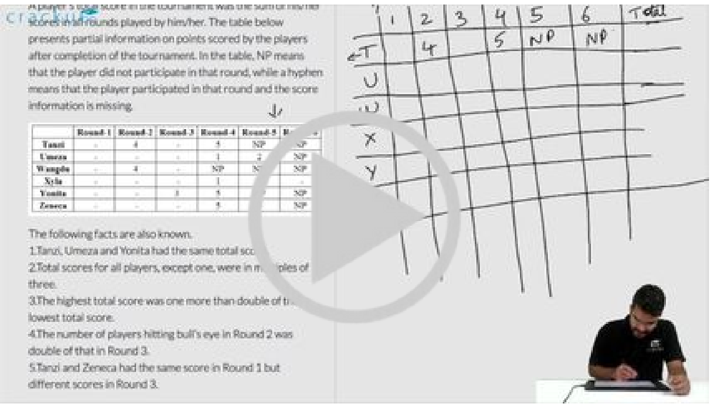
What was Zeneca's total score?

A 21

B 22

C 23

D 24



VIDEO SOLUTION

Instructions

Five restaurants, coded R1, R2, R3, R4 and R5 gave integer ratings to five gig workers -

Ullas, Vasu, Waman, Xavier and Yusuf, on a scale of 1 to 5.

The means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.
The summary statistics of these ratings for the five workers is given below.

	Ullas	Vasu	Waman	Xavier	Yusuf
Mean rating	2.2	3.8	3.4	3.6	2.6
Median rating	2	4	4	4	3
Modal rating	2	4	5	5	1 and 4
Range of rating*	3	3	4	4	3

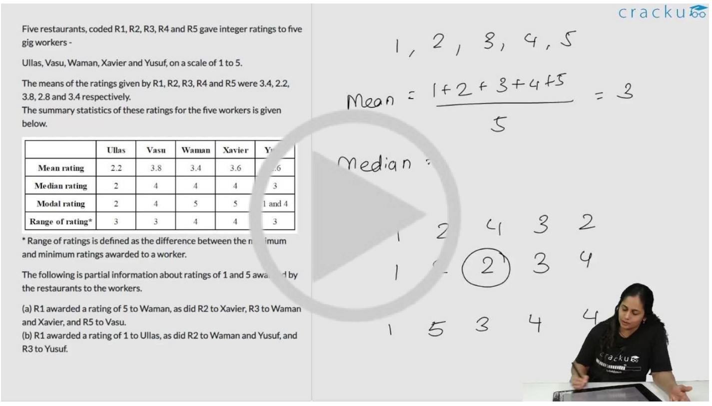
* Range of ratings is defined as the difference between the maximum and minimum ratings awarded to a worker.

The following is partial information about ratings of 1 and 5 awarded by the restaurants to the workers.

(a) R1 awarded a rating of 5 to Waman, as did R2 to Xavier, R3 to Waman and Xavier, and R5 to Vasu.
(b) R1 awarded a rating of 1 to Ullas, as did R2 to Waman and Yusuf, and R3 to Yusuf.

Question 23

What is the median of the ratings given by R3 to the five workers?



VIDEO SOLUTION

Instructions

In a coaching class, some students register online, and some others register offline. No student registers both online and offline; hence the total registration number is the sum of online and offline registrations. The following facts and table pertain to these registration numbers for the five months - January to May of 2023. The table shows the minimum, maximum, median registration numbers of these five months, separately for online, offline and total number of registrations. The following additional facts are known.

1. In every month, both online and offline registration numbers were multiples of 10.
2. In January, the number of offline registrations was twice that of online registrations.
3. In April, the number of online registrations was twice that of offline registrations.
4. The number of online registrations in March was the same as the number of offline registrations in February.
5. The number of online registrations was the largest in May.

	Minimum	Maximum	Median
Online	40	100	80
Offline	30	80	50
Total	110	130	120

Question 24

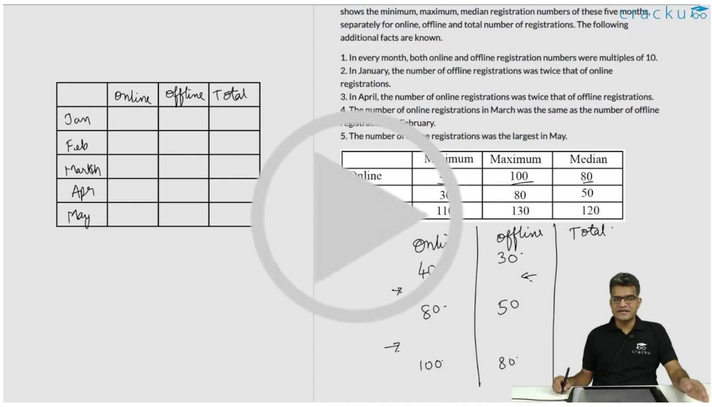
What best can be concluded about the number of offline registrations in February?

A 80

B 50 or 80

C 30 or 50 or 80

D 50



VIDEO SOLUTION

Top-500 Free CAT Questions (With Solutions)

Instructions

Five restaurants, coded R1, R2, R3, R4 and R5 gave integer ratings to five gig workers -

Ullas, Vasu, Waman, Xavier and Yusuf, on a scale of 1 to 5.

The means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.
The summary statistics of these ratings for the five workers is given below.

	Ullas	Vasu	Waman	Xavier	Yusuf
Mean rating	2.2	3.8	3.4	3.6	2.6
Median rating	2	4	4	4	3
Modal rating	2	4	5	5	1 and 4
Range of rating*	3	3	4	4	3

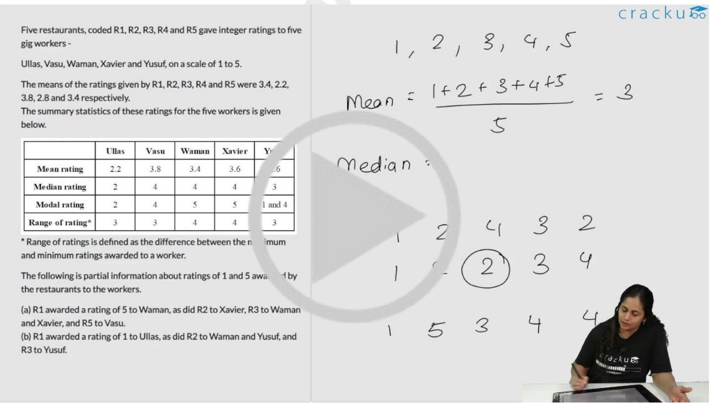
* Range of ratings is defined as the difference between the maximum and minimum ratings awarded to a worker.

The following is partial information about ratings of 1 and 5 awarded by the restaurants to the workers.

(a) R1 awarded a rating of 5 to Waman, as did R2 to Xavier, R3 to Waman and Xavier, and R5 to Vasu.
(b) R1 awarded a rating of 1 to Ullas, as did R2 to Waman and Yusuf, and R3 to Yusuf.

Question 25

How many individual ratings cannot be determined from the above information?



VIDEO SOLUTION

Instructions

Odsville has five firms - Alfloo, Bzygoo, Czechy, Drjbna and Elavalaki. Each of these firms was founded in some year and also closed down a few years later.

Each firm raised Rs. 1 crore in its first and last year of existence. The amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down. No firm raised the same amount of money in two consecutive years. Each annual increase and decrease was either by Rs. 1 crore or by Rs. 2 crores. The table below provides partial information about the five firms.

Firm	First year of existence	Last year of existence	Total amount raised (Rs. Crores)
Alfloo	2009	2016	21
Bzygoo	2012	2015	
Czechy	2013		9
Drjbna	2011	2015	10
Elavalaki	2010		13

Question 26

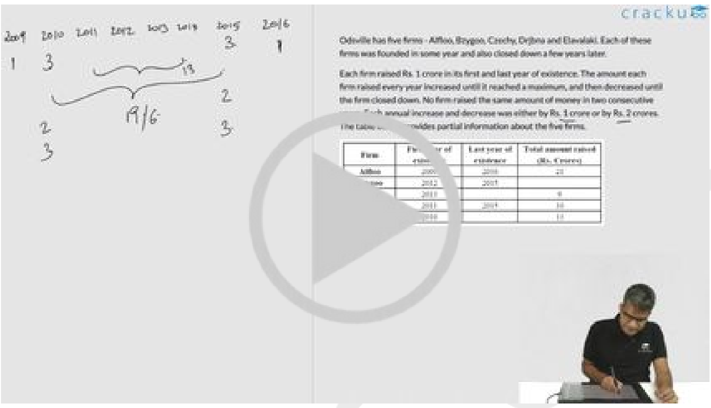
For which firm(s) can the amounts raised by them be concluded with certainty in each year?

A Only Bzygoo and Czechy and Drjbna

B Only Czechy and Drjbna

C Only Drjbna

D Only Czechy



VIDEO SOLUTION

Instructions

In an election several candidates contested for a constituency. In any constituency, the winning candidate was the one who polled the highest number of votes, the first runner up was the one who polled the second highest number of votes, the second runner up was the one who polled the third highest number of votes, and so on. There were no ties (in terms of number of votes polled by the candidates) in any of the constituencies in this election. In an electoral system, a security deposit is the sum of money that a candidate is required to pay to the election commission before he or she is permitted to contest. Only the defeated candidates (i.e., one who is not the winning candidate) who fail to secure more than one sixth of the valid votes polled in the constituency, lose their security deposits.

The following table provides some incomplete information about votes polled in four constituencies: A, B, C and D, in this election .

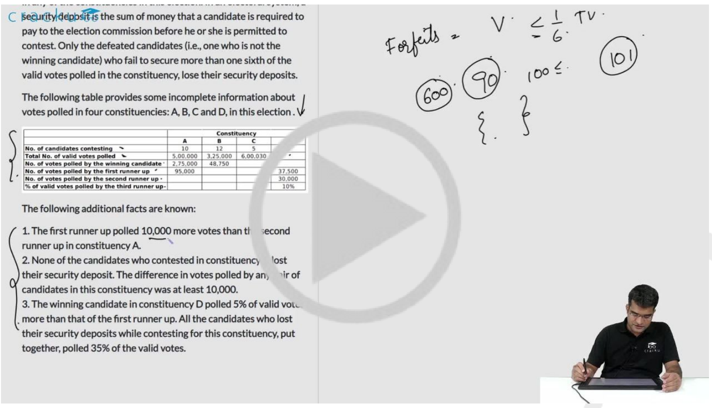
	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	
No. of votes polled by the winning candidate	2,75,000	48,750		
No. of votes polled by the first runner up	95,000			37,500
No. of votes polled by the second runner up				30,000
% of valid votes polled by the third runner up				10%

The following additional facts are known:

1. The first runner up polled 10,000 more votes than the second runner up in constituency A.
2. None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000.
3. The winning candidate in constituency D polled 5% of valid votes more than that of the first runner up. All the candidates who lost their security deposits while contesting for this constituency, put together, polled 35% of the valid votes.

Question 27

How many candidates who contested in constituency B lost their security deposit?



VIDEO SOLUTION

Free CAT Study Material

Instructions

The table given below shows the amount, in grams, of carbohydrate, protein, fat and all other nutrients, per 100 grams of nutrients in seven food grains. The first column shows the food grain category and the second column its codename. The table has some missing values.

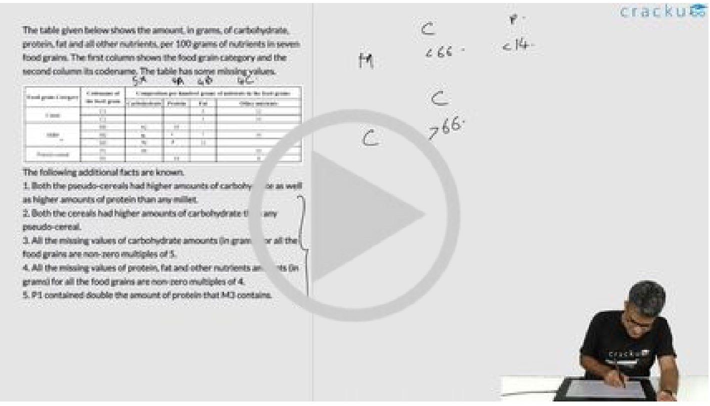
Food grain Category	Codename of the food grain	Composition per hundred grams of nutrients in the food grains			
		Carbohydrate	Protein	Fat	Other nutrients
Cereal	C1			0	12
	C2			3	10
Millet	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereal	P1	66			10
	P2		14		8

The following additional facts are known.
1. Both the pseudo-cereals had higher amounts of carbohydrate as well as higher amounts of protein than any millet.

2. Both the cereals had higher amounts of carbohydrate than any pseudo-cereal.
3. All the missing values of carbohydrate amounts (in grams) for all the food grains are non-zero multiples of 5.
4. All the missing values of protein, fat and other nutrients amounts (in grams) for all the food grains are non-zero multiples of 4.
5. P1 contained double the amount of protein that M3 contains.

Question 28

What is the median of the number of grams of protein in 100 grams of nutrients among these food grains?



VIDEO SOLUTION

Instructions

The year is 2089. Beijing, London, New York, and Paris are in contention to host the 2096 Olympics. The eventual winner is determined through several rounds of voting by members of the IOC with each member representing a different city. All the four cities in contention are also represented in IOC.

Round	Total Votes Cast	Maximum Votes Cast		Eliminated	
		City	No. of Votes	City	No. of Votes
1		London	30	New York	12
2	83	Paris	32	Beijing	21
3	75				

- In any round of voting, the city receiving the lowest number of votes in that round gets eliminated. The survivor after the last round of voting gets to host the event.
- A member is allowed to cast votes for at most two different cities in all rounds of voting combined. (Hence, a member becomes ineligible to cast a vote in a given round if both the cities (s)he voted for in earlier rounds are out of contention in that round of voting.)
- A member is also ineligible to cast a vote in a round if the city (s)he represents is in contention in that round of voting.
- As long as the member is eligible, (s)he must vote and vote for only one candidate city in any round of voting.The following incomplete table shows the information on cities that received the maximum and minimum votes in different rounds, the number of votes cast in their favour, and the total votes that were cast in those rounds.It is also known that:
- All those who voted for London and Paris in round 1, continued to vote for the samecities in subsequent rounds as long as these cities were in contention. 75% of those whovoted for Beijing in round 1, voted for Beijing in round 2 as well.
- Those who voted for New York in round 1, voted either for Beijing or Paris in round 2.
- The difference in votes cast for the two contending cities in the last round was 1.
- 50% of those who voted for Beijing in round 1, voted for Paris in round 3.

Question 29

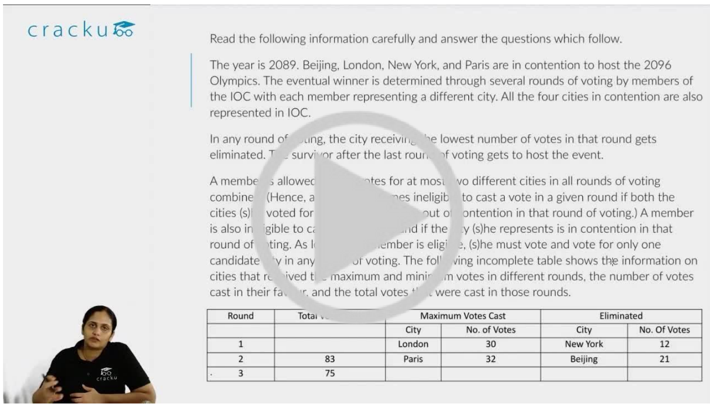
What is the number of votes cast for Paris in round 1?

A 16

B 18

C 22

D 24



VIDEO SOLUTION

Instructions

Funky Pizzeria was required to supply pizzas to three different parties. The total number of pizzas it had to deliver was 800, 70% of which were to be delivered to Party 3 and the rest equally divided between Party 1 and Party 2.

Pizzas could be of Thin Crust (T) or Deep Dish (D) variety and come in either Normal Cheese (NC) or Extra Cheese (EC) versions. Hence, there are four types of pizzas: T-NC, T-EC, D-NC and D-EC. Partial information about proportions of T and NC pizzas ordered by the three parties is given below:

	Thin crust (T)	Normal cheese (NC)
Party 1	0.6	
Party 2	0.55	0.3
Party 3		0.65
Total	0.375	0.52

Question 30

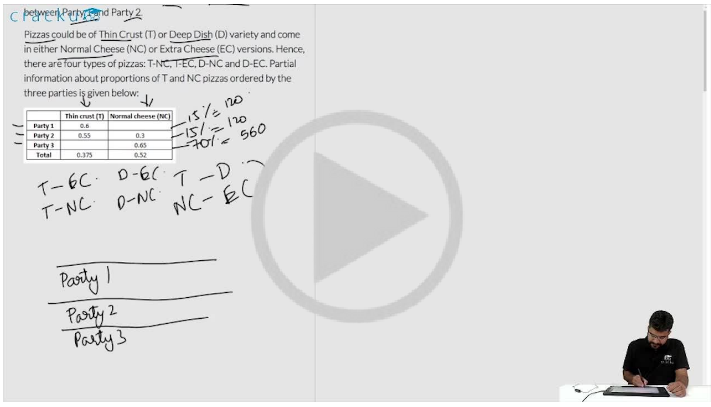
How many Normal Cheese pizzas were required to be delivered to Party 1?

A 104

B 84

C 16

D 196



VIDEO SOLUTION

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Question 31

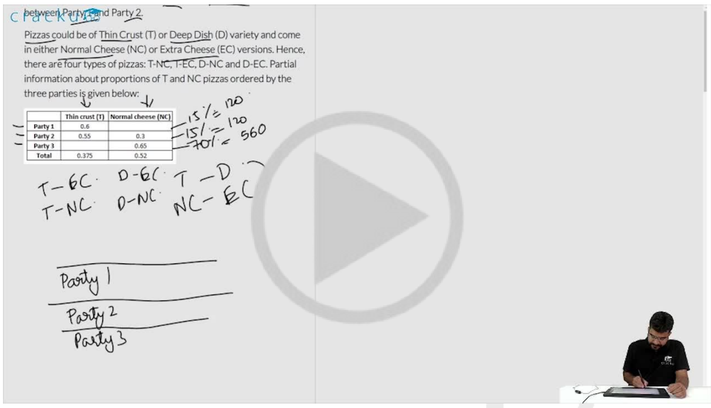
How many Thin Crust pizzas were to be delivered to Party 3?

A 398

B 162

C 96

D 364



VIDEO SOLUTION

Instructions

Five restaurants, coded R1, R2, R3, R4 and R5 gave integer ratings to five gig workers -

Ullas, Vasu, Waman, Xavier and Yusuf, on a scale of 1 to 5.

The means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively. The summary statistics of these ratings for the five workers is given below.

	Ullas	Vasu	Waman	Xavier	Yusuf
Mean rating	2.2	3.8	3.4	3.6	2.6
Median rating	2	4	4	4	3
Modal rating	2	4	5	5	1 and 4
Range of rating*	3	3	4	4	3

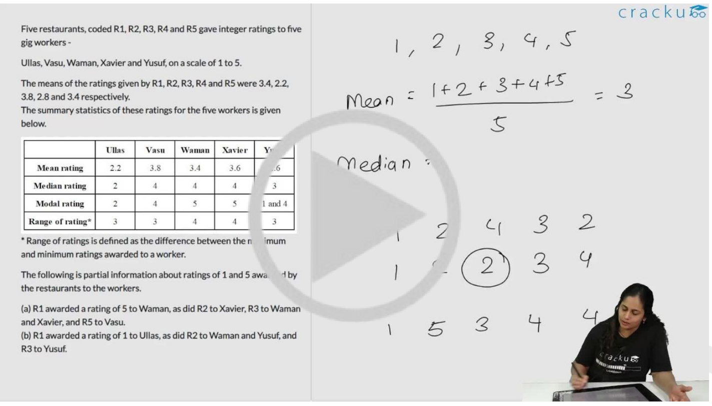
* Range of ratings is defined as the difference between the maximum and minimum ratings awarded to a worker.

The following is partial information about ratings of 1 and 5 awarded by the restaurants to the workers.

(a) R1 awarded a rating of 5 to Waman, as did R2 to Xavier, R3 to Waman and Xavier, and R5 to Vasu.
(b) R1 awarded a rating of 1 to Ullas, as did R2 to Waman and Yusuf, and R3 to Yusuf.

Question 32

To how many workers did R2 give a rating of 4?



VIDEO SOLUTION

Question 33

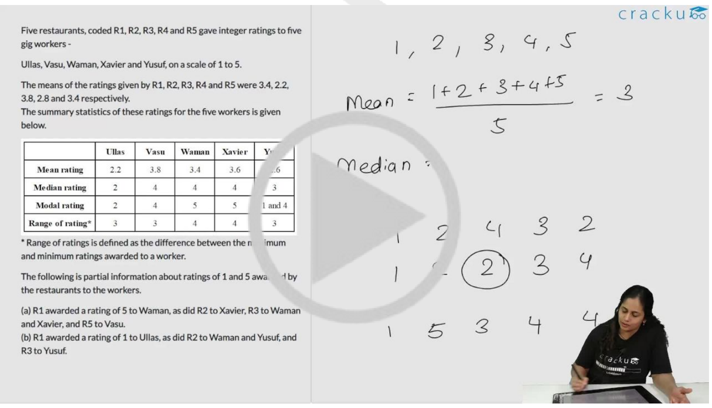
Which among the following restaurants gave its median rating to exactly one of the workers?

A R2

B R5

C R4

D R3



VIDEO SOLUTION

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Instructions

Answer the following questions based on the information given below:

For admission to various affiliated colleges, a university conducts a written test with four different sections, each with a maximum of 50 marks. The following table gives the aggregate as well as the sectional cut-off marks fixed by six different colleges affiliated to the university. A student will get admission only if he/she gets marks greater than or equal to the cut-off marks in each of the sections and his/her aggregate marks are at least equal to the aggregate cut-off marks as specified by the college.

	Sectional Cut-Off Marks				Aggregate Cut - off Marks
	Section A	Section B	Section C	Section D	
College 1	42	42	42		176
College 2		45	45		175
College 3			46		171
College 4	43			45	178
College 5	45		43		180
College 6		41		44	176

Question 34

Aditya did not get a call from even a single college. What could be the maximum aggregate marks obtained by him?

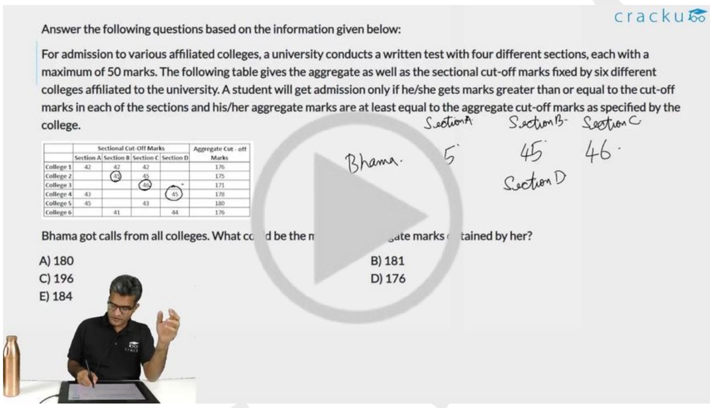
A 181

B 176

C 184

D 196

E 190



VIDEO SOLUTION

Question 35

Charlie got calls from two colleges. What could be the minimum marks obtained by him in a section?

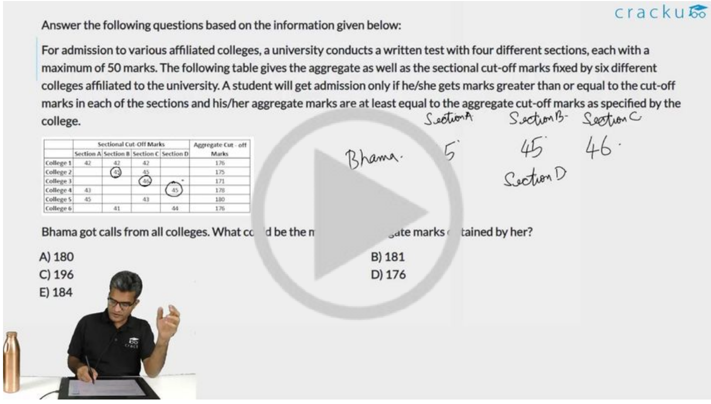
A 0

B 21

C 25

D 35

E 41



VIDEO SOLUTION

Question 36

Bhama got calls from all colleges. What could be the minimum aggregate marks obtained by her?

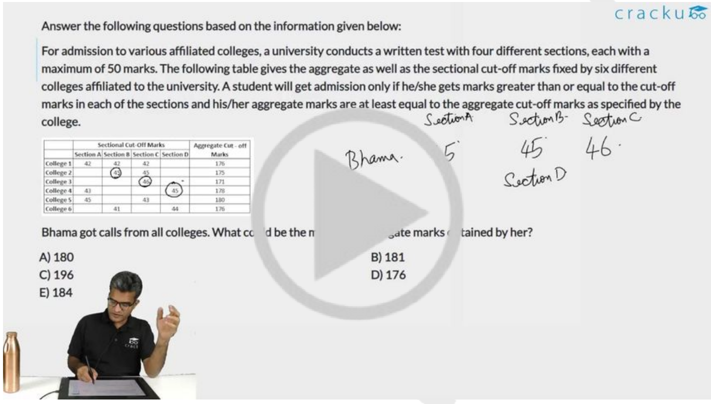
A 180

B 181

C 196

D 176

E 184



VIDEO SOLUTION

Free Videos for CAT Preparation

Instructions

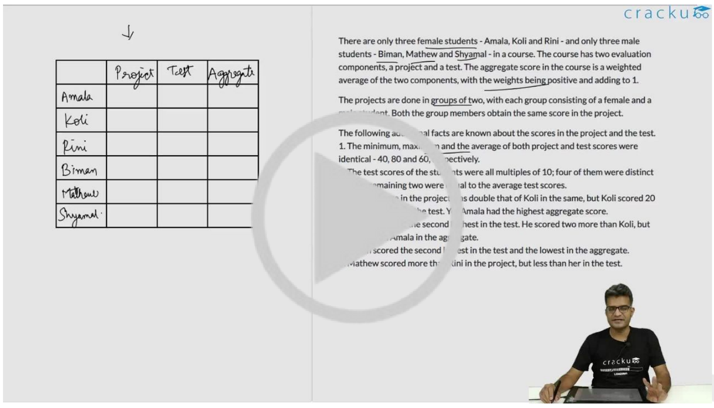
There are only three female students - Amala, Koli and Rini - and only three male students - Biman, Mathew and Shyamal - in a course. The course has two evaluation components, a project and a test. The aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

The projects are done in groups of two, with each group consisting of a female and a male student. Both the group members obtain the same score in the project.

The following additional facts are known about the scores in the project and the test.
1. The minimum, maximum and the average of both project and test scores were identical - 40, 80 and 60, respectively.
2. The test scores of the students were all multiples of 10; four of them were distinct and the remaining two were equal to the average test scores.
3. Amala’s score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Yet Amala had the highest aggregate score.
4. Shyamal scored the second highest in the test. He scored two more than Koli, but two less than Amala in the aggregate.
5. Biman scored the second lowest in the test and the lowest in the aggregate.
6. Mathew scored more than Rini in the project, but less than her in the test.

Question 37

What was Rini’s score in the project?



VIDEO SOLUTION

Instructions

A company administers a written test comprising of three sections of 20 marks each - Data Interpretation (DI), Written English (WE) and General Awareness (GA), for recruitment. A composite score for a candidate (out of 80) is calculated by doubling her marks in DI and adding it to the sum of her marks in the other two sections. Candidates who score less than 70% marks in two or more sections are disqualified. From among the rest, the four with the highest composite scores are recruited. If four or less candidates qualify, all who qualify are recruited.

Ten candidates appeared for the written test. Their marks in the test are given in the table below. Some marks in the table are missing, but the following facts are known:

1. No two candidates had the same composite score.
2. Ajay was the unique highest scorer in WE.
3. Among the four recruited, Geeta had the lowest composite score.
4. Indu was recruited.
5. Danish, Harini, and Indu had scored the same marks the in GA.
6. Indu and Jatin both scored 100% in exactly one section and Jatin’s composite score was 10 more than Indu’s.

Candidate	Marks out of 20		
	DI	WE	GA
Ajay	8		16
Bala		9	11
Chetna	19	4	12
Danish	8	15	
Ester	12	18	16
Falak	15	7	10
Geeta	14		6
Harini	5		
Indu		8	
Jatin		16	14

Question 38

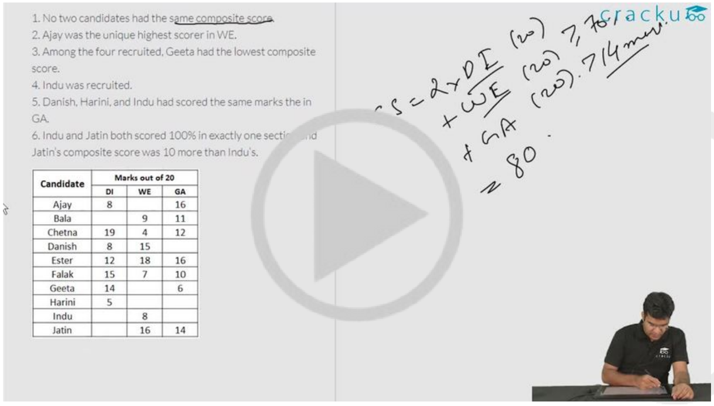
Which of the following statements MUST be FALSE?

A Bala scored same as Jatin in DI

B Harini’s composite score was less than that of Falak

C Bala’s composite score was less than that of Ester

D Chetna scored more than Bala in DI



VIDEO SOLUTION

Instructions

In an election several candidates contested for a constituency. In any constituency, the winning candidate was the one who polled the highest number of votes, the first runner up was the one who polled the second highest number of votes, the second runner up was the one who polled the third highest number of votes, and so on. There were no ties (in terms of number of votes polled by the candidates) in any of the constituencies in this election. In an electoral system, a security deposit is the sum of money that a candidate is required to pay to the election commission before he or she is permitted to contest. Only the defeated candidates (i.e., one who is not the winning candidate) who fail to secure more than one sixth of the valid votes polled in the constituency, lose their security deposits.

The following table provides some incomplete information about votes polled in four constituencies: A, B, C and D, in this election .

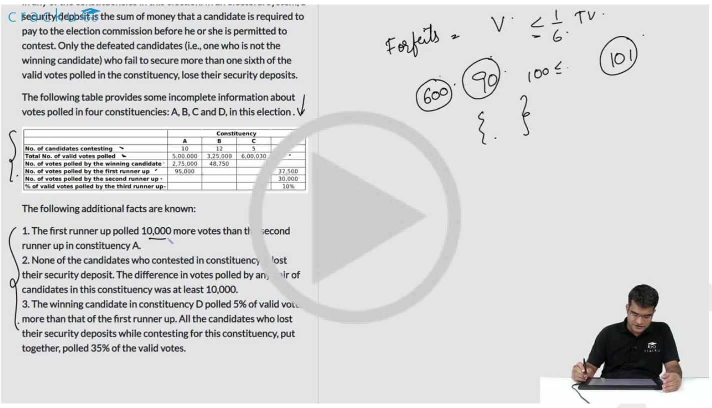
	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	
No. of votes polled by the winning candidate	2,75,000	48,750		
No. of votes polled by the first runner up	95,000			37,500
No. of votes polled by the second runner up				30,000
% of valid votes polled by the third runner up				10%

The following additional facts are known:

1. The first runner up polled 10,000 more votes than the second runner up in constituency A.
2. None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000.
3. The winning candidate in constituency D polled 5% of valid votes more than that of the first runner up. All the candidates who lost their security deposits while contesting for this constituency, put together, polled 35% of the valid votes.

Question 39

What is the percentage of votes polled in total by all the candidates who lost their security deposits while contesting for constituency A?



VIDEO SOLUTION

CAT Percentile Predictor

Instructions

A company administers a written test comprising of three sections of 20 marks each - Data Interpretation (DI), Written English (WE) and General Awareness (GA), for recruitment. A composite score for a candidate (out of 80) is calculated by doubling her marks in DI and adding it to the sum of her marks in the other two sections. Candidates who score less than 70% marks in two or more sections are disqualified. From among the rest, the four with the highest composite scores are recruited. If four or less candidates qualify, all who qualify are recruited.

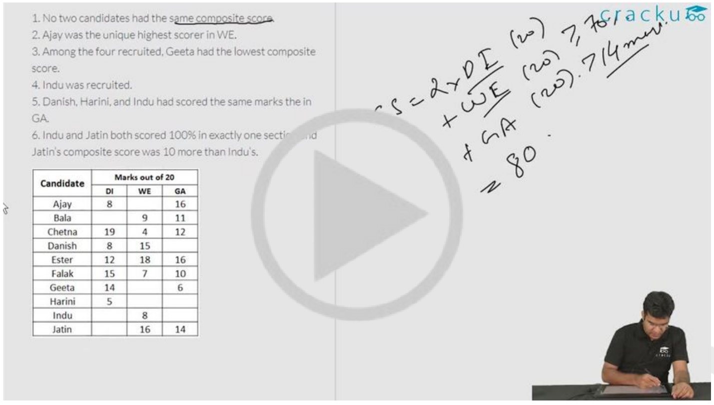
Ten candidates appeared for the written test. Their marks in the test are given in the table below. Some marks in the table are missing, but the following facts are known:

1. No two candidates had the same composite score.
2. Ajay was the unique highest scorer in WE.
3. Among the four recruited, Geeta had the lowest composite score.
4. Indu was recruited.
5. Danish, Harini, and Indu had scored the same marks the in GA.
6. Indu and Jatin both scored 100% in exactly one section and Jatin's composite score was 10 more than Indu's.

Candidate	Marks out of 20		
	DI	WE	GA
Ajay	8		16
Bala		9	11
Chetna	19	4	12
Danish	8	15	
Ester	12	18	16
Falak	15	7	10
Geeta	14		6
Harini	5		
Indu		8	
Jatin		16	14

Question 40

If all the candidates except Ajay and Danish had different marks in DI, and Bala's composite score was less than Chetna's composite score, then what is the maximum marks that Bala could have scored in DI?



VIDEO SOLUTION

Instructions

Odsville has five firms - Alfloo, Bzygoo, Czechy, Drjbna and Elavalaki. Each of these firms was founded in some year and also closed down a few years later.

Each firm raised Rs. 1 crore in its first and last year of existence. The amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down. No firm raised the same amount of money in two consecutive years. Each annual increase and decrease was either by Rs. 1 crore or by Rs. 2 crores. The table below provides partial information about the five firms.

Firm	First year of existence	Last year of existence	Total amount raised (Rs. Crores)
Alfloo	2009	2016	21
Bzygoo	2012	2015	
Czechy	2013		9
Drjbna	2011	2015	10
Elavalaki	2010		13

Question 41

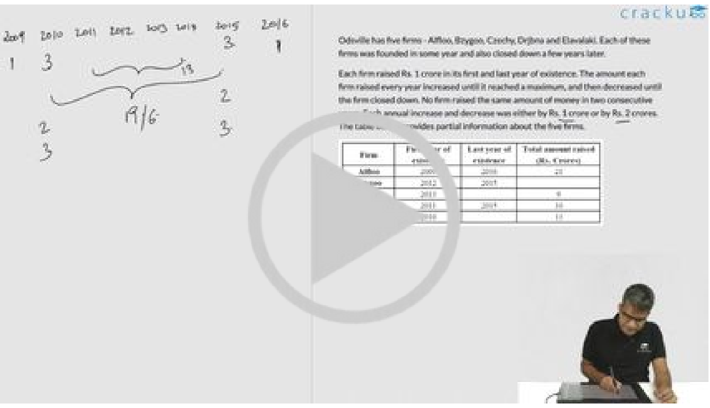
What best can be concluded about the total amount of money raised in 2015?

A It is either Rs. 7 crores or Rs. 8 crores or Rs. 9 crores.

B It is exactly Rs. 8 crores.

C It is either Rs. 7 crores or Rs. 8 crores.

D It is either Rs. 8 crores or Rs. 9 crores.



VIDEO SOLUTION

Question 42

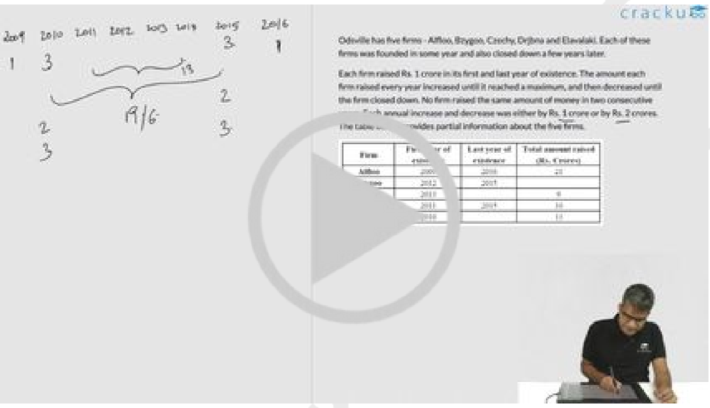
If Elavalaki raised Rs. 3 crores in 2013, then what is the smallest possible total amount of money (in Rs. crores) that could have been raised by all the companies in 2012?

A 12

B 9

C 11

D 10



VIDEO SOLUTION

Instructions

In a coaching class, some students register online, and some others register offline. No student registers both online and offline; hence the total registration number is the sum of online and offline registrations. The following facts and table pertain to these registration numbers for the five months - January to May of 2023. The table shows the minimum, maximum, median registration numbers of these five months, separately for online, offline and total number of registrations. The following additional facts are known.

1. In every month, both online and offline registration numbers were multiples of 10.
2. In January, the number of offline registrations was twice that of online registrations.
3. In April, the number of online registrations was twice that of offline registrations.
4. The number of online registrations in March was the same as the number of offline registrations in February.
5. The number of online registrations was the largest in May.

	Minimum	Maximum	Median
Online	40	100	80
Offline	30	80	50
Total	110	130	120

Question 43

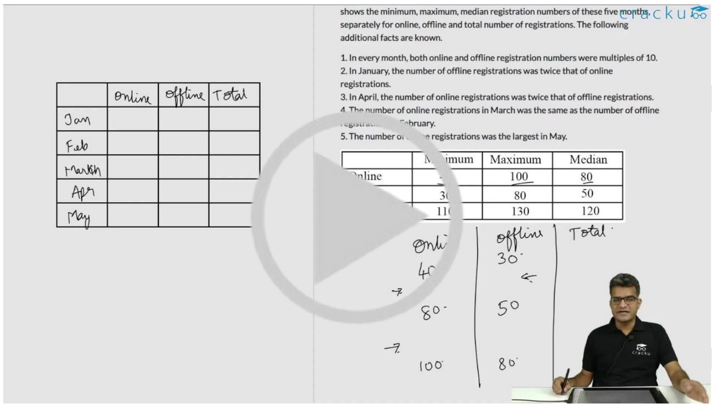
Which of the following statements can be true?
I. The number of offline registrations was the smallest in May.
II. The total number of registrations was the smallest in February.

A Both I and II

B Only II

C Neither I nor II

D Only I



VIDEO SOLUTION

Instructions

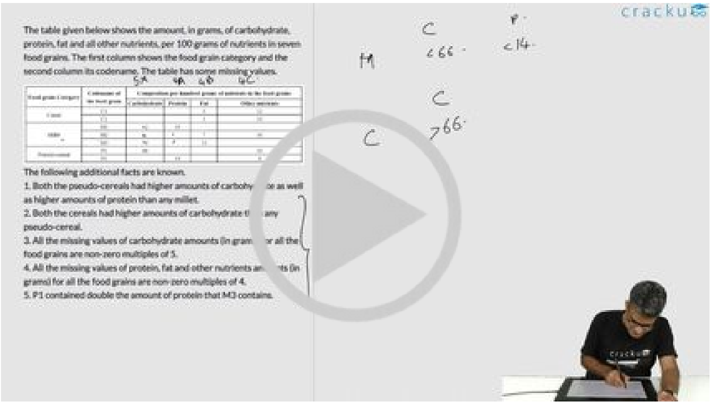
The table given below shows the amount, in grams, of carbohydrate, protein, fat and all other nutrients, per 100 grams of nutrients in seven food grains. The first column shows the food grain category and the second column its codename. The table has some missing values.

Food grain Category	Codename of the food grain	Composition per hundred grams of nutrients in the food grains			
		Carbohydrate	Protein	Fat	Other nutrients
Cereal	C1			0	12
	C2			3	10
Millet	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereal	P1	66			10
	P2		14		8

The following additional facts are known.
1. Both the pseudo-cereals had higher amounts of carbohydrate as well as higher amounts of protein than any millet.
2. Both the cereals had higher amounts of carbohydrate than any pseudo-cereal.
3. All the missing values of carbohydrate amounts (in grams) for all the food grains are non-zero multiples of 5.
4. All the missing values of protein, fat and other nutrients amounts (in grams) for all the food grains are non-zero multiples of 4.
5. P1 contained double the amount of protein that M3 contains.

Question 44

How many foodgrains had a higher amount of carbohydrate per 100 grams ofnutrients than M1?



VIDEO SOLUTION

Instructions

In an election several candidates contested for a constituency. In any constituency, the winning candidate was the one who polled the highest number of votes, the first runner up was the one who polled the second highest number of votes, the second runner up was the one who polled the third highest number of votes, and so on. There were no ties (in terms of number of votes polled by the candidates) in any of the constituencies in this election. In an electoral system, a security deposit is the sum of money that a candidate is required to pay to the election commission before he or she is permitted to contest. Only the defeated candidates (i.e., one who is not the winning candidate) who fail to secure more than one sixth of the valid votes polled in the constituency, lose their security deposits.

The following table provides some incomplete information about votes polled in four constituencies: A, B, C and D, in this election .

	Constituency			
	A	**B**	**C**	**D**
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	
No. of votes polled by the winning candidate	2,75,000	48,750		
No. of votes polled by the first runner up	95,000			37,500
No. of votes polled by the second runner up				30,000
% of valid votes polled by the third runner up				10%

The following additional facts are known:

1. The first runner up polled 10,000 more votes than the second runner up in constituency A.
2. None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000.
3. The winning candidate in constituency D polled 5% of valid votes more than that of the first runner up. All the candidates who lost their security deposits while contesting for this constituency, put together, polled 35% of the valid votes.

Question 45

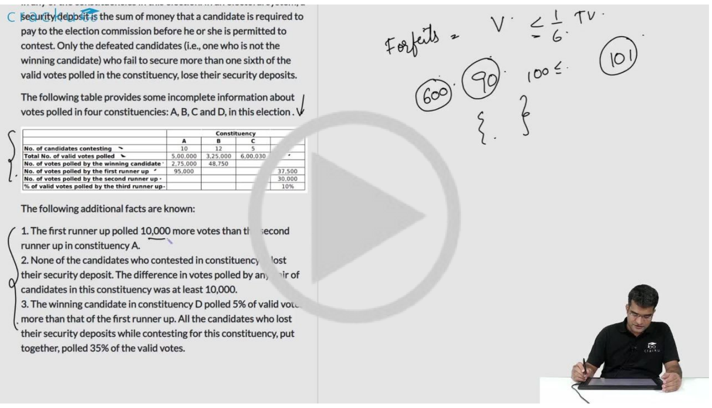
What BEST can be concluded about the number of votes polled by the winning candidate in constituency C?

A 1,40,010

B between 1,40,005 and 1,40,010

C less than 2,00,010

D 1,40,006



VIDEO SOLUTION

Data Interpretation for CAT Questions (download pdf)

Question 46

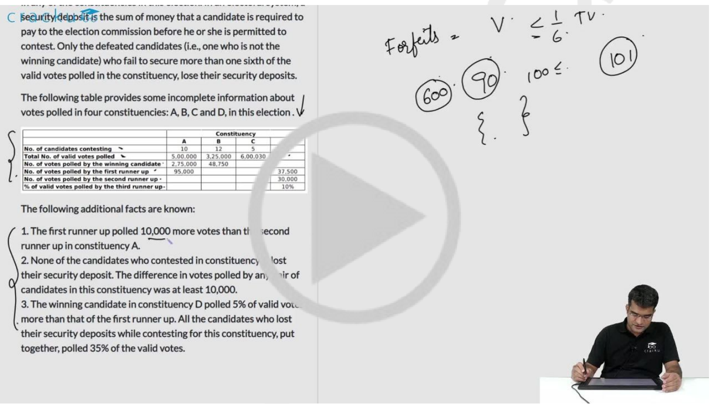
The winning margin of a constituency is defined as the difference of votes polled by the winner and that of the first runner up. Which of the following CANNOT be the list of constituencies, in increasing order of winning margin?

A D, B, C, A

B B, D, C, A

C B, C, D, A

D D, C, B, A



VIDEO SOLUTION

Instructions

An agency entrusted to accredit colleges looks at four parameters: faculty quality (F), reputation (R), placement quality (P), and infrastructure (I). The four parameters are used to arrive at an overall score, which the agency uses to give an accreditation to the colleges. In each parameter, there are five possible letter grades given, each carrying certain points: A (50 points), B (40 points), C (30 points), D (20 points), and F (0 points). The overall score for a college is the weighted sum of the points scored in the four parameters. The weights of the parameters are 0.1, 0.2, 0.3 and 0.4 in some order, but the order is not disclosed. Accreditation is awarded based on the following scheme:

Range	Accreditation
Overall score ≥ 45	AAA
35 ≤ Overall score < 45	BAA
25 ≤ Overall score < 35	BBA
15 ≤ Overall score < 25	BBB
Overall score < 15	Junk

Eight colleges apply for accreditation, and receive the following grades in the four parameters (F, R, P, and I):

	F	R	P	I
A-one	A	A	A	B
Best Ed	B	C	D	D
Cosmopolitan	B	D	D	C
Dominance	D	D	B	C
Education Aid	A	A	B	A
Fancy	A	A	B	B
Global	C	F	D	D
High Q	C	D	D	B

It is further known that in terms of overall scores:
1. High Q is better than Best Ed;
2. Best Ed is better than Cosmopolitan; and
3. Education Aid is better than A-one.

Question 47

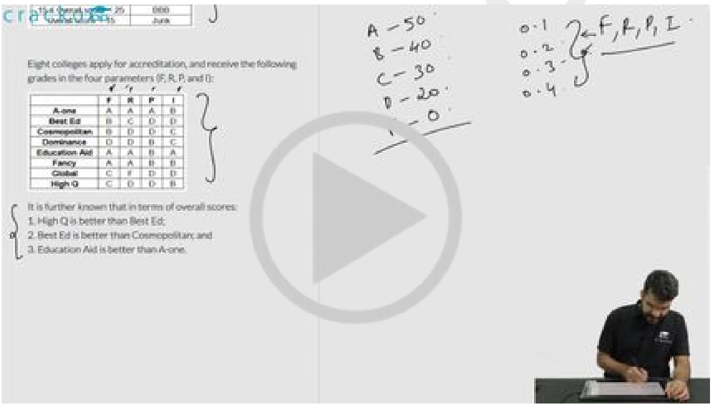
What is the weight of the faculty quality parameter?

A 0.2

B 0.3

C 0.4

D 0.1



VIDEO SOLUTION

Question 48

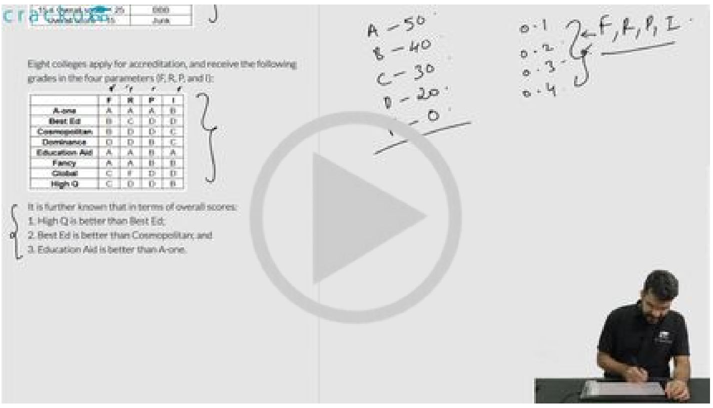
How many colleges have overall scores between 31 and 40, both inclusive?

A 0

B 2

C 1

D 3



VIDEO SOLUTION

Logical Reasoning for CAT Questions (download pdf)

Instructions

Amudha, Bharatan, Chandran, Dhinesh, Ezhil, Fani and Gowtham are seven people in a town. Any pair of them could either be strangers, acquaintances, or friends. All relationships are mutual. For example, if Amudha is a friend of Bharatan, then Bharatan is also a friend of Amudha. Similarly, if Amudha is a stranger to Bharatan, then Bharatan is also a stranger to Amudha.

Partial information about the number of friends, acquaintances, and strangers of each of these people among them is given in the table below.

	No. of Friends	No. of Acquaintances	No. of Strangers
Amudha		1	4
Bharatan			
Chandran		1	
Dhinesh			2
Ezhil			1
Fani	1		
Gowtham		3	2

The following additional facts are also known.
1. Amudha, Bharatan, and Chandran are mutual strangers.
2. Amudha, Dhinesh, and Fani are Ezil's friends.
3. Chandran and Gowtham are friends.
4. Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance of Bharatan is a friend of Amudha.
5. Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan.

Question 49

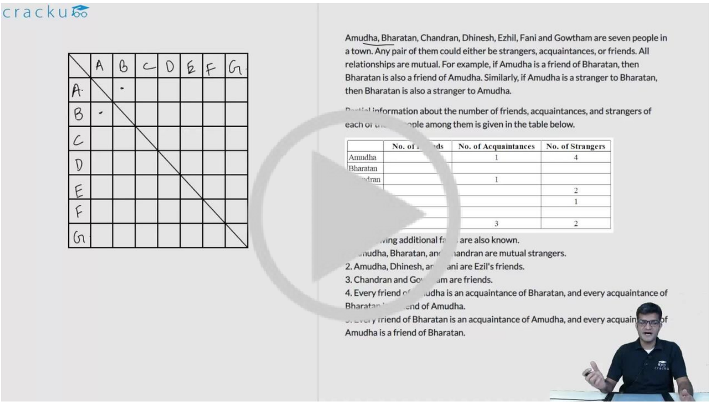
Which of these pairs share the same type of relationship?

A (Amudha, Gowtham) and (Ezhil, Fani)

B (Bharatan, Chandran) and (Dhinesh, Ezhil)

C (Chandran, Ezhil) and (Dhinesh, Gowtham)

D (Bharatan, Ezhil) and (Fani, Gowtham)



VIDEO SOLUTION

Question 50

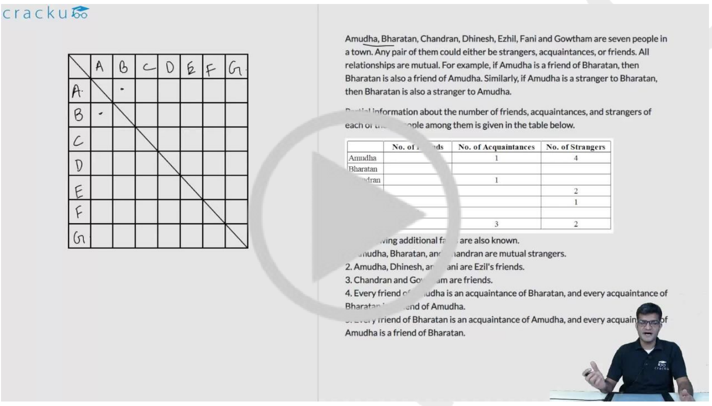
Who is an acquaintance of Chandran?

A Dhinesh

B Fani

C Ezhil

D Bharatan



VIDEO SOLUTION

Instructions

Odsville has five firms - Alfloo, Bzygoo, Czechy, Drjbna and Elavalaki. Each of these firms was founded in some year and also closed down a few years later.

Each firm raised Rs. 1 crore in its first and last year of existence. The amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down. No firm raised the same amount of money in two consecutive years. Each annual increase and decrease was either by Rs. 1 crore or by

Rs. 2 crores. The table below provides partial information about the five firms.

Firm	First year of existence	Last year of existence	Total amount raised (Rs. Crores)
Alfloo	2009	2016	21
Bzygoo	2012	2015	
Czechy	2013		9
Drjbna	2011	2015	10
Elavalaki	2010		13

Question 51

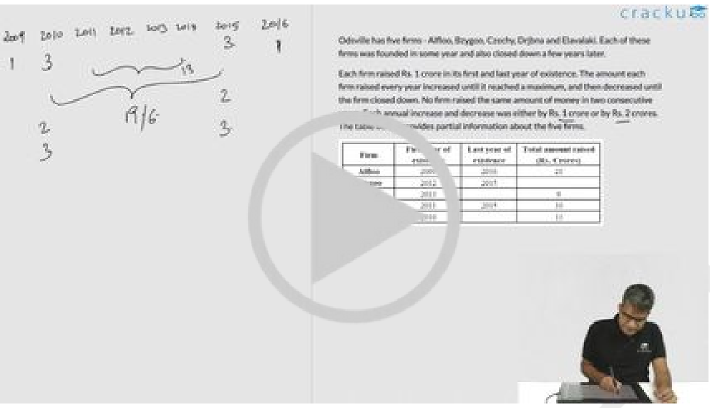
If the total amount of money raised in 2014 is Rs. 12 crores, then which of the following is not possible?

A Bzygoo raised the same amount of money as Elavalaki in 2013.

B Alfloo raised the same amount of money as Drjbna in 2013.

C Alfloo raised the same amount of money as Bzygoo in 2014.

D Bzygoo raised more money than Elavalaki in 2014.



VIDEO SOLUTION

Instructions

Amudha, Bharatan, Chandran, Dhinesh, Ezhil, Fani and Gowtham are seven people in a town. Any pair of them could either be strangers, acquaintances, or friends. All relationships are mutual. For example, if Amudha is a friend of Bharatan, then Bharatan is also a friend of Amudha. Similarly, if Amudha is a stranger to Bharatan, then Bharatan is also a stranger to Amudha.

Partial information about the number of friends, acquaintances, and strangers of each of these people among them is given in the table below.

	No. of Friends	No. of Acquaintances	No. of Strangers
Amudha		1	4
Bharatan			
Chandran		1	
Dhinesh			2
Ezhil			1
Fani	1		
Gowtham		3	2

The following additional facts are also known.
1. Amudha, Bharatan, and Chandran are mutual strangers.
2. Amudha, Dhinesh, and Fani are Ezil's friends.
3. Chandran and Gowtham are friends.
4. Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance of Bharatan is a friend of Amudha.
5. Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan.

Question 52

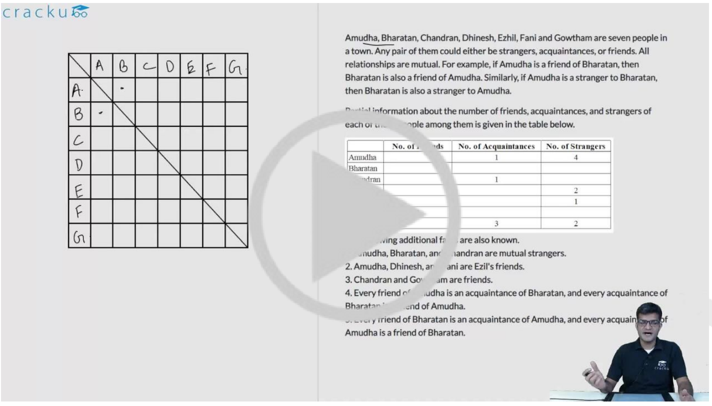
Who are Gowtham's acquaintances?

A Dhinesh, Ezhil and Fani

B Amudha, Dhinesh and Fani

C Bharatan, Dhinesh and Ezhil

D Amudha, Bharatan and Fani



VIDEO SOLUTION

Instructions

Five restaurants, coded R1, R2, R3, R4 and R5 gave integer ratings to five gig workers -

Ullas, Vasu, Waman, Xavier and Yusuf, on a scale of 1 to 5.

The means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.
The summary statistics of these ratings for the five workers is given below.

	Ullas	Vasu	Waman	Xavier	Yusuf
Mean rating	2.2	3.8	3.4	3.6	2.6
Median rating	2	4	4	4	3
Modal rating	2	4	5	5	1 and 4
Range of rating*	3	3	4	4	3

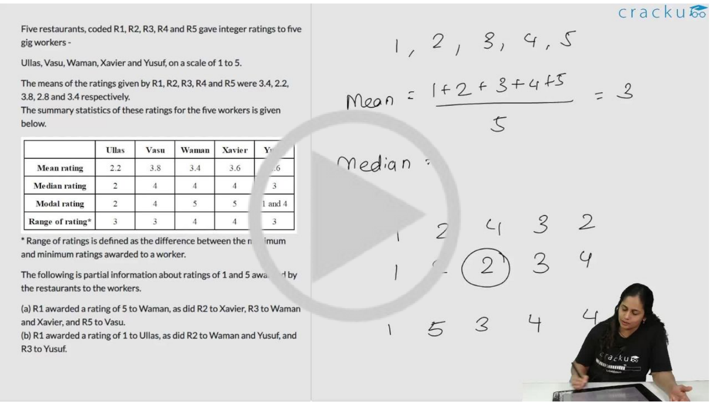
* Range of ratings is defined as the difference between the maximum and minimum ratings awarded to a worker.

The following is partial information about ratings of 1 and 5 awarded by the restaurants to the workers.

(a) R1 awarded a rating of 5 to Waman, as did R2 to Xavier, R3 to Waman and Xavier, and R5 to Vasu.
(b) R1 awarded a rating of 1 to Ullas, as did R2 to Waman and Yusuf, and R3 to Yusuf.

Question 53

What rating did R1 give to Xavier?



VIDEO SOLUTION

Instructions

DIRECTIONS for the following four questions:

The proportion of male students and the proportion of vegetarian students in a school are given below. The school has a total of 800 students, 80% of whom are in the Secondary Section and rest equally divided between Class 11 and 12.

	Male (M)	Vegetarian (V)
Class 12	0.6	
Class 11	0.55	0.5
Secondary Section		0.55
Total	0.475	0.53

Question 54

In the Secondary Section, 50% of the males are vegetarian. Which of the following statements is correct?

A Except vegetarian males, all other groups have same number of students.

B Except non-vegetarian males, all other groups have same number of students.

C Except vegetarian females, all other groups have same number of students.

D Except non-vegetarian females, all other groups have same number of students.

E All of the above groups have the same number of students.

VIEW SOLUTION

Question 55

What is the percentage of vegetarian students in Class 12?

A 40

B 45

C 50

D 55

E 60

VIEW SOLUTION

Question 56

In Class 12, twenty five per cent of the vegetarians are male. What is the difference between the number of female vegetarians and male non-vegetarians?

A less than 8

B 10

C 12

D 14

E 16

VIEW SOLUTION

Question 57

What is the percentage of male students in the secondary section?

A 40

B 45

C 50

D 55

E 60

VIEW SOLUTION

Instructions

The table given below shows the amount, in grams, of carbohydrate, protein, fat and all other nutrients, per 100 grams of nutrients in seven food grains. The first column shows the food grain category and the second column its codename. The table has some missing values.

Food grain Category	Codename of the food grain	Composition per hundred grams of nutrients in the food grains			
		Carbohydrate	Protein	Fat	Other nutrients
Cereal	C1			0	12
	C2			3	10
Millet	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereal	P1	66			10
	P2		14		8

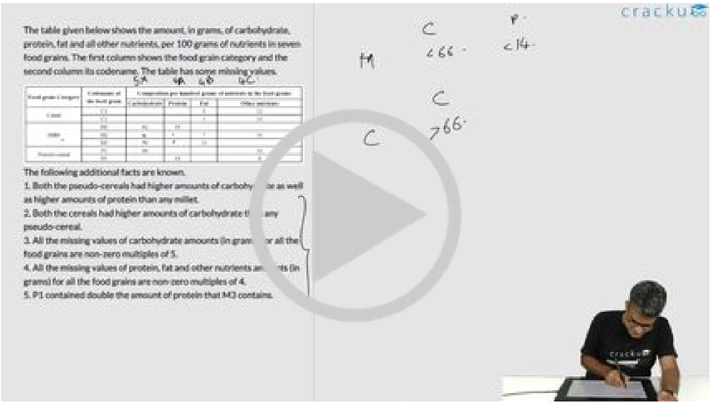
The following additional facts are known.

1. Both the pseudo-cereals had higher amounts of carbohydrate as well as higher amounts of protein than any millet.

2. Both the cereals had higher amounts of carbohydrate than any pseudo-cereal.
3. All the missing values of carbohydrate amounts (in grams) for all the food grains are non-zero multiples of 5.
4. All the missing values of protein, fat and other nutrients amounts (in grams) for all the food grains are non-zero multiples of 4.
5. P1 contained double the amount of protein that M3 contains.

Question 58

How many grams of protein were there in 100 grams of nutrients in M2?



VIDEO SOLUTION

Instructions

The year is 2089. Beijing, London, New York, and Paris are in contention to host the 2096 Olympics. The eventual winner is determined through several rounds of voting by members of the IOC with each member representing a different city. All the four cities in contention are also represented in IOC.

Round	Total Votes Cast	Maximum Votes Cast		Eliminated	
		City	No. of Votes	City	No. of Votes
1		London	30	New York	12
2	83	Paris	32	Beijing	21
3	75				

- In any round of voting, the city receiving the lowest number of votes in that round gets eliminated. The survivor after the last round of voting gets to host the event.
- A member is allowed to cast votes for at most two different cities in all rounds of voting combined. (Hence, a member becomes ineligible to cast a vote in a given round if both the cities (s)he voted for in earlier rounds are out of contention in that round of voting.)
- A member is also ineligible to cast a vote in a round if the city (s)he represents is in contention in that round of voting.
- As long as the member is eligible, (s)he must vote and vote for only one candidate city in any round of voting.The following incomplete table shows the information on cities that received the maximum and minimum votes in different rounds, the number of votes cast in their favour, and the total votes that were cast in those rounds.It is also known that:
- All those who voted for London and Paris in round 1, continued to vote for the samecities in subsequent rounds as long as these cities were in contention. 75% of those whovoted for Beijing in round 1, voted for Beijing in round 2 as well.
- Those who voted for New York in round 1, voted either for Beijing or Paris in round 2.
- The difference in votes cast for the two contending cities in the last round was 1.
- 50% of those who voted for Beijing in round 1, voted for Paris in round 3.

Question 59

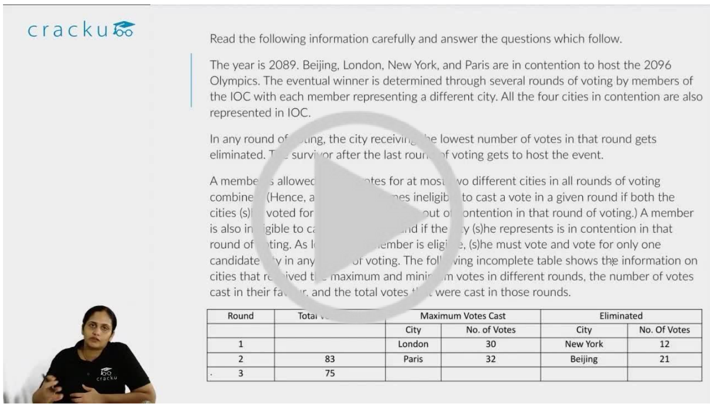
What percentage of members from among those who voted for Beijing in round 2 and were eligible to vote in round 3, voted for London?

A 33.33

B 38.10

C 50

D 66.67



VIDEO SOLUTION

Instructions

In an election several candidates contested for a constituency. In any constituency, the winning candidate was the one who polled the highest number of votes, the first runner up was the one who polled the second highest number of votes, the second runner up was the one who polled the third highest number of votes, and so on. There were no ties (in terms of number of votes polled by the candidates) in any of the constituencies in this election. In an electoral system, a security deposit is the sum of money that a candidate is required to pay to the election commission before he or she is permitted to contest. Only the defeated candidates (i.e., one who is not the winning candidate) who fail to secure more than one sixth of the valid votes polled in the constituency, lose their security deposits.

The following table provides some incomplete information about votes polled in four constituencies: A, B, C and D, in this election .

	Constituency			
	A	**B**	**C**	**D**
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	
No. of votes polled by the winning candidate	2,75,000	48,750		
No. of votes polled by the first runner up	95,000			37,500
No. of votes polled by the second runner up				30,000
% of valid votes polled by the third runner up				10%

The following additional facts are known:

1. The first runner up polled 10,000 more votes than the second runner up in constituency A.
2. None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000.
3. The winning candidate in constituency D polled 5% of valid votes more than that of the first runner up. All the candidates who lost their security deposits while contesting for this constituency, put together, polled 35% of the valid votes.

Question 60

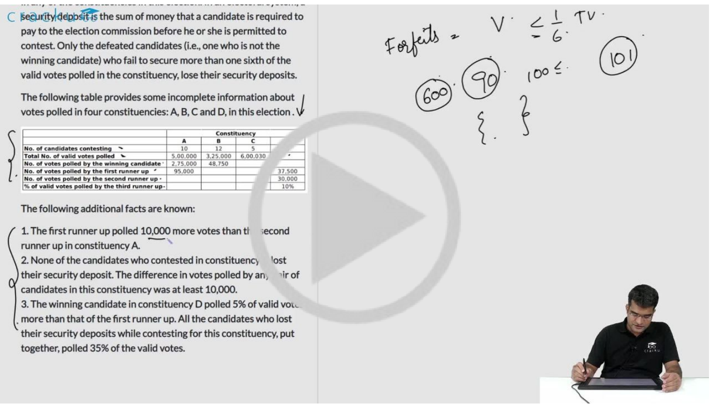
For all the four constituencies taken together, what was the approximate number of votes polled by all the candidates who lost their security deposit expressed as a percentage of the total valid votes from these four constituencies?

A 38.25%

B 23.54%

C 23.91%

D 32.00%



VIDEO SOLUTION

Enroll for Excellent CAT/MBA courses

Question 61

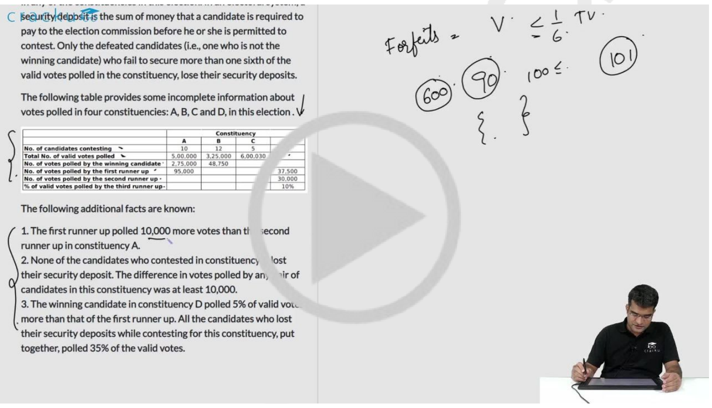
What was the number of valid votes polled in constituency D?

A 1,25,000

B 1,50,000

C 1,75,000

D 62,500



VIDEO SOLUTION

Instructions

A chain of departmental stores has outlets in Delhi, Mumbai, Bengaluru and Kolkata. The sales are categorized by its three departments - 'Apparel', 'Electronics', and 'Home Décor'. An Accountant has been asked to prepare a summary of the 2018 and 2019 sales amounts for an internal report. He has collated partial information and prepared the following table.

Sales Amounts (Crore Rupees)								
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparels	-	-	-	-	-	-	-	54
Electronics	78	98	82	102	90	70	80	100
HomeDecor	-	100	-	72	-	80	-	54

The following additional information is known.
1. The sales amounts in the Apparel departments were the same for Delhi and Kolkata in 2018.
2. The sales amounts in the Apparel departments were the same for Mumbai and Bengaluru in 2018. This sales amount matched the sales amount in the Apparel department for Delhi in 2019.
3. The sales amounts in the Home Décor departments were the same for Mumbai and Kolkata in 2018.
4. The sum of the sales amounts of four Electronics departments increased by the same amount as the sum of the sales amounts of four Apparel departments from 2018 to 2019.
5. The total sales amounts of the four Home Décor departments increased by Rs 70 Crores from 2018 to 2019.
6. The sales amounts in the Home Décor departments of Delhi and Bengaluru each increased by Rs 20 Crores from 2018 to 2019.
7. The sales amounts in the Apparel departments of Delhi and Bengaluru each increased by the same amount in 2019 from 2018. The sales amounts in the Apparel departments of Mumbai and Kolkata also each increased by the same amount in 2019 from 2018.
8. The sales amounts in the Apparel departments of Delhi, Kolkata and Bengaluru in 2019 followed an Arithmetic Progression.

Question 62

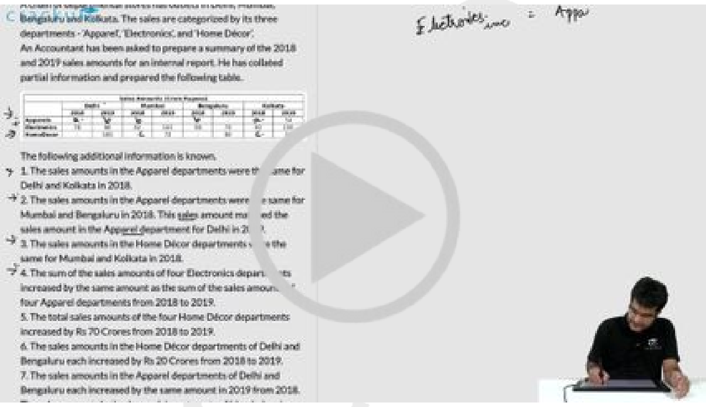
What was the total sales amount, in Crore Rupees, in 2019 for the chain of departmental stores?

A 600

B 900

C 150

D 750



VIDEO SOLUTION

Question 63

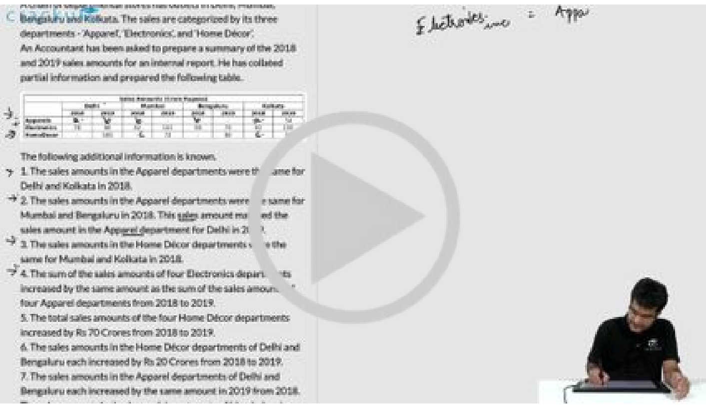
Among all the 12 departments (i.e., the 3 departments in each of the 4 cities), what was the maximum percentage increase in sales amount from 2018 to 2019?

A 25

B 28

C 75

D 50



VIDEO SOLUTION

Question 64

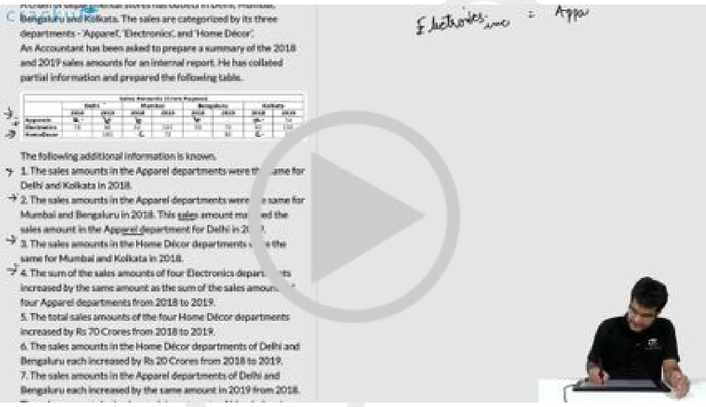
What was the increase in sales amount, in Crore Rupees, in the Apparel department of Mumbai from 2018 to 2019?

A 12

B 8

C 5

D 10



VIDEO SOLUTION

Question 65

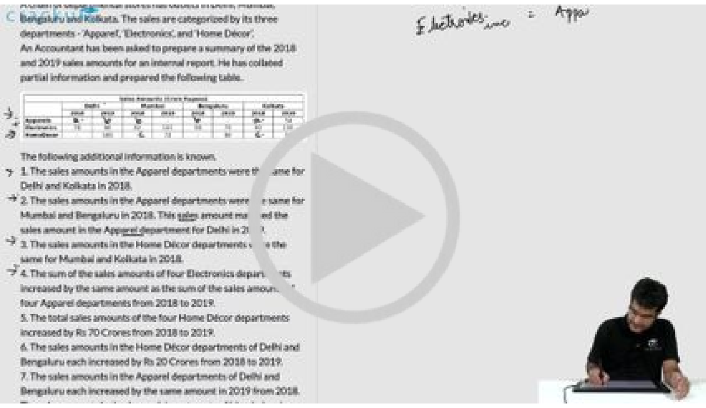
In Home Décor departments of which cities were the sales amounts the highest in 2018 and 2019, respectively?

A Bengaluru and Delhi

B Mumbai and Mumbai

C Mumbai and Delhi

D Delhi and Delhi



VIDEO SOLUTION

Instructions

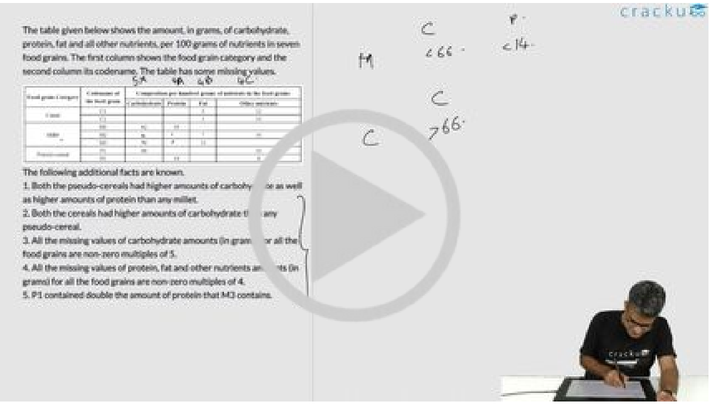
The table given below shows the amount, in grams, of carbohydrate, protein, fat and all other nutrients, per 100 grams of nutrients in seven food grains. The first column shows the food grain category and the second column its codename. The table has some missing values.

Food grain Category	Codename of the food grain	Composition per hundred grams of nutrients in the food grains			
		Carbohydrate	Protein	Fat	Other nutrients
Cereal	C1			0	12
	C2			3	10
Millet	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereal	P1	66			10
	P2		14		8

The following additional facts are known.
1. Both the pseudo-cereals had higher amounts of carbohydrate as well as higher amounts of protein than any millet.
2. Both the cereals had higher amounts of carbohydrate than any pseudo-cereal.
3. All the missing values of carbohydrate amounts (in grams) for all the food grains are non-zero multiples of 5.
4. All the missing values of protein, fat and other nutrients amounts (in grams) for all the food grains are non-zero multiples of 4.
5. P1 contained double the amount of protein that M3 contains.

Question 66

How many grams of other nutrients were there in 100 grams of nutrients in M3?



VIDEO SOLUTION

Instructions

There are only three female students - Amala, Koli and Rini - and only three male students - Biman, Mathew and Shyamal - in a course. The course has two evaluation components, a project and a test. The aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

The projects are done in groups of two, with each group consisting of a female and a male student. Both the group members obtain the same score in the project.

The following additional facts are known about the scores in the project and the test.
1. The minimum, maximum and the average of both project and test scores were identical - 40, 80 and 60, respectively.
2. The test scores of the students were all multiples of 10; four of them were distinct and the remaining two were equal to the average test scores.
3. Amala's score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Yet Amala had the highest aggregate score.
4. Shyamal scored the second highest in the test. He scored two more than Koli, but two less than Amala in the aggregate.
5. Biman scored the second lowest in the test and the lowest in the aggregate.
6. Mathew scored more than Rini in the project, but less than her in the test.

Question 67

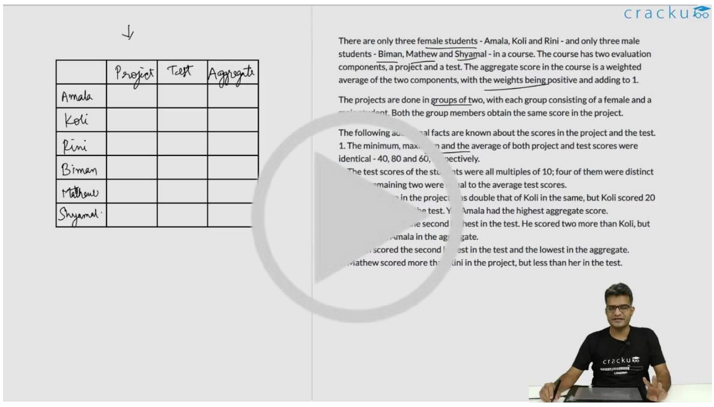
What was the weight of the test component?

A 0.60

B 0.50

C 0.75

D 0.40



VIDEO SOLUTION

Instructions

Amudha, Bharatan, Chandran, Dhinesh, Ezhil, Fani and Gowtham are seven people in a town. Any pair of them could either be strangers, acquaintances, or friends. All relationships are mutual. For example, if Amudha is a friend of Bharatan, then Bharatan is also a friend of Amudha. Similarly, if Amudha is a stranger to Bharatan, then Bharatan is also a stranger to Amudha.

Partial information about the number of friends, acquaintances, and strangers of each of these people among them is given in the table below.

	No. of Friends	No. of Acquaintances	No. of Strangers
Amudha		1	4
Bharatan			
Chandran		1	
Dhinesh			2
Ezhil			1
Fani	1		
Gowtham		3	2

The following additional facts are also known.
1. Amudha, Bharatan, and Chandran are mutual strangers.
2. Amudha, Dhinesh, and Fani are Ezil's friends.
3. Chandran and Gowtham are friends.
4. Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance of Bharatan is a friend of Amudha.
5. Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan.

Question 68

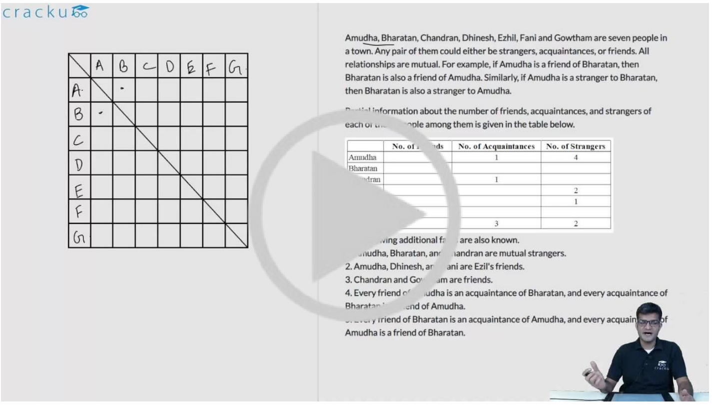
Who is an acquaintance of Amudha?

A Dhinesh

B Fani

C Gowtham

D Ezhil



VIDEO SOLUTION

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.

The Dean's office recently scanned student results into the central computer system. When their character reading software cannot read something, it leaves the space blank. The scanner output read as follows:

Name	Finance	Marketing	Statistics	Strategy	Operations	GPA
Aparna		B	F			1.4
Bikas	D	D	F	F		
Chandra		D	A	F	F	2.4
Deepak	A	B		D	D	3.2
Fazal	D	F		B	D	2.4
Gowri	C	C	A		B	3.8
Hari		B	A		D	2.8
Ismet			B		A	
Jagdeep	A	A	B		C	3.8
Kunal	F		A	F	F	1.8
Leena	B	A		B	F	3.2
Manab			A	B	B	
Nisha	A	D	B	A	F	3.6
Osman	C		B	B	A	4.6
Preeti	F	D		D		3.2
Rahul	A	C	A		F	4.2
Sameer		C	F	B		
Tara	B					2.4
Utkarsh			F	C	A	3.0
Vipul	A		C	C	F	2.4

In the grading system, A, B, C, D, and F grades fetch 6, 4, 3, 2, and 0 grade points respectively. The Grade Point Average (GPA) is the arithmetic mean of the grade points obtained in the five subjects. For example Nisha's GPA is (6 + 2 + 4 + 6 + 0) / 5 = 3.6. Some additional facts are also known about the students' grades. These are

(i) Vipul obtained the same grade in Marketing as Aparna obtained in Finance and Strategy.

(ii) Fazal obtained the same grade in Strategy as Utkarsh did in Marketing.

(iii) Tara received the same grade in exactly three courses.

Question 69

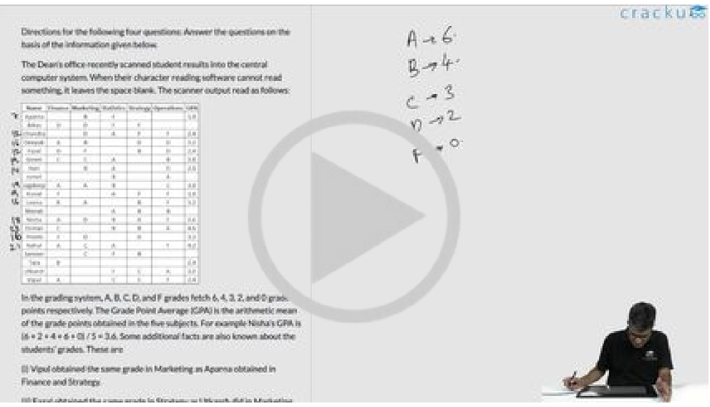
In operations, Tara could have received the same grade as

A Ismet

B Hari

C Jagdeep

D Manab



VIDEO SOLUTION

Instructions

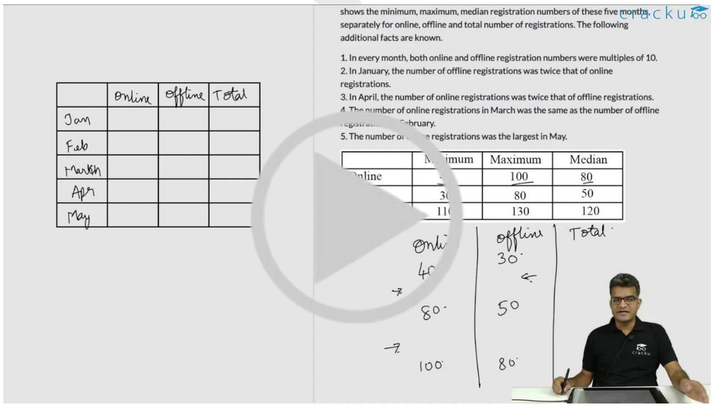
In a coaching class, some students register online, and some others register offline. No student registers both online and offline; hence the total registration number is the sum of online and offline registrations. The following facts and table pertain to these registration numbers for the five months - January to May of 2023. The table shows the minimum, maximum, median registration numbers of these five months, separately for online, offline and total number of registrations. The following additional facts are known.

1. In every month, both online and offline registration numbers were multiples of 10.
2. In January, the number of offline registrations was twice that of online registrations.
3. In April, the number of online registrations was twice that of offline registrations.
4. The number of online registrations in March was the same as the number of offline registrations in February.
5. The number of online registrations was the largest in May.

	Minimum	Maximum	Median
Online	40	100	80
Offline	30	80	50
Total	110	130	120

Question 70

What was the total number of registrations in April?



VIDEO SOLUTION

Question 71

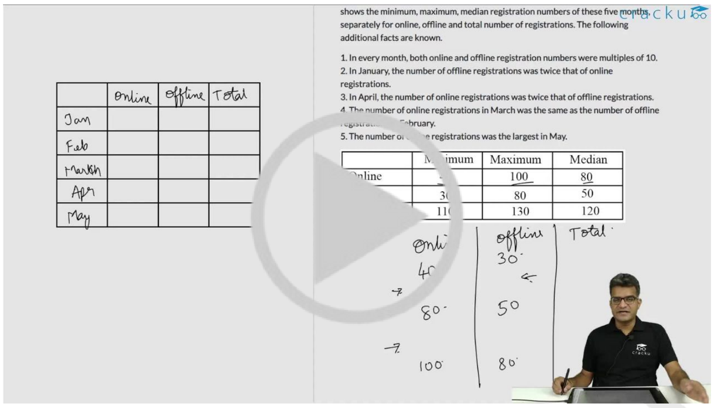
Which pair of months definitely had the same total number of registrations?
I. January and April
II. February and May

A Both I and II

B Only II

C Only I

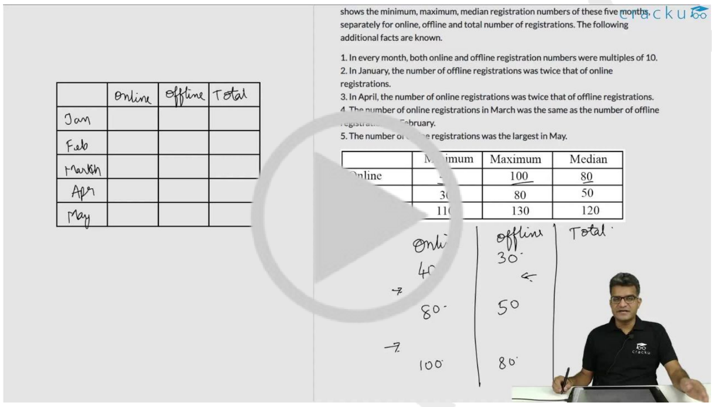
D Neither I nor II



VIDEO SOLUTION

Question 72

What was the number of online registrations in January?



VIDEO SOLUTION

Instructions

There are only three female students - Amala, Koli and Rini - and only three male students - Biman, Mathew and Shyamal - in a course. The course has two evaluation components, a project and a test. The aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

The projects are done in groups of two, with each group consisting of a female and a male student. Both the group members obtain the same score in the project.

The following additional facts are known about the scores in the project and the test.
1. The minimum, maximum and the average of both project and test scores were identical - 40, 80 and 60, respectively.
2. The test scores of the students were all multiples of 10; four of them were distinct and the remaining two were equal to the average test scores.
3. Amala's score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Yet Amala had the highest aggregate score.
4. Shyamal scored the second highest in the test. He scored two more than Koli, but two less than Amala in the aggregate.
5. Biman scored the second lowest in the test and the lowest in the aggregate.
6. Mathew scored more than Rini in the project, but less than her in the test.

Question 73

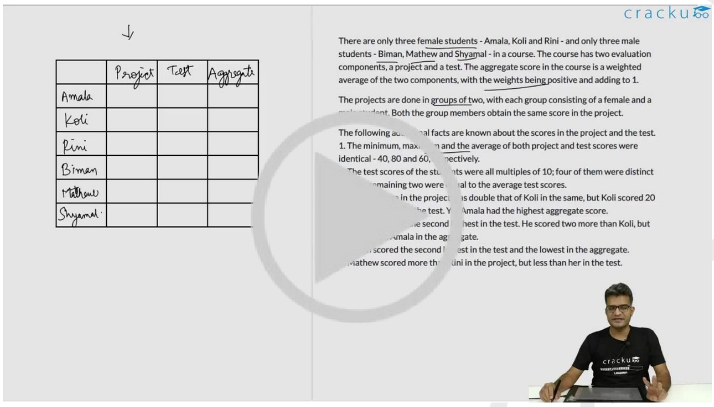
What was the maximum aggregate score obtained by the students?

A 68

B 80

C 62

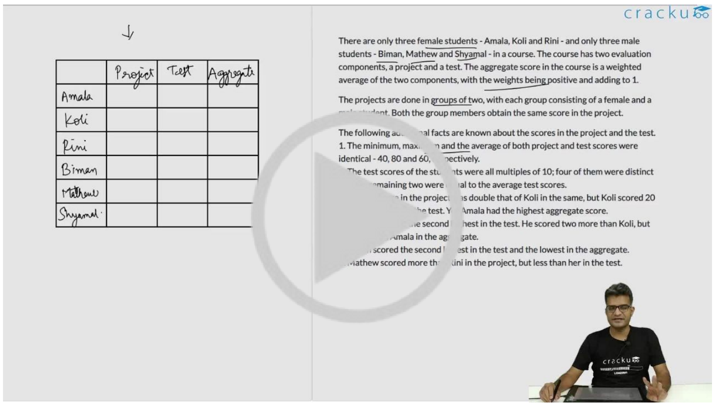
D 66



VIDEO SOLUTION

Question 74

What was Mathew's score in the test?



VIDEO SOLUTION

Question 75

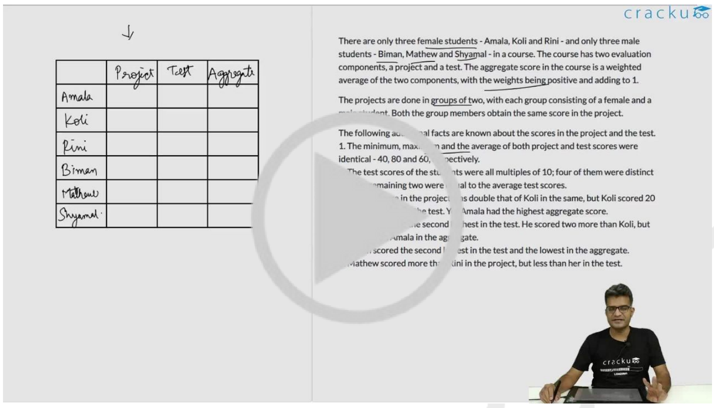
Which of the following pairs of students were part of the same project team?
i) Amala and Biman
ii) Koli and Mathew

A Only ii)

B Only i)

C Neither i) nor ii)

D Both i) and ii)



VIDEO SOLUTION

Answers

1.**A**	2.**B**	3.**14**	4.**A**	5.**A**	6.**D**	7.**D**	8.**4**
9.**A**	10.**D**	11.**B**	12.**C**	13.**D**	14.**D**	15.**48**	16.**3**
17.**A**	18.**D**	19.**C**	20.**3**	21.**17**	22.**D**	23.**4**	24.**D**
25.**0**	26.**B**	27.**11**	28.**12**	29.**D**	30.**C**	31.**B**	32.**0**
33.**C**	34.**C**	35.**C**	36.**B**	37.**60**	38.**A**	39.**9**	40.**13**
41.**C**	42.**C**	43.**D**	44.**5**	45.**D**	46.**C**	47.**D**	48.**A**
49.**D**	50.**B**	51.**A**	52.**A**	53.**3**	54.**C**	55.**A**	56.**E**
57.**B**	58.**12**	59.**D**	60.**C**	61.**C**	62.**B**	63.**D**	64.**A**
65.**D**	66.**24**	67.**A**	68.**A**	69.**D**	70.**120**	71.**A**	72.**40**
73.**A**	74.**40**	75.**C**					

Explanations

1.**A**

Let preeti obtain x and y in stastistics and operation respectively. Now we know that GPA = 3.2 .

So (x+y+2+2) = 16 .

So x+y = 12. for this only one value is possible x=y=6 .

So A grade .

VIDEO SOLUTION

2.**B**

We are given that Party 3 received 70% of total pizzas,therefore, number of pizzas received by Party 3 = $\frac{70}{100} \times 800$ = 560

Remaining 240 pizzas are equally divided among party 1 and party 2 hence we can say that each of Party 1 and Party 2 received 120 pizzas.

We know that all of the pizza can be classified into a total of 4 types. Hence, on drawing a table which can accommodate all of the cases:

	Thin Crust (T)			Deep Dish (D)		
	T - NC	T - EC	Total	D - NC	D - EC	Total
Party 1 **(120)**						
Party 2 **(120)**						
Party 3 **(560)**						
Total **(600)**						

Total number of Thin Crust pizzas = 0.375*800 = 300. Therefore, total number of Deep Dish pizzas = 800 - 300 = 500.

Out of 120 pizzas that Party 1 received, 60% were of Thin Crust type hence, total number of Thin Crust pizza received by Party 1 = 0.6*120 = 72. Consequently Party 1, must have received 42 Deep Dish type pizzas.

Out of 120 pizzas that Party 2 received, 55% were of Thin Crust type hence, total number of Thin Crust pizza received by Party 2 = 0.55*120 = 66. Consequently Party 1, must have received 54 Deep Dish type pizzas.

Therefore, total number of Thin Crust pizzas ordered by Party 3 = Total Thin Crust pizzas ordered - Thin Crust pizzas ordered by Party 1 - Thin Crust pizzas ordered by Party 2

$\Rightarrow$ 300 - 72 - 66 = 162

Hence number of Deep Dish type of pizzas order by Party 3 = 560 - 162 = 398

	Thin Crust (T)			Deep Dish (D)		
	T - NC	T - EC	Total	D - NC	D - EC	Total
Party 1 **(120)**			**72**			**48**
Party 2 **(120)**			**66**			**54**
Party 3 **(560)**			**162**			**398**
Total **(800)**			**300**			**500**

Number of Normal Cheese pizzas require to be delivered to Party 2 = 0.3*120 = 36
It is given that 50% of these Normal Cheese pizzas were of Thin Crust variety, then We can say that remaining 50% were of Deep Dish variety. We can find out each of 4 types of pizzas require to be delivered to Party 2.

	Thin Crust (T)			Deep Dish (D)		
	T - NC	T - EC	Total	D - NC	D - EC	Total
Party 1 **(120)**			**72**			**48**
Party 2 **(120)**	**18**	**48**	**66**	**18**	**36**	**54**
Party 3 **(560)**			**162**			**398**
Total **(800)**			**300**			**500**

Hence, the difference between the numbers of T-EC and D-EC pizzas to be delivered to Party 2 = 48 - 36 = 12

Therefore, option B is the correct answer.

VIDEO SOLUTION

3. **14**
It is given that Indu and Jatin both scored 100% in exactly one section. We can say that Jatin scored 100% marks in DI. Therefore, Jatin's composite score = 2*20+16+14 = 70

It is given that Jatin’s composite score was 10 more than Indu’s. Therefore, we can say that Indu's composite score = 70 - 10 = 60.

Indu also scored 100% in exactly one section.

Case 1: Indu scored 100% marks in DI.

If Indu scored 100% marks in DI, then Indu's score in GA = 60 - 2*20 - 8 = 12 which is less than 70% of maximum possible marks. Indu already has less than 70% in WE, therefore we Indu can't be recruited . Hence, we can reject this case.

Consequently, we can say that Indu scored 100% marks in WE. Therefore, Indu's score in DI = $\frac{60-8-20}{2}$ = 16

It is also given that Danish, Harini, and Indu had scored the same marks the in GA.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8		16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14		6	
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

We are given that, among the four recruited, Geeta had the lowest composite score.

Maximum composite score that Geeta can get = 2*14 + 6 + 20 = 54 {Assuming 100% marks in WE}. Since, Geeta was recruitedat a composite score of 54 or less we can say that Ester was definitely recruited.

It is given that no two candidates had the same composite score. We can see that Chetna's composite score is 54. Hence, Geeta can't have a composite score of 54. Therefore, we can say that Geeta's composite score is 53 or less.

We already know the four people(Jatin, Indu, Geeta, Ester) which were recruited. Hence, we cab say that Danish was rejected at a composite score of 51. Hence, we can say that Geeta's composite score in 52 or more.

Consequently, we can say that Geeta's composite score if either 52 or 53. Therefore we can say that Geeta scored either 18 {52-(2*14+6)} or 19 {53-(2*14+6)} marks in WE.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8		16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14	18/19	6	52/53
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

Ajay was the unique highest scorer in WE.

Case 1: Geeta scored 19 marks in WE.

We can say that if Geeta scored 19 marks in WE, then Ajay scored 20 marks in DI. In that case Ajay's composite score = 2*8 + 20 + 16 = 52. Which is a possible case.

Case 1: Geeta scored 18 marks in WE.

We can say that if Geeta scored 18 marks in WE, then Ajay can score either 19 or 20 marks in DI.

If Ajay scored 20 marks in DI then in that case Ajay's composite score = 2*8 + 20 + 16 = 52 which will be same as Geeta's composite score. Hence, we can say that in this case Ajay can't score 20 marks.

If Ajay scored 19 marks in DI then in that case Ajay's composite score = 2*8 + 19 + 16 = 51 which will be same as Danish's composite score. Hence, we can say that in this case Ajay can't score 19 marks.

Therefore, we can say that case 2 is not possible at all.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8	20	16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14	19	6	53
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

It is given that all the candidates scored different marks in WE.

We can see that Ajay, Geeta and Ester has already scored 20, 19 and 18 marks in WE. Therefore, Harini can score a maximum of 17 marks in WE. If Harini's score in WE is 17, then Harini's composite score = 2*5+17+20 = 47 which is same as Falak's composite score. Hence, we can say that Harini can't score 17 marks in WE. Jatin and Danish have already scored 16 and 15 marks respectively.

Therefore, we can say that the maximum marks that Harini could have scored in WE = 14.

VIDEO SOLUTION

4. **A**

From given conditions we find that :

	London(l)	Paris(p)	Beijing(b)	NY(n)	Total
Round1 (r1)	30	24	16	12	82
Round2(r2)	30	24(p r1) +3 (n r1) +4 (B r1) +1(n - representative)	12 (b r1) +9(n r1)	0	83
Round3(r3)	30+8 (Beijing r1 , r2)	24(p r1,r2) +4(b r1 ;p r2) +4(b r1,r2) +3 (n r1, p r2) +1 (n-representative) +1(b-representative)	0	0	75

We can see that - IOC member from New York must have voted for Paris in round 2.(IOC member - representative). Hence a is true. Also we can see that IOC member from Beijing voted for paris in round 3.Hence b is false. So answer is option A.

VIDEO SOLUTION

5. **A**

	Round-1	Round-2	Round-3	Round-4	Round-5	Round-6
Tanzi	-	4	-	5	NP	NP
Umeza	-	-	-	1	2	NP
Wangdu	-	4	-	NP	NP	NP
Xyla	-	-	-	1	5	-
Yonita	-	-	3	5	NP	NP
Zeneca	-	-	-	5	5	NP

It is given that every bull's eye score in the first three rounds gave a player one additional chance to shoot in the bonus rounds, Rounds 4 to 6, which means Tanzi scored Bull's eye only once in the first 3 rounds because she participated only once in round 4 to 6. Similarly, Umeza scored Bull's eye exactly 2 times in the first 3 rounds. Wangdu did not score Bull's eye in the first three rounds and so on.

Now from 1, Tanzi, Umeza and Yonita had the same total score.

So, Total score of Tanzi will be 4+5+5+a=14+a, (She scored Bull's eye(a score of 5) in exactly one round and a is the unknown score)

Total score of Umeza = 1+2+5+5+b = 13+b (She scored Bull's eye(a score of 5) in exactly 2 rounds and b is the unknown score)

Total score of Yonita = 3+5+5+c=13+c (She scored Bull's eye(a score of 5) in exactly one round and c is the unknown score)

Now 14+a=13+b=13+c,

Also it is given that total scores for all players, except one, were in multiples of three, so these three will have to be a multiple of 3.

So, (a,b,c) can be either (1,2,2) or (4,5,5) in the same order. But the value (5,5) for b and c is not possible. (Umeza scored Bull's eye in exactly 2 rounds and Yonita in exactly 1 round)

Hence, a=1,b=2 and c=2. So each of Tanzi, Umeza and Yonita had total score of 15.

Tabulating the data, we have

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	1/5	4	1/5	5	NP	NP	15
Umeza	5/5/2	5/5/2	5/5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2/5	2/5	3	5	NP	NP	15
Zeneca	-	-	-	5	5	NP	-

From 5, Tanzi and Zeneca had the same score in Round 1 but different scores in Round 3.

Zeneca score Bull's eye 2 times in round 1 to 3. If Tanzi scored 1 in round 1, then Zeneca also has to score 1 in round 1, which means both Tanzi and Zeneca scores in round 3 will be 5, which violates 5. Hence Tanzi scored 5 in round 1 and Zeneca also scored the same in round 1.So the new table is:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	5/5/2	5/5/2	5/5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2/5	2/5	3	5	NP	NP	15
Zeneca	5	5/(2/3/4)	5/(2/3/4)	5	5	NP	-

From 4, the number of players hitting bull's eye in Round 2 was double of that in Round 3.

So, in round 3 either 1 or 2 Bull's eye can be scored and in round 2, 2 or 4 Bull's eye can be scored.

Case 1: If only 1 Bull's eye is scored in the round 3, then in round 3 Umeza will score 2 and Zeneca will score 2/3/4 in round 3, which means both will score 5 in round 2. So minimum Bull's eye in round 2 will be 3. (Umeza, Zeneca and Xyla)

Hence this case is rejected.

Case 2: 2 Bull's eye were scored in round 3 and 4 Bull's eye were scored in round 2. So in round 2 Umeza, Yonita and Zeneca scored 5. This can be tabulated as:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	5/2	5	5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	(2/3/4)	5	5	NP	-

In round 3, 2 Bull's eye can only be scored by Xyla and Umeza.

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	2	5	5	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	(2/3/4)	5	5	NP	22-24

The highest scorer can be either Xyla or Zeneca. The lowest scorer will be Wangdu.

1.Consider Zeneca is the highest scorer.

From 3, the highest total score was one more than double of the lowest total score. So the only possible score for Zeneca is 23 and that for Wangdu is 11. (11*2+1=23)

But this will violate condition 2, since both Zeneca and Wangdu do not have their scores as multiples of three in this case.

Hence, Xyla will be the highest scorer. The only possible total score for Xyla will be 25, and that for Wangdu is 12(4+4+4). (12*2+1=25)

Since Xyla already has non-multiple of 3 as total score. Zeneca will have 24 as the total score. The complete table is:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	2	5	5	1	2	NP	15
Wangdu	4	4	4	NP	NP	NP	12
Xyla	5	5	5	1	5	4	25
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	4	5	5	NP	24

The highest score is 25.

VIDEO SOLUTION

CAT Previous Papers PDF

6. **D**

	Round-1	Round-2	Round-3	Round-4	Round-5	Round-6
Tanzi	-	4	-	5	NP	NP
Umeza	-	-	-	1	2	NP
Wangdu	-	4	-	NP	NP	NP
Xyla	-	-	-	1	5	-
Yonita	-	-	3	5	NP	NP
Zeneca	-	-	-	5	5	NP

It is given that every bull’s eye score in the first three rounds gave a player one additional chance to shoot in the bonus rounds, Rounds 4 to 6, which means Tanzi scored Bull's eye only once in the first 3 rounds because she participated only once in round 4 to 6. Similarly, Umeza scored Bull's eye exactly 2 times in the first 3 rounds. Wangdu did not score Bull's eye in the first three rounds and so on.

Now from 1, Tanzi, Umeza and Yonita had the same total score.

So, Total score of Tanzi will be 4+5+5+a=14+a, (She scored Bull's eye(a score of 5) in exactly one round and a is the unknown score)

Total score of Umeza = 1+2+5+5+b = 13+b (She scored Bull's eye(a score of 5) in exactly 2 rounds and b is the unknown score)

Total score of Yonita = 3+5+5+c=13+c (She scored Bull's eye(a score of 5) in exactly one round and c is the unknown score)

Now 14+a=13+b=13+c,

Also it is given that total scores for all players, except one, were in multiples of three, so these three will have to be a multiple of 3.

So, (a,b,c) can be either (1,2,2) or (4,5,5) in the same order. But the value (5,5) for b and c is not possible. (Umeza scored Bull's eye in exactly 2 rounds and Yonita in exactly 1 round)

Hence, a=1,b=2 and c=2. So each of Tanzi, Umeza and Yonita had total score of 15.

Tabulating the data, we have

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	1/5	4	1/5	5	NP	NP	15
Umeza	5/5/2	5/5/2	5/5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2/5	2/5	3	5	NP	NP	15
Zeneca	-	-	-	5	5	NP	-

From 5, Tanzi and Zeneca had the same score in Round 1 but different scores in Round 3.

Zeneca score Bull's eye 2 times in round 1 to 3. If Tanzi scored 1 in round 1, then Zeneca also has to score 1 in round 1, which means both Tanzi and Zeneca scores in round 3 will be 5, which violates 5. Hence Tanzi scored 5 in round 1 and Zeneca also scored the same in round 1.So the new table is:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	5/5/2	5/5/2	5/5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2/5	2/5	3	5	NP	NP	15
Zeneca	5	5/(2/3/4)	5/(2/3/4)	5	5	NP	-

From 4, the number of players hitting bull's eye in Round 2 was double of that in Round 3.

So, in round 3 either 1 or 2 Bull's eye can be scored and in round 2, 2 or 4 Bull's eye can be scored.

Case 1: If only 1 Bull's eye is scored in the round 3, then in round 3 Umeza will score 2 and Zeneca will score 2/3/4 in round 3, which means both will score 5 in round 2. So minimum Bull's eye in round 2 will be 3. (Umeza, Zeneca and Xyla)

Hence this case is rejected.

Case 2: 2 Bull's eye were scored in round 3 and 4 Bull's eye were scored in round 2. So in round 2 Umeza, Yonita and Zeneca scored 5. This can be tabulated as:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	5/2	5	5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	(2/3/4)	5	5	NP	-

In round 3, 2 Bull's eye can only be scored by Xyla and Umeza.

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	2	5	5	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	(2/3/4)	5	5	NP	22-24

The highest scorer can be either Xyla or Zeneca. The lowest scorer will be Wangdu.

1.Consider Zeneca is the highest scorer.

From 3, the highest total score was one more than double of the lowest total score. So the only possible score for Zeneca is 23 and that for Wangdu is 11. (11*2+1=23)

But this will violate condition 2, since both Zeneca and Wangdu do not have their scores as multiples of three in this case.

Hence, Xyla will be the highest scorer. The only possible total score for Xyla will be 25, and that for Wangdu is 12(4+4+4). (12*2+1=25)

Since Xyla already has non-multiple of 3 as total score. Zeneca will have 24 as the total score. The complete table is:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	2	5	5	1	2	NP	15
Wangdu	4	4	4	NP	NP	NP	12
Xyla	5	5	5	1	5	4	25
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	4	5	5	NP	24

Xyla was the highest scorer.

VIDEO SOLUTION

7. **D**

	Round-1	Round-2	Round-3	Round-4	Round-5	Round-6
Tanzi	-	4	-	5	NP	NP
Umeza	-	-	-	1	2	NP
Wangdu	-	4	-	NP	NP	NP
Xyla	-	-	-	1	5	-
Yonita	-	-	3	5	NP	NP
Zeneca	-	-	-	5	5	NP

It is given that every bull's eye score in the first three rounds gave a player one additional chance to shoot in the bonus rounds, Rounds 4 to 6, which means Tanzi scored Bull's eye only once in the first 3 rounds because she participated only once in round 4 to 6. Similarly, Umeza scored Bull's eye exactly 2 times in the first 3 rounds. Wangdu did not score Bull's eye in the first three rounds and so on.

Now from 1, Tanzi, Umeza and Yonita had the same total score.

So, Total score of Tanzi will be 4+5+5+a=14+a, (She scored Bull's eye(a score of 5) in exactly one round and a is the unknown score)

Total score of Umeza = 1+2+5+5+b = 13+b (She scored Bull's eye(a score of 5) in exactly 2 rounds and b is the unknown score)

Total score of Yonita = 3+5+5+c=13+c (She scored Bull's eye(a score of 5) in exactly one round and c is the unknown score)

Now 14+a=13+b=13+c,

Also it is given that total scores for all players, except one, were in multiples of three, so these three will have to be a multiple of 3.

So, (a,b,c) can be either (1,2,2) or (4,5,5) in the same order. But the value (5,5) for b and c is not possible. (Umeza scored Bull's eye in exactly 2 rounds and Yonita in exactly 1 round)

Hence, a=1,b=2 and c=2. So each of Tanzi, Umeza and Yonita had total score of 15.

Tabulating the data, we have

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	1/5	4	1/5	5	NP	NP	15
Umeza	5/5/2	5/5/2	5/5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2/5	2/5	3	5	NP	NP	15
Zeneca	-	-	-	5	5	NP	-

From 5, Tanzi and Zeneca had the same score in Round 1 but different scores in Round 3.

Zeneca score Bull's eye 2 times in round 1 to 3. If Tanzi scored 1 in round 1, then Zeneca also has to score 1 in round 1, which means both Tanzi and Zeneca scores in round 3 will be 5, which violates 5. Hence Tanzi scored 5 in round 1 and Zeneca also scored the same in round 1.So the new table is:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	5/5/2	5/5/2	5/5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2/5	2/5	3	5	NP	NP	15
Zeneca	5	5/(2/3/4)	5/(2/3/4)	5	5	NP	-

From 4, the number of players hitting bull's eye in Round 2 was double of that in Round 3.

So, in round 3 either 1 or 2 Bull's eye can be scored and in round 2, 2 or 4 Bull's eye can be scored.

Case 1: If only 1 Bull's eye is scored in the round 3, then in round 3 Umeza will score 2 and Zeneca will score 2/3/4 in round 3, which means both will score 5 in round 2. So minimum Bull's eye in round 2 will be 3. (Umeza, Zeneca and Xyla)

Hence this case is rejected.

Case 2: 2 Bull's eye were scored in round 3 and 4 Bull's eye were scored in round 2. So in round 2 Umeza, Yonita and Zeneca scored 5. This can be tabulated as:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	5/2	5	5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	(2/3/4)	5	5	NP	-

In round 3, 2 Bull's eye can only be scored by Xyla and Umeza.

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	2	5	5	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	(2/3/4)	5	5	NP	22-24

The highest scorer can be either Xyla or Zeneca. The lowest scorer will be Wangdu.

1.Consider Zeneca is the highest scorer.

From 3, the highest total score was one more than double of the lowest total score. So the only possible score for Zeneca is 23 and that for Wangdu is 11. (11*2+1=23)

But this will violate condition 2, since both Zeneca and Wangdu do not have their scores as multiples of three in this case.

Hence, Xyla will be the highest scorer. The only possible total score for Xyla will be 25, and that for Wangdu is 12(4+4+4). (12*2+1=25)

Since Xyla already has non-multiple of 3 as total score. Zeneca will have 24 as the total score. The complete table is:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	2	5	5	1	2	NP	15
Wangdu	4	4	4	NP	NP	NP	12
Xyla	5	5	5	1	5	4	25
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	4	5	5	NP	24

Tanzi scored 1 in round 3.

VIDEO SOLUTION

8. **4**

Since A, B, C are mutual strangers,(B, C)are strangers for A, (A, C) are strangers for B, (A, B) are strangers for C.

Since the total number of strangers+ acquaintances+ friends for any among the 7 is 6.

The number of friends for Amudha is 1, the number of friends for Gowtham is 1.

Using statement 3 Chandran and Gowtham are friends.

Using statement 2: Amudha, Dinesh, and Fani are Ezil's friends. Similarly, Ezil is a friend of Amudha, Dhinesh, and Fani.

Using statement 4 Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan, Hence the number of acquaintances of Bharatan is equal to the number of friends of Amudha.

Using statement 5 Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance Bharatan is a friend of Amudha, Hence the number of acquaintances of Amudha is equal to the number of friends of Bharatan.

Hence Bharatan has one friend, 1 Acquaintance, 4 strangers.

For Amudha we are yet to find a relationship with Dhinesh, Fani, and Gowtham. Any among the three can be the stranger for Amudha, considering the three different cases.

Case 1 :

Considering Fani as an acquaintance of Amudha, then Dhinesh and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Fani is an acquaintance of Amudha, hence Amudha is an Acquaintance of Fani, Fani is a friend of Bharathan and hence Bharathan is a friend of Fani. But Fani has only one friend and Ezhil is already a friend of Fani.

Hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1, (F)	4 (B, C, D, G)
Bharatan	1, (F)	1, (E)	4, (A, C, D, G)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 2 :

Considering Gowtham as an acquaintance of Amudha, then Dhinesh and Fani are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan. Since Gowtham is a friend of Bharatan, Bharatan must be a friend of Gowtham. But Gowtham can only have one friend and it already mentioned that Chandran is a friend of Gowtham and hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (G)	4 (B, C, D, F)
Bharatan	1, (G)	1, (E)	4, (A, C, D, F)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 3 :

Considering Dhinesh as an acquaintance of Amudha, then Fani and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Since Fani, Gowtham are strangers to Amudha, Bharatan. Amudha, Bharatan are strangers to Fani, Gowtham.

The 2 strangers to Gowtham and his only friend are known. Hence his three acquaintances are (Dhinesh, Ezhil, Fani).

Hence Gowtham is an acquaintance of Dhinesh, Ezhil, and Fani.

Dhinesh is an acquaintance of Amudha and hence Amudha must be an acquaintance of Dhinesh.

Dhinesh is a friend of Bharatan and hence Bharatan is a friend of Dhinesh.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4 (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	G	1	(A, B)
Dhinesh	E, B	G, A	2
Ezhil	A, D, F	G	1
Fani	1 (E)	G	A, B
Gowtham	1 (C)	3 (D, E, F)	2 (A, B)

Ezhil is an acquaintance of Bharatan and hence Bharatan is an acquaintance of Ezhil. The only stranger to Ezhil who is left is Chandran. Hence Ezhil is a stranger to Chandran.

The two strangers to Dhinesh who are left are Chandran and Fani. Chandran is a stranger to Dhinesh and hence Dhinesh is a stranger to Chandran.

The only acquaintance of Chandran who is left is Fani.

The remaining relationships with Fani are Dhinesh and Chandran. Dhinesh is a stranger to Fani and Chandran is an acquaintance of Fani.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4, (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	1, (G)	1 (F)	4, (A, B, E, D)
Dhinesh	2, (E, B)	2, (G, A)	2, (C, F)
Ezhil	3, (A, D, F)	2, (G, B)	1, (C).
Fani	1 (E)	2, (G, C)	3, (A, B, D)
Gowtham	1 (C)	3 (D, E, F)	2, (A, B)

Ezhil has Amudha, Dhinesh, Fani as his friends. Dinesh has Bharatan as his friend. Hence a total of 4 (Amudha, Bharatan, Dhinesh, and Fani) are his friends or friend of a friend

VIDEO SOLUTION

9. **A**

We are given that Party 3 received 70% of total pizzas,therefore, number of pizzas received by Party 3 = $\frac{70}{100} \times 800$ = 560

Remaining 240 pizzas are equally divided among party 1 and party 2 hence we can say that each of Party 1 and Party 2 received 120 pizzas.

We know that all of the pizza can be classified into a total of 4 types. Hence, on drawing a table which can accommodate all of the cases:

	Thin Crust (T)			Deep Dish (D)		
	T - NC	**T - EC**	**Total**	**D - NC**	**D - EC**	**Total**
Party 1 **(120)**						
Party 2 **(120)**						
Party 3 **(560)**						
Total **(600)**						

Total number of Thin Crust pizzas = 0.375*800 = 300. Therefore, total number of Deep Dish pizzas = 800 - 300 = 500.

Out of 120 pizzas that Party 1 received, 60% were of Thin Crust type hence, total number of Thin Crust pizza received by Party 1 = 0.6*120 = 72. Consequently Party 1, must have received 42 Deep Dish type pizzas.

Out of 120 pizzas that Party 2 received, 55% were of Thin Crust type hence, total number of Thin Crust pizza received by Party 2 = 0.55*120 = 66. Consequently Party 1, must have received 54 Deep Dish type pizzas.

Therefore, total number of Thin Crust pizzas ordered by Party 3 = Total Thin Crust pizzas ordered - Thin Crust pizzas ordered by Party 1 - Thin Crust pizzas ordered by Party 2

$\Rightarrow$ 300 - 72 - 66 = 162

Hence number of Deep Dish type of pizzas order by Party 3 = 560 - 162 = 398

	Thin Crust (T)			Deep Dish (D)		
	T - NC	T - EC	Total	D - NC	D - EC	Total
Party 1 **(120)**			**72**			**48**
Party 2 **(120)**			**66**			**54**
Party 3 **(560)**			**162**			**398**
Total **(800)**			**300**			**500**

Total number of Normal Cheese pizzas require to be delivered = 0.52*800 = 416
Number of Normal Cheese pizzas require to be delivered to Party 2 = 0.3*120 = 36
Number of Normal Cheese pizzas require to be delivered to Party 3 = 0.65*560 = 364
Therefore, total number of Normal Cheese pizzas require to be delivered to Party 1 = Total Normal Cheese pizzas to be delivered - Normal Cheese pizzas require to be delivered to Party 2 - Normal Cheese pizzas require to be delivered to Party 3

$\Rightarrow$ 416 - 36 - 364 = 16

It is given that 25% of these 16 Normal Cheese pizzas were of Deep Dish type, hence the number of D- NC type pizza require to be delivered to Party 1 = 0.25*16 = 4

Consequently, the number of T- NC type pizza require to be delivered to Party 1 = 16 - 4 = 12

We can find out each type of pizza that is required to be delivered to Party 1.

	Thin Crust (T)			Deep Dish (D)		
	T - NC	T - EC	Total	D - NC	D - EC	Total
Party 1 **(120)**	**12**	**60**	**72**	**4**	**44**	**48**
Party 2 **(120)**			**66**			**54**
Party 3 **(560)**			**162**			**398**
Total **(800)**			**300**			**500**

Cost Price of a T-EC pizza = Rs. 500

Cost Price of a D-EC pizza = Rs. 550

Cost Price of a T-NC pizza = $\frac{3}{5} \times 550$ = Rs. 330

Cost Price of a D-NC pizza = $\frac{3}{5} \times 550$ = Rs. 330

Therefore the total bill amount for Party 1 = 12*330 + 60*500 + 4*330 + 44*550 = Rs. 59480

Therefore, option A is the correct answer.

VIDEO SOLUTION

10. **D**
It is given that Indu and Jatin both scored 100% in exactly one section. We can say that Jatin scored 100% marks in DI. Therefore, Jatin's composite score = 2*20+16+14 = 70

It is given that Jatin's composite score was 10 more than Indu's. Therefore, we can say that Indu's composite score = 70 - 10 = 60.

Indu also scored 100% in exactly one section.

Case 1: Indu scored 100% marks in DI.

If Indu scored 100% marks in DI, then Indu's score in GA = 60 - 2*20 - 8 = 12 which is less than 70% of maximum possible marks. Indu already has less than 70% in WE, therefore we Indu can't be recruited . Hence, we can reject this case.

Consequently, we can say that Indu scored 100% marks in WE. Therefore, Indu's score in DI = $\frac{60-8-20}{2}$ = 16

It is also given that Danish, Harini, and Indu had scored the same marks the in GA.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8		16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14		6	
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

We are given that, among the four recruited, Geeta had the lowest composite score.

Maximum composite score that Geeta can get = 2*14 + 6 + 20 = 54 {Assuming 100% marks in WE}. Since, Geeta was recruitedat a composite score of 54 or less we can say that Ester was definitely recruited.

It is given that no two candidates had the same composite score. We can see that Chetna's composite score is 54. Hence, Geeta can't have a composite score of 54. Therefore, we can say that Geeta's composite score is 53 or less.

We already know the four people(Jatin, Indu, Geeta, Ester) which were recruited. Hence, we cab say that Danish was rejected at a composite score of 51. Hence, we can say that Geeta's composite score in 52 or more.

Consequently, we can say that Geeta's composite score if either 52 or 53. Therefore we can say that Geeta scored either 18 {52-(2*14+6)} or 19 {53-(2*14+6)} marks in WE.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8		16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14	18/19	6	52/53
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

Ajay was the unique highest scorer in WE.

Case 1: Geeta scored 19 marks in WE.

We can say that if Geeta scored 19 marks in WE, then Ajay scored 20 marks in DI. In that case Ajay's composite score = 2*8 + 20 + 16 = 52. Which is a possible case.

Case 1: Geeta scored 18 marks in WE.

We can say that if Geeta scored 18 marks in WE, then Ajay can score either 19 or 20 marks in DI.

If Ajay scored 20 marks in DI then in that case Ajay's composite score = 2*8 + 20 + 16 = 52 which will be same as Geeta's composite score. Hence, we can say that in this case Ajay can't score 20 marks.

If Ajay scored 19 marks in DI then in that case Ajay's composite score = 2*8 + 19 + 16 = 51 which will be same as Danish's composite score. Hence, we can say that in this case Ajay can't score 19 marks.

Therefore, we can say that case 2 is not possible at all.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8	20	16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14	19	6	53
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

Let us check all the statement one by one.

Statement 1: Jatin's composite score was more than that of Danish. We can see that this statement is correct.
Statement 2: Indu scored less than Chetna in DI. We can see that Indu scored 16 marks in DI whereas Chetna scored 19 marks in DI. Hence, we can say that this statement is also correct.
Statement 3: Jatin scored more than Indu in GA. We can see that Jatin scored 14 marks in GA whereas Indu scored 20 marks in GA. Hence, we can say that this statement is incorrect.

Hence, we can say that option D is the correct answer.

VIDEO SOLUTION

11. **B**
Knowing the GPA we can find gawris grade in strategy as

16 + x=19; x=3. So her grade was C.

Now considering hari's grade 12+x+y=14 ; so x+y=2.

So he must have got D or F in strategy. Hence in Strategy, Gowri's grade point was higher than that obtained by hari.

VIDEO SOLUTION

12. **C**
Considering utkarsh's grades we get , x+y = 6 . So possibilities of his missing grades are (4,2) or (6,0). Now according to given condition, his grades in marketing are same as faizal's in strategy and faizal got grade B in strategy . hence utkarsh got 4 in marketing and 2 in finance i.e. grade D . Hence option C.

VIDEO SOLUTION

13. **D**
The average income is calculated as show in the table given below.

	Total Earning				Total Days				Average		
Emp No.	Complex	Medium	Simple	Total	Complex	Medium	Simple	Total	Complex	Medium	Simple
2001147	82.98	0	636.53	719.51	3	0	23	26	27.66	-	27.67522
2001148	51.53	0	461.73	513.26	3.3	1.67	16	21	15.61515	0	28.85813
2001149	171.10	0.61	79.10	250.81	5.5	4	8.5	18	31.10909	0.1525	9.305882
2001150	100.47	0.01	497.47	597.95	6	4.67	7.33	18	16.745	0.002141328	67.86767
2001151	594.43	159.64	0	754.06	9.67	13.33	0	23	61.47156	11.975994	-
2001156	83.70	0	6	89.70	8	0	1	9	10.4625	-	6
2001158	472.31	109.73	0	582.04	1.39	9.61	0	11	339.7914	11.41831426	-
2001164	402.25	735.22	213.67	1351.14	5.27	12.07	0.67	18	76.32827	60.91300746	318.9104
2001170	576.57	0	0	576.57	21	0	0	21	27.45571	-	-
2001171	286.47	6.10	0	292.57	8.38	4.25	0.38	13	34.18496	1.435294118	0
2001172	512.10	117.46	0	629.56	10	8.5	3.5	22	51.21	13.81882353	0
2001173	1303.88	0	0	1303.88	25.5	0	0.5	26	51.13255	-	0
2001174	1017.94	0	0	1017.9	26	0	0	26	39.15154	-	-
2001179	46.56	776.19	0	822.75	2	19	0	21	23.28	40.85210526	-
2001180	116.40	1262.79	0	1379.19	5	19	0	24	23.28	66.46263158	-

From the table given we can make out that number of employees whose average earning per day in complex operations is more than average earning per day in medium operations for 8 employees (2001148, 2001149, 2001150, 2001151, 2001158, 2001164, 2001171, 2001172)

VIEW SOLUTION

14. **D**

Let us consider that there are 'm' members totally,there are 4 cities so total number of voters in round 1 will be m - 4, as 4 members of the cities are not allowed to vote.In the same way we have in total m - 3 members who will vote in the 2nd round.For the 3rd round, we have m - 2 - r members voting, r - number of of voters who voted for NY and Beijing in Round 1 and 2. According to given conditions these voters can't vote again. Also ,m - 3 = 83. So in total there were m = 86 members. And 83 people voted in 1st round and it is given that there are 75 votes in 3rd round . So we have r=9 i.e. 9 voters who voted for Ny in 2st round and beijing in 2nd round.
Also it is given that Ny voters in Round 1 voted either for Beijing or Paris and there are 9 voters who voted for NY in round 1, voted for Beijing in round 2.So 9*100/12 = 75% . hence option D.

VIDEO SOLUTION

15. **48**

It is given that: High Q > Best Ed > Cosmopolitan and Education Aid > A-one

We can say that High Q > Cosmopolitan

We can see that both High Q and Cosmopolitan got same points in reputation (R) and placement quality (P). High Q received more points in infrastructure (I) than Cosmopolitan whereas Cosmopolitan received more points in faculty Quality (F) than High Q.

Hence, we can say that Infrastructure's weight should be greater than Faculty quality. i.e.   I > F

Similarly, We can see that both Best Ed and Cosmopolitan got same points in faculty Quality (F) and placement quality (P). Best Ed received more points in reputation (R) than Cosmopolitan whereas Cosmopolitan received more points in infrastructure (I) than Best Ed.

Hence, we can say that reputation's weight should be greater than infrastructure. i.e.   R > I

Similarly, We can see that both Education Aid and A-one got same points in faculty Quality (F) and reputation (R). Education Aid received more points in infrastructure (I) than A-one whereas A-one received more points in placement quality (P) than Education Aid.

Hence, we can say that reputation's weight should be greater than infrastructure. i.e.   I > P

So basically there are two possible cases: R > I > P > F or R > I > F > P

Case 1:   Order of weights assigned = R > I > P > F

R = 0.4, I = 0.3. P = 0.2, F = 0.1

In this case overall score received by Best Ed = 0.1*40+0.4*30+0.2*20+0.3*20 = 26

In this case overall score received by High Q = 0.1*30+0.4*20+0.2*20+0.3*40 = 27

We can see that High Q's overall score is higher than Best Ed. Hence, this is a possible case.

Case 2: Order of weights assigned = R > I > F > P

R = 0.4, I = 0.3. P = 0.1, F = 0.2

In this case overall score received by Best Ed = 0.2*40+0.4*30+0.1*20+0.3*20 = 28

In this case overall score received by High Q = 0.2*30+0.4*20+0.1*20+0.3*40 = 28

We can see that High Q's overall score is not greater than the overall score received Best Ed. Hence, this case is not possible.

Now that we know the weight of each parameter, we can calculate the overall score and accreditation received by each college.

	F (0.1)	R (0.4)	P (0.2)	I (0.3)	Overall score	Accreditation
A-one	50	50	50	40	47	AAA
Best Ed	40	30	20	20	26	BBA
Cosmopolitan	40	20	20	30	25	BBA
Dominance	20	20	40	30	27	BBA
Education Aid	50	50	40	50	48	AAA
Fancy	50	50	40	40	45	AAA
Global	30	0	20	20	13	Junk
High Q	30	20	20	40	27	BBA

From the table we can see that Education Aid scored the highest overall score = 48.

VIDEO SOLUTION

16. **3**

It is given that: High Q > Best Ed > Cosmopolitan and Education Aid > A-one

We can say that High Q > Cosmopolitan

We can see that both High Q and Cosmopolitan got same points in reputation (R) and placement quality (P). High Q received more points in infrastructure (I) than Cosmopolitan whereas Cosmopolitan received more points in faculty Quality (F) than High Q.

Hence, we can say that Infrastructure's weight should be greater than Faculty quality. i.e. I > F

Similarly, We can see that both Best Ed and Cosmopolitan got same points in faculty Quality (F) and placement quality (P). Best Ed received more points in reputation (R) than Cosmopolitan whereas Cosmopolitan received more points in infrastructure (I) than Best Ed.

Hence, we can say that reputation's weight should be greater than infrastructure. i.e. R > I

Similarly, We can see that both Education Aid and A-one got same points in faculty Quality (F) and reputation (R). Education Aid received more points in infrastructure (I) than A-one whereas A-one received more points in placement quality (P) than Education Aid.

Hence, we can say that reputation's weight should be greater than infrastructure. i.e. I > P

So basically there are two possible cases: R > I > P > F or R > I > F > P

Case 1: Order of weights assigned = R > I > P > F

R = 0.4, I = 0.3. P = 0.2, F = 0.1

In this case overall score received by Best Ed = 0.1*40+0.4*30+0.2*20+0.3*20 = 26

In this case overall score received by High Q = 0.1*30+0.4*20+0.2*20+0.3*40 = 27

We can see that High Q's overall score is higher than Best Ed. Hence, this is a possible case.

Case 2: Order of weights assigned = R > I > F > P

R = 0.4, I = 0.3. P = 0.1, F = 0.2

In this case overall score received by Best Ed = 0.2*40+0.4*30+0.1*20+0.3*20 = 28

In this case overall score received by High Q = 0.2*30+0.4*20+0.1*20+0.3*40 = 28

We can see that High Q's overall score is not greater than the overall score received Best Ed. Hence, this case is not possible.

Now that we know the weight of each parameter, we can calculate the overall score and accreditation received by each college.

	F (0.1)	R (0.4)	P (0.2)	I (0.3)	Overall score	Accreditation
A-one	50	50	50	40	47	AAA
Best Ed	40	30	20	20	26	BBA
Cosmopolitan	40	20	20	30	25	BBA
Dominance	20	20	40	30	27	BBA
Education Aid	50	50	40	50	48	AAA
Fancy	50	50	40	40	45	AAA
Global	30	0	20	20	13	Junk
High Q	30	20	20	40	27	BBA

From the table, we can see that three received the accreditation of AAA.

VIDEO SOLUTION

17. **A**

We need to find the person who has the highest average per-day earnings in medium operations.

So, we divide the total income in medium operations by the total number of day in these operations.

Employee no. 2001180 earned 1262.9 by working 19 daysfor medium operation which is highest. Hence option A.

VIEW SOLUTION

18. **D**

80% of attendance comes out to 20 days.

We can see from the table that there are 7 such employees who have earned more than Rs. 600 and having more than 80% of attendance.

Hence option D.

VIEW SOLUTION

19. **C**

Average income can be found by dividing the total income by total number of days.
The average income is calculated as show in the table given below.

	Total Earning				Total Days				Average		
Emp No.	Complex	Medium	Simple	Total	Complex	Medium	Simple	Total	Complex	Medium	Simple
2001147	82.98	0	636.53	719.51	3	0	23	26	27.66	-	27.67522
2001148	51.53	0	461.73	513.26	3.3	1.67	16	21	15.61515	0	28.85813
2001149	171.10	0.61	79.10	250.81	5.5	4	8.5	18	31.10909	0.1525	9.305882
2001150	100.47	0.01	497.47	597.95	6	4.67	7.33	18	16.745	0.002141328	67.86767
2001151	594.43	159.64	0	754.06	9.67	13.33	0	23	61.47156	11.975994	-
2001156	83.70	0	6	89.70	8	0	1	9	10.4625	-	6
2001158	472.31	109.73	0	582.04	1.39	9.61	0	11	339.7914	11.41831426	-
2001164	402.25	735.22	213.67	1351.14	5.27	12.07	0.67	18	76.32827	60.91300746	318.9104
2001170	576.57	0	0	576.57	21	0	0	21	27.45571	-	-
2001171	286.47	6.10	0	292.57	8.38	4.25	0.38	13	34.18496	1.435294118	0
2001172	512.10	117.46	0	629.56	10	8.5	3.5	22	51.21	13.81882353	0
2001173	1303.88	0	0	1303.88	25.5	0	0.5	26	51.13255	-	0
2001174	1017.94	0	0	1017.9	26	0	0	26	39.15154	-	-
2001179	46.56	776.19	0	822.75	2	19	0	21	23.28	40.85210526	-
2001180	116.40	1262.79	0	1379.19	5	19	0	24	23.28	66.46263158	-

Using this formula, we can find that there are 5 employees (Employee IDs: 1151, 1158, 1164, 1172, and 1173) who earned more than Rs 50 per day.

VIEW SOLUTION

20. **3**

Since A, B, C are mutual strangers,(B, C)are strangers for A, (A, C) are strangers for B, (A, B) are strangers for C.

Since the total number of strangers+ acquaintances+ friends for any among the 7 is 6.

The number of friends for Amudha is 1, the number of friends for Gowtham is 1.

Using statement 3 Chandran and Gowtham are friends.

Using statement 2: Amudha, Dinesh, and Fani are Ezil's friends. Similarly, Ezil is a friend of Amudha, Dhinesh, and Fani.

Using statement 4 Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan, Hence the number of acquaintances of Bharatan is equal to the number of friends of Amudha.

Using statement 5 Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance Bharatan is a friend of Amudha, Hence the number of acquaintances of Amudha is equal to the number of friends of Bharatan.

Hence Bharatan has one friend, 1 Acquaintance, 4 strangers.

For Amudha we are yet to find a relationship with Dhinesh, Fani, and Gowtham. Any among the three can be the stranger for Amudha, considering the three different cases.

Case 1 :

Considering Fani as an acquaintance of Amudha, then Dhinesh and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Fani is an acquaintance of Amudha, hence Amudha is an Acquaintance of Fani, Fani is a friend of Bharathan and hence Bharathan is a friend of Fani. But Fani has only one friend and Ezhil is already a friend of Fani.

Hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1, (F)	4 (B, C, D, G)
Bharatan	1, (F)	1, (E)	4, (A, C, D, G)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 2 :

Considering Gowtham as an acquaintance of Amudha, then Dhinesh and Fani are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.Since Gowtham is a friend of Bharatan, Bharatan must be a friend of Gowtham. But Gowtham can only have one friend and it already mentioned that Chandran is a friend of Gowtham and hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (G)	4 (B, C, D, F)
Bharatan	1, (G)	1, (E)	4, (A, C, D, F)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 3 :

Considering Dhinesh as an acquaintance of Amudha, then Fani and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Since Fani, Gowtham are strangers to Amudha, Bharatan. Amudha, Bharatan are strangers to Fani, Gowtham.

The 2 strangers to Gowtham and his only friend are known. Hence his three acquaintances are (Dhinesh, Ezhil, Fani).

Hence Gowtham is an acquaintance of Dhinesh, Ezhil, and Fani.

Dhinesh is an acquaintance of Amudha and hence Amudha must be an acquaintance of Dhinesh.

Dhinesh is a friend of Bharatan and hence Bharatan is a friend of Dhinesh.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4 (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	G	1	(A, B)
Dhinesh	E, B	G, A	2
Ezhil	A, D, F	G	1
Fani	1 (E)	G	A, B
Gowtham	1 (C)	3 (D, E, F)	2 (A, B)

Ezhil is an acquaintance of Bharatan and hence Bharatan is an acquaintance of Ezhil. The only stranger to Ezhil who is left is Chandran. Hence Ezhil is a stranger to Chandran.

The two strangers to Dhinesh who are left are Chandran and Fani. Chandran is a stranger to Dhinesh and hence Dhinesh is a stranger to Chandran.

The only acquaintance of Chandran who is left is Fani.

The remaining relationships with Fani are Dhinesh and Chandran. Dhinesh is a stranger to Fani and Chandran is an acquaintance of Fani.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4, (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	1, (G)	1 (F)	4, (A, B, E, D)
Dhinesh	2, (E, B)	2, (G, A)	2, (C, F)
Ezhil	3, (A, D, F)	2, (G, B)	1, (C).
Fani	1 (E)	2, (G, C)	3, (A, B, D)
Gowtham	1 (C)	3 (D, E, F)	2, (A, B)

Ezhil has a total of 3 friends.

21. **17**

In this set, we are told that the amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down and no firm raised the same amount of money in two consecutive years.

The increase or decrease can be ± 1 or ± 2. => (1)

We are also told that each firm raised Rs. 1 crore in its first and last year of existence

Consider A:

It raised money for 8 years

=> The raising pattern looks like follows:

1, a, b, c, d, e, f, 1 => where a, b, c,..,, f are the unknown amounts raised.

Also a + b + c + d + e + f = 21 - 2 = 19.

We can observe that 19/6 is slightly greater than 3 => The average amount raised should be around 3.

If a = 3 and f = 3 => b + c + d + e = 13 (not possible) as the minimum case would be (4, 5, 6, 4) => Not possible.

If a = 3 and f = 2 => b + c + d + e = 14 (not possible) as the minimum case would be (4, 5, 4, 3) => Not possible.

=> a = 2 and f = 2 => b + c + d + e = 15 the minimum case is (3, 4, 5, 3) or (3, 5, 4, 3) which gives a sum of 15.

So, the possible cases for A are:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

Consider B:

The patterns looks as follows:

1, a, b, 1

If a = 2, b has to be equal to 3 to satisfy (1)

if a = 3, b has to be equal to 2 to satisfy (1)

=> The possible cases for B are:

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

Consider C:

The pattern looks as follows:

1, ..., 1

Let us assume there are 2 gaps between => a + b = 7 (Not possible) as maximum case would be 1, 3, 2, 1

Let us assume there are 3 gaps between => a + b + c = 7, the minimum case possible is 1, 2, 3, 2, 1 => Satisfies.

Now, if there are 4 gaps => a + b + c + d = 7 => The average value is 7/4 which is less than 2 => Not possible.

=> The possible cases for C are:

C	2013	2014	2015	2016	2017
	1	2	3	2	1

Consider D:

The pattern looks as follows:

1, a, b, c, 1

=> a + b + c = 8

When a = 2 and c = 2 => b = 4 => 2, 4, 2 => Satisfies.

When a = 2 and c = 3, b should be 3 (Not satisfying (1))

When a = 3 and c = 3, b should be 2 (Not satisfying (1))

=> The possible cases for D are:

D	2011	2012	2013	2014	2015
	1	2	4	2	1

Consider E:

The pattern looks as follows:

1,.....,1

For 1 or 2 gaps, we can't get a sum of 11.

Assume 3 gaps => a + b + c = 11, the maximum case is 3, 5, 3 => Satisfies.

Now, assume 4 gaps

=> a + b + c + d = 11, the minimum case is 2, 3, 4, 2 or 2, 4, 3, 2 which satisfies (1) and 2 + 3 + 4 + 2 = 11.

=> The possible cases for E are:

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

In summary, the possible cases for all 5 companies is:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

C	2013	2014	2015	2016	2017
	1	2	3	2	1

D	2011	2012	2013	2014	2015
	1	2	4	2	1

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

Maximum money raised in 2013 is 5 + 3 + 1 + 4 + 4 = 17.

VIDEO SOLUTION

22. **D**

	Round-1	Round-2	Round-3	Round-4	Round-5	Round-6
Tanzi	-	4	-	5	NP	NP
Umeza	-	-	-	1	2	NP
Wangdu	-	4	-	NP	NP	NP
Xyla	-	-	-	1	5	-
Yonita	-	-	3	5	NP	NP
Zeneca	-	-	-	5	5	NP

It is given that every bull's eye score in the first three rounds gave a player one additional chance to shoot in the bonus rounds, Rounds 4 to 6, which means Tanzi scored Bull's eye only once in the first 3 rounds because she participated only once in round 4 to 6. Similarly, Umeza scored Bull's eye exactly 2 times in the first 3 rounds. Wangdu did not score Bull's eye in the first three rounds and so on.

Now from 1, Tanzi, Umeza and Yonita had the same total score.

So, Total score of Tanzi will be 4+5+5+a=14+a, (She scored Bull's eye(a score of 5) in exactly one round and a is the unknown score)

Total score of Umeza = 1+2+5+5+b = 13+b (She scored Bull's eye(a score of 5) in exactly 2 rounds and b is the unknown score)

Total score of Yonita = 3+5+5+c=13+c (She scored Bull's eye(a score of 5) in exactly one round and c is the unknown score)

Now 14+a=13+b=13+c,

Also it is given that total scores for all players, except one, were in multiples of three, so these three will have to be a multiple of 3.

So, (a,b,c) can be either (1,2,2) or (4,5,5) in the same order. But the value (5,5) for b and c is not possible. (Umeza scored Bull's eye in exactly 2 rounds and Yonita in exactly 1 round)

Hence, a=1,b=2 and c=2. So each of Tanzi, Umeza and Yonita had total score of 15.

Tabulating the data, we have

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	1/5	4	1/5	5	NP	NP	15
Umeza	5/5/2	5/5/2	5/5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2/5	2/5	3	5	NP	NP	15
Zeneca	-	-	-	5	5	NP	-

From 5, Tanzi and Zeneca had the same score in Round 1 but different scores in Round 3.

Zeneca score Bull's eye 2 times in round 1 to 3. If Tanzi scored 1 in round 1, then Zeneca also has to score 1 in round 1, which means both Tanzi and Zeneca scores in round 3 will be 5, which violates 5. Hence Tanzi scored 5 in round 1 and Zeneca also scored the same in round 1.So the new table is:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	5/5/2	5/5/2	5/5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2/5	2/5	3	5	NP	NP	15
Zeneca	5	5/(2/3/4)	5/(2/3/4)	5	5	NP	-

From 4, the number of players hitting bull's eye in Round 2 was double of that in Round 3.

So, in round 3 either 1 or 2 Bull's eye can be scored and in round 2, 2 or 4 Bull's eye can be scored.

Case 1: If only 1 Bull's eye is scored in the round 3, then in round 3 Umeza will score 2 and Zeneca will score 2/3/4 in round 3, which means both will score 5 in round 2. So minimum Bull's eye in round 2 will be 3. (Umeza, Zeneca and Xyla)

Hence this case is rejected.

Case 2: 2 Bull's eye were scored in round 3 and 4 Bull's eye were scored in round 2. So in round 2 Umeza, Yonita and Zeneca scored 5. This can be tabulated as:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	5/2	5	5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	(2/3/4)	5	5	NP	-

In round 3, 2 Bull's eye can only be scored by Xyla and Umeza.

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	2	5	5	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	(2/3/4)	5	5	NP	22-24

The highest scorer can be either Xyla or Zeneca. The lowest scorer will be Wangdu.

1.Consider Zeneca is the highest scorer.

From 3, the highest total score was one more than double of the lowest total score. So the only possible score for Zeneca is 23 and that for Wangdu is 11. (11*2+1=23)

But this will violate condition 2, since both Zeneca and Wangdu do not have their scores as multiples of three in this case.

Hence, Xyla will be the highest scorer. The only possible total score for Xyla will be 25, and that for Wangdu is 12(4+4+4). (12*2+1=25)

Since Xyla already has non-multiple of 3 as total score. Zeneca will have 24 as the total score. The complete table is:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	2	5	5	1	2	NP	15
Wangdu	4	4	4	NP	NP	NP	12
Xyla	5	5	5	1	5	4	25
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	4	5	5	NP	24

Zeneca total score is 24.

VIDEO SOLUTION

23. **4**

Given that the means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

=> The sum of ratings given by R1, R2, R3 R4, R5 are 5*means = 17, 11, 19, 14, and 17 respectively.

Similarly the sum of ratings received by U, V, W, X and Y are 5*means = 11, 19, 17, 18, and 13 respectively.

Also capturing the absolute data given in the partial information (a) and (b) and representing as a table, we get:

	R1	R2	R3	R4	R5	Total
U	1					11
V					5	19
W	5	1	5			17
X		5	5			18
Y		1	1			13
Total	17	11	19	14	17	

Now,

Consider U

Given median = 2, mode = 2 and range = 3

=> His ratings should be of the form 1, a , 2, b, 4 => 1 + 2 + 4 + a + b = 11 => a + b = 4. For mode = 2 => a = b = 2

=> U's ratings are 1, 2, 2, 2, 4.

Consider V

Given median = 4, mode = 4 and range = 3

=> His ratings should be of the form 2, a, 4, b, 5 => 2 + 4 + 5 + a + b = 19 => a + b = 8 => For mode = 4 => a = b = 4

=> V's ratings are 2, 4, 4, 4, 5.

Consider W

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => 1 + a + 4 + 5 + 5 = 17 => a = 2

=> W's ratings are 1, 2, 4, 5, 5.

Consider X

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => a + 1 + 4 + 5 + 5 = 18 => a = 3

=> X's ratings are 1, 3, 4, 5, 5

Consider Y

Given median = 3, mode = 1 & 4, Range = 3

=> His ratings are 1, 1, 3, 4, 4.

Capturing this data in the table, we get:

	R1	R2	R3	R4	R5	Total	Entries
U	1					11	2,2,2,4
V					5	19	2,4,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R3 => The two missing entries should add up to 19 - 1 - 5 - 5 = 8, (only possibility is 4 + 4) => We can fill the row "U" and 4 in the row "V"

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V			4		5	19	2,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R2 => Missing entry should be 11 - 2 - 1 - 5 - 1 = 2

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Consider column R1, the missing elements should add up to 17 - 5 - 4 - 1 = 7 (3 + 4 or 4 + 3) ----(1)

Consider R5, the missing elements should add up to 10 => 2 + 4 + 4 or 4 + 3 + 3 (not possible) as (1) requires a 3.

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X		5	5		4	18	1,3
Y		1	1		4	13	3,4
Total	17	11	19	14	17		

Now, we can fill column R1 as 3 + 4 and the remaining in column R4 and we can get the complete table

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X	3	5	5	1	4	18	
Y	4	1	1	3	4	13	
Total	17	11	19	14	17		

=> Ratings give by R3 are 1, 4, 4, 5, 5 => Median = 4.

VIDEO SOLUTION

24. **D**

Given that in every month, both online and offline registration numbers were multiples of 10.

From (2), in Jan, the number of offline registrations was twice that of online registrations.

=> If x is number of online registrations => 2x is the number of offline registrations => 3x is the total number of registrations.

According to the data given in the table => 3x should lie between the minimum and maximum total number of registrations. => x = 40 (as x should also be a multiple of 10)

=> In Jan => (40,80) are the online and offline registrations respectively.

Similarly from (3) => In Apr (80,40) are the online and offline registrations respectively.

From-5, the number of online registrations is highest in may => In may there are 100 online registrations. The lowest possible number of offline registrations is 30 and maximum possible total registrations is 130 => In May (100,30) are the online and offline registrations respectively.

Let us assume, 'x' to be the number of offline registrations in May = number of online registrations in March.

Let us capture all this data in a table:

Month	Online	Offline	Total
Jan	40	80	120
Feb	y	x	
Mar	x	z	
Apr	80	40	120
May	100	30	130

From the table given in the question, 50 is the median for Offline data

=> x should lie between 50 and 80 (included)

For 80 to be the median for the online data => y lie between 80 and 100 (included).

Now, consider Feb => Minimum value of y + x = 80 + 50 = 130 (which is the maximum value possible of the total possible registrations)

=> x = 50 and y = 80

Since, 110 is the minimum number of total registrations, the only possibility is in March => 50 + z = 110 => z = 60.

Now, filling the complete table we get,

Month	Online	Offline	Total
Jan	40	80	120
Feb	80	50	130
Mar	50	60	110
Apr	80	40	120
May	100	30	130

The number of offline registrations in Feb is 50.

VIDEO SOLUTION

25. **0**

Given that the means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

=> The sum of ratings given by R1, R2, R3 R4, R5 are 5*means = 17, 11, 19, 14, and 17 respectively.

Similarly the sum of ratings received by U, V, W, X and Y are 5*means = 11, 19, 17, 18, and 13 respectively.

Also capturing the absolute data given in the partial information (a) and (b) and representing as a table, we get:

	R1	R2	R3	R4	R5	Total
U	1					11
V					5	19
W	5	1	5			17
X		5	5			18
Y		1	1			13
Total	17	11	19	14	17	

Now,

Consider U

Given median = 2, mode = 2 and range = 3

=> His ratings should be of the form 1, a , 2, b, 4 => 1 + 2 + 4 + a + b = 11 => a + b = 4. For mode = 2 => a = b = 2

=> U's ratings are 1, 2, 2, 2, 4.

Consider V

Given median = 4, mode = 4 and range = 3

=> His ratings should be of the form 2, a, 4, b, 5 => 2 + 4 + 5 + a + b = 19 => a + b = 8 => For mode = 4 => a = b = 4

=> V's ratings are 2, 4, 4, 4, 5.

Consider W

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => 1 + a + 4 + 5 + 5 = 17 => a = 2

=> W's ratings are 1, 2, 4, 5, 5.

Consider X

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => a + 1 + 4 + 5 + 5 = 18 => a = 3

=> X's ratings are 1, 3, 4, 5, 5

Consider Y

Given median = 3, mode = 1 & 4, Range = 3

=> His ratings are 1, 1, 3, 4, 4.

Capturing this data in the table, we get:

	R1	R2	R3	R4	R5	Total	Entries
U	1					11	2,2,2,4
V					5	19	2,4,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R3 => The two missing entries should add up to 19 - 1 - 5 - 5 = 8, (only possibility is 4 + 4) => We can fill the row "U" and 4 in the row "V"

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V			4		5	19	2,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R2 => Missing entry should be 11 - 2 - 1 - 5 - 1 = 2

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Consider column R1, the missing elements should add up to 17 - 5 - 4 - 1 = 7 (3 + 4 or 4 + 3) ----(1)

Consider R5, the missing elements should add up to 10 => 2 + 4 + 4 or 4 + 3 + 3 (not possible) as (1) requires a 3.

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X		5	5		4	18	1,3
Y		1	1		4	13	3,4
Total	17	11	19	14	17		

Now, we can fill column R1 as 3 + 4 and the remaining in column R4 and we can get the complete table

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X	3	5	5	1	4	18	
Y	4	1	1	3	4	13	
Total	17	11	19	14	17		

=> All ratings can be determined uniquely => 0.

VIDEO SOLUTION

26. **B**

In this set, we are told that the amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down and no firm raised the same amount of money in two consecutive years.

The increase or decrease can be ± 1 or ± 2. => (1)

We are also told that each firm raised Rs. 1 crore in its first and last year of existence

Consider A:

It raised money for 8 years

=> The raising pattern looks like follows:

1, a, b, c, d, e, f, 1 => where a, b, c,..,, f are the unknown amounts raised.

Also a + b + c + d + e + f = 21 - 2 = 19.

We can observe that 19/6 is slightly greater than 3 => The average amount raised should be around 3.

If a = 3 and f = 3 => b + c + d + e = 13 (not possible) as the minimum case would be (4, 5, 6, 4) => Not possible.

If a = 3 and f = 2 => b + c + d + e = 14 (not possible) as the minimum case would be (4, 5, 4, 3) => Not possible.

=> a = 2 and f = 2 => b + c + d + e = 15 the minimum case is (3, 4, 5, 3) or (3, 5, 4, 3) which gives a sum of 15.

So, the possible cases for A are:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

Consider B:

The patterns looks as follows:

1, a, b, 1

If a = 2, b has to be equal to 3 to satisfy (1)

if a = 3, b has to be equal to 2 to satisfy (1)

=> The possible cases for B are:

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

Consider C:

The pattern looks as follows:

1, ..., 1

Let us assume there are 2 gaps between => a + b = 7 (Not possible) as maximum case would be 1, 3, 2, 1

Let us assume there are 3 gaps between => a + b + c = 7, the minimum case possible is 1, 2, 3, 2, 1 => Satisfies.

Now, if there are 4 gaps => a + b + c + d = 7 => The average value is 7/4 which is less than 2 => Not possible.

=> The possible cases for C are:

C	2013	2014	2015	2016	2017
	1	2	3	2	1

Consider D:

The pattern looks as follows:

1, a, b, c, 1

=> a + b + c = 8

When a = 2 and c = 2 => b = 4 => 2, 4, 2 => Satisfies.

When a = 2 and c = 3, b should be 3 (Not satisfying (1))

When a = 3 and c = 3, b should be 2 (Not satisfying (1))

=> The possible cases for D are:

D	2011	2012	2013	2014	2015
	1	2	4	2	1

Consider E:

The pattern looks as follows:

1,.....,1

For 1 or 2 gaps, we can't get a sum of 11.

Assume 3 gaps => a + b + c = 11, the maximum case is 3, 5, 3 => Satisfies.

Now, assume 4 gaps

=> a + b + c + d = 11, the minimum case is 2, 3, 4, 2 or 2, 4, 3, 2 which satisfies (1) and 2 + 3 + 4 + 2 = 11.

=> The possible cases for E are:

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

In summary, the possible cases for all 5 companies is:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

C	2013	2014	2015	2016	2017
	1	2	3	2	1

D	2011	2012	2013	2014	2015
	1	2	4	2	1

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

We see that only for C and D, we can conclude the amounts raised with certainty.

VIDEO SOLUTION

27. **11**

It's given in the question that the first runner up polled 10,000 more votes than the second runner up in constituency A. Now the first runner up has got 95000 votes, hence the second runner up will get 85000 votes.

Now the remaining votes will be 500000-275000-95000-85000=45000

From 2, None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000 => the person who got 5th highest votes must have got > 600030/6 => $\geq$ 100006. Since it is also given that the difference of votes is $\geq$ 10000, the only possible case is winner, 1st runner up, 2nd runner up, 3rd runner up, 4th runner up must have got 140006,130006,120006,110006,100006 respectively which sums upto exactly 600030.

Let the number of votes polled in D be 100x.

From 3, The winning candidate in constituency D must have got 5x+37500

The table now looks like:

	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	100x
No. of votes polled by the winning candidate	2,75,000	48,750	140006	37500+5x
No. of votes polled by the first runner up	95,000		130006	37,500
No. of votes polled by the second runner up	85000		120006	30,000
% of valid votes polled by the third runner up			16.67%	10%

In constituency B, the mark for not losing the security deposit is 1/6(325000) or 54,167.

But winner himself/herself got < 54167 => all the other candidates lost their security deposits.

11 is the correct answer,

VIDEO SOLUTION

28. **12**
The set's starting point is that the sum of each row must be 100

Clue 3 tells us that all the missing elements in the Carbs column are multiple of 5. Similarly, clue 4 tells us that the missing elements of the other three columns are multiples of 4.

Note that we look at clue 1, which says that the carbs in C1 and C2 must be more than any carbs in any pseudo cereal.
We have the reference point of P1 at 66

Trying to fill in for C1:
The possible values are 75, 80, 85, 90, 95
Since 12 grams is of other nutrients, we can eliminate 90 and 95.
If we take 85, 85+12 = 97, which would leave 3 grams of protein, but from clue 4, we know this has to be a multiple of 4.

Taking 75, 75+12 = 87 would leave 13 grams of protein, which, too, is eliminated.
Leaving only 80 grams of carb in C1 and 8 grams of protein.

Similar logic is to be applied for C2; we have 13 grams already present, so the values of 90 and 95 are eliminated.
Taking 85 carbs would give 2 grams of protein, which can be eliminated.
80 and 70 would also not work for the same reason.
Leavin has only 75 grams of carbs, leaving 12 grams of protein.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2		14		8

Using clues 1 and 2 together, we can determine that the carbs in P2 must be less than 75 but at least greater than 62.
The only possible values are 65 and 70
Putting 65 grams of carbs in P2 gives us 100-65-14-8 = 13 grams of protein, which is invalid.

Putting 70 grams of carbs in P2 gives 100-70-14-8 = 8 grams of protein.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

Clue 1 gives us that the protein in M2 should be less than that present in either of the pseudo-cereals, which we right now have a baseline of 14.
So the protein in M2 (and M3) can be 12, 8, 4, 0

The protein and carbs in M2 should add up to 100-7-16 = 77
This is only possible if the protein count is 12, giving carbs as 65

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

The protein in M3 can be 0, 4, 8 or 12

We are given in clue five that the protein in P1 is double that in M3.
The protein in P1 thus can be 0, 8, 16 or 24
Since this P1 protein also has to be more than M1 and M2 protein, it can not be 0 or 8
Leaving only 16 or 24 as the valid values.

The protein and fats in P1 must add up to 100-66-10 = 24
If P1 had 24 grams of protein, then it would have 0 grams of fat, but in clue 4, we are given that all missing fats are **non-zero multiples of 4**
Hence, the only possible protein value in P1 is 16, with 8 grams of fats. Giving 8 grams of protein in M3 and 24 grams of others in M3.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56	8	12	24
Pseudo-cereals	P1	66	16	8	10
	P2	70	14	8	8

The proteins in the foodgrains, when arranged in ascending order, are: 8, 8, 10, 12, 12,14, 16
Giving the median as 12.

VIDEO SOLUTION

29. **D**

According to given condition 75% of voters who voted for Beijing in round 1, voted for Beijing in round 2. There are 21-9 = 12 voters who voted for Beijing in round 2 also Beijing in round 1. Hence total voters for beijing in round 1 is 16. Son in round 1 total 82 voters , out of which beijing -16 voters,london - 30 , new york - 12. Hence total voters for paris 82-16-30-12 = 24. Hence option D .

VIDEO SOLUTION

30. **C**

We are given that Party 3 received 70% of total pizzas,therefore, number of pizzas received by Party 3 = $\frac{70}{100} \times 800$ = 560

Remaining 240 pizzas are equally divided among party 1 and party 2 hence we can say that each of Party 1 and Party 2 received 120 pizzas.

We know that all of the pizza can be classified into a total of 4 types. Hence, on drawing a table which can accommodate all of the cases:

	Thin Crust (T)			**Deep Dish (D)**		
	T - NC	**T - EC**	**Total**	**D - NC**	**D - EC**	**Total**
Party 1 **(120)**						
Party 2 **(120)**						
Party 3 **(560)**						
Total **(600)**						

Total number of Thin Crust pizzas = 0.375*800 = 300. Therefore, total number of Deep Dish pizzas = 800 - 300 = 500.

Out of 120 pizzas that Party 1 received, 60% were of Thin Crust type hence, total number of Thin Crust pizza received by Party 1 = 0.6*120 = 72. Consequently Party 1, must have received 42 Deep Dish type pizzas.

Out of 120 pizzas that Party 2 received, 55% were of Thin Crust type hence, total number of Thin Crust pizza received by Party 2 = 0.55*120 = 66. Consequently Party 1, must have received 54 Deep Dish type pizzas.

Therefore, total number of Thin Crust pizzas ordered by Party 3 = Total Thin Crust pizzas ordered - Thin Crust pizzas ordered by Party 1 - Thin Crust pizzas ordered by Party 2

$\Rightarrow$ 300 - 72 - 66 = 162

Hence number of Deep Dish type of pizzas order by Party 3 = 560 - 162 = 398

	Thin Crust (T)			**Deep Dish (D)**		
	T - NC	**T - EC**	**Total**	**D - NC**	**D - EC**	**Total**
Party 1 **(120)**			**72**			**48**
Party 2 **(120)**			**66**			**54**
Party 3 **(560)**			**162**			**398**
Total **(800)**			**300**			**500**

Total number of Normal Cheese pizzas require to be delivered = 0.52*800 = 416
Number of Normal Cheese pizzas require to be delivered to Party 2 = 0.3*120 = 36
Number of Normal Cheese pizzas require to be delivered to Party 3 = 0.65*560 = 364

Therefore, total number of Normal Cheese pizzas require to be delivered to Party 1 = Total Normal Cheese pizzas to be delivered - Normal Cheese pizzas require to be delivered to Party 2 - Normal Cheese pizzas require to be delivered to Party 3

$\Rightarrow$ 416 - 36 - 364 = 16

Hence, option C is the correct answer.

VIDEO SOLUTION

31. **B**

We are given that Party 3 received 70% of total pizzas,therefore, number of pizzas received by Party 3 = $\frac{70}{100} \times 800$ = 560

Remaining 240 pizzas are equally divided among party 1 and party 2 hence we can say that each of Party 1 and Party 2 received 120 pizzas.

We know that all of the pizza can be classified into a total of 4 types. Hence, on drawing a table which can accommodate all of the cases:

	Thin Crust (T)			**Deep Dish (D)**		
	T - NC	**T - EC**	**Total**	**D - NC**	**D - EC**	**Total**
Party 1 **(120)**						
Party 2 **(120)**						
Party 3 **(560)**						
Total **(800)**						

Total number of Thin Crust pizzas = 0.375*800 = 300. Therefore, total number of Deep Dish pizzas = 800 - 300 = 500.

Out of 120 pizzas that Party 1 received, 60% were of Thin Crust type hence, total number of Thin Crust pizza received by Party 1 = 0.6*120 = 72. Consequently Party 1, must have received 48 Deep Dish type pizzas.

Out of 120 pizzas that Party 2 received, 55% were of Thin Crust type hence, total number of Thin Crust pizza received by Party 2 = 0.55*120 = 66. Consequently Party 1, must have received 54 Deep Dish type pizzas.

Therefore, total number of Thin Crust pizzas ordered by Party 3 = Total Thin Crust pizzas ordered - Thin Crust pizzas ordered by Party 1 - Thin Crust pizzas ordered by Party 2

$\Rightarrow$ 300 - 72 - 66 = 162

Hence, option B is the correct answer.

VIDEO SOLUTION

32. **0**

Given that the means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

=> The sum of ratings given by R1, R2, R3 R4, R5 are 5*means = 17, 11, 19, 14, and 17 respectively.

Similarly the sum of ratings received by U, V, W, X and Y are 5*means = 11, 19, 17, 18, and 13 respectively.

Also capturing the absolute data given in the partial information (a) and (b) and representing as a table, we get:

	R1	R2	R3	R4	R5	Total
U	1					11
V					5	19
W	5	1	5			17
X		5	5			18
Y		1	1			13
Total	17	11	19	14	17	

Now,

Consider U

Given median = 2, mode = 2 and range = 3

=> His ratings should be of the form 1, a , 2, b, 4 => 1 + 2 + 4 + a + b = 11 => a + b = 4. For mode = 2 => a = b = 2

=> U's ratings are 1, 2, 2, 2, 4.

Consider V

Given median = 4, mode = 4 and range = 3

=> His ratings should be of the form 2, a, 4, b, 5 => 2 + 4 + 5 + a + b = 19 => a + b = 8 => For mode = 4 => a = b = 4

=> V's ratings are 2, 4, 4, 4, 5.

Consider W

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => 1 + a + 4 + 5 + 5 = 17 => a = 2

=> W's ratings are 1, 2, 4, 5, 5.

Consider X

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => a + 1 + 4 + 5 + 5 = 18 => a = 3

=> X's ratings are 1, 3, 4, 5, 5

Consider Y

Given median = 3, mode = 1 & 4, Range = 3

=> His ratings are 1, 1, 3, 4, 4.

Capturing this data in the table, we get:

	R1	R2	R3	R4	R5	Total	Entries
U	1					11	2,2,2,4
V					5	19	2,4,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R3 => The two missing entries should add up to 19 - 1 - 5 - 5 = 8, (only possibility is 4 + 4) => We can fill the row "U" and 4 in the row "V"

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V			4		5	19	2,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R2 => Missing entry should be 11 - 2 - 1 - 5 - 1 = 2

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Consider column R1, the missing elements should add up to 17 - 5 - 4 - 1 = 7 (3 + 4 or 4 + 3) ----(1)

Consider R5, the missing elements should add up to 10 => 2 + 4 + 4 or 4 + 3 + 3 (not possible) as (1) requires a 3.

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X		5	5		4	18	1,3
Y		1	1		4	13	3,4
Total	17	11	19	14	17		

Now, we can fill column R1 as 3 + 4 and the remaining in column R4 and we can get the complete table

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X	3	5	5	1	4	18	
Y	4	1	1	3	4	13	
Total	17	11	19	14	17		

R2 gave ratings of 1, 1, 2, 2, 5 => He gave 4 to 0 workers => 0 is the answer.

VIDEO SOLUTION

33. **C**

Given that the means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

=> The sum of ratings given by R1, R2, R3 R4, R5 are 5*means = 17, 11, 19, 14, and 17 respectively.

Similarly the sum of ratings received by U, V, W, X and Y are 5*means = 11, 19, 17, 18, and 13 respectively.

Also capturing the absolute data given in the partial information (a) and (b) and representing as a table, we get:

	R1	R2	R3	R4	R5	Total
U	1					11
V					5	19
W	5	1	5			17
X		5	5			18
Y		1	1			13
Total	17	11	19	14	17	

Now,

Consider U

Given median = 2, mode = 2 and range = 3

=> His ratings should be of the form 1, a , 2, b, 4 => 1 + 2 + 4 + a + b = 11 => a + b = 4. For mode = 2 => a = b = 2

=> U's ratings are 1, 2, 2, 2, 4.

Consider V

Given median = 4, mode = 4 and range = 3

=> His ratings should be of the form 2, a, 4, b, 5 => 2 + 4 + 5 + a + b = 19 => a + b = 8 => For mode = 4 => a = b = 4

=> V's ratings are 2, 4, 4, 4, 5.

Consider W

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => 1 + a + 4 + 5 + 5 = 17 => a = 2

=> W's ratings are 1, 2, 4, 5, 5.

Consider X

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => a + 1 + 4 + 5 + 5 = 18 => a = 3

=> X's ratings are 1, 3, 4, 5, 5

Consider Y

Given median = 3, mode = 1 & 4, Range = 3

=> His ratings are 1, 1, 3, 4, 4.

Capturing this data in the table, we get:

	R1	R2	R3	R4	R5	Total	Entries
U	1					11	2,2,2,4
V					5	19	2,4,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R3 => The two missing entries should add up to 19 - 1 - 5 - 5 = 8, (only possibility is 4 + 4) => We can fill the row "U" and 4 in the row "V"

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V			4		5	19	2,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R2 => Missing entry should be 11 - 2 - 1 - 5 - 1 = 2

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Consider column R1, the missing elements should add up to 17 - 5 - 4 - 1 = 7 (3 + 4 or 4 + 3) ----(1)

Consider R5, the missing elements should add up to 10 => 2 + 4 + 4 or 4 + 3 + 3 (not possible) as (1) requires a 3.

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X		5	5		4	18	1,3
Y		1	1		4	13	3,4
Total	17	11	19	14	17		

Now, we can fill column R1 as 3 + 4 and the remaining in column R4 and we can get the complete table

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X	3	5	5	1	4	18	
Y	4	1	1	3	4	13	
Total	17	11	19	14	17		

=> R2 median rating is 2 => given to 2 workers

=> R5 median rating is 4 => given to 2 workers

=> R4 median rating is 3 => given to only 1 worker.

=> R3 median rating is 4 => given to 2 workers.

VIDEO SOLUTION

34. **C**

Since Aditya didn't get a call from any of the colleges, so for each college, he either didn't clear one of the sectional cut-offs or he didn't clear the aggregate cutoff or both. If he didn't clear one of the sectional cut-offs, then for that section he scored less marks than the least cut-off among the given cut-offs of all the colleges.

For example, for section A, it is given that the cut-offs for colleges 1, 4 and 5 are 42, 43 and 45 respectively. The least cut-off among them is 42. So, in order to not clear the sectional cut-off of section A for colleges 1, 4 and 5, he should have scored less than 42.

Similarly for section B, in order to not clear the sectional cut-off for colleges 1, 2 and 6, he should have scored less than 41.
Similarly for section C, in order to not clear the sectional cut-off for colleges 1, 2, 3 and 5, he should have scored less than 42.
Similarly for section D, in order to not clear the sectional cut-off for colleges 4 and 6, he should have scored less than 44.

We can see that if we consider section C and D then we will cover all six colleges. So the maximum score that one can score in section C and D despite that he misses the sectional cut off are 41 and 43 respectively. Also in order to maximize the total we can assume that the person scored full marks in the both section A and B.

Maximum marks obtained by Aditya such that he doesn't get any calls = 41 + 43 + 50 + 50 = 184
Hence, option C is the correct answer.

VIDEO SOLUTION

35. **C**

We have to minimum marks obtained by him in a section which is possible only when Charlie must have scored full marks in remaining 3 sections. Along with that we have to ensure that Charlie gets enough marks to clear aggregate cut off as well.

For college 1, the minimum marks that Charlie needs to get a call = 176. To get the minimum marks in any section he has to score full marks in remaining 3 sections. Since he has to clear the cut off in sections A, B and C we can say that he can score a minimum marks in section D.

Minimum marks in section D despite which he will get call from college 1 = 176 - 3*50 = 26.
Similarly, minimum marks in one section out of A and D despite which he will get call from college 2 = 175 - 3*50 = 25.
Similarly, minimum marks in one section out of A, B and Ddespite which he will get call from college 3 = 171 - 3*50 = 21.
Similarly, minimum marks in one section out of B and C despite which he will get call from college 4 = 178 - 3*50 = 28.
Similarly, minimum marks in one section out of B and D despite which he will get call from college 5 = 180 - 3*50 = 30.
Similarly, minimum marks in one section out of A and C despite which he will get call from college 6 = 176 - 3*50 = 26.

We can see that Charlie need to score at least 25 marks in one section out of A and D to get call from exactly 2 colleges. Therefore, option C is correct answer.

VIDEO SOLUTION

36. **B**

We can see that Bhama got calls from all the college hence it is essential for her to clear all 4 sectionals along with overall cutoff. That essentially means that she has to score marks which are greater or equal to the greatest cut-off for that section across the six colleges. from the table we can see that to clear section A the cut offs for colleges 1, 4 and 5 are 42, 43 and 45 respectively. Hence, in order to clear the sectional cut-off of section A for all the colleges, she should have scored at least 45 marks.
Similarly, the minimum marks required sectional cut-off for section B for all colleges = Max{42, 42, 41} = 45.
Similarly, the minimum marks required sectional cut-off for section C for all colleges = Max{42, 45, 46, 43} = 46.
Similarly, the minimum marks required sectional cut-off for section D for all colleges = Max{45, 44} = 45.
So total minimum aggregate marks required is 45+45+46+45 = 181.
Note we have to check for aggregate cut-off as well for all 6 colleges. We can see that 181 is also more than aggregate cut-off marks of all six colleges. Therefore option B is the correct answer.

VIDEO SOLUTION

37. **60**
It is given that there are only three female students - Amala, Koli, and Rini - and only three male students - Biman, Mathew, and Shyamal - in a course.

It is also known that the aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

Let the project score component be x, which implies the test score component will be (1-x). The projects are done in groups of two, with each group consisting of a female and a male student, which implies there are three groups for the project. It is also known that both the group members obtain the same score in the project. The score obtained in the project is 40, 60, and 80, respectively.

Therefore, we can say that each female student will consist of a different group, and no two male students or female students will be in the same group.

For the test scores, there are six scores given for six students among which four are distinct and the remaining two are average scores, which is 60. It is also known that the maximum score possible is 80, and the minimum score is 40.

Hence, the distinct scores are 80, 70, 50, and 40 (since all the test scores are multiple of 10), and the remaining two scores are 60, and 60, respectively.

From point 3, we know that Amala’s score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Hence, we can say the score obtained by Amala in the project is 80, and the score obtained by Koli is 40, which implies the score obtained by Rini in the project is 60. Now, Koli scored 20 more than Amala in the test, which implies the score obtained by Koli can be either 80, 70, or 60.

The score obtained by them is given below:

Students	Test scores	Project scores
Amala	40/50/60	80
Koli	60/70/80	40
Rini		60
Biman		
Mathew		
Shyamal		

It is known that Amala had the highest aggregate score, and Shyamal scored the second highest on the test. He scored two more than Koli, but two less than Amala in the aggregate.

Hence, the score obtained by Shyamal in the test is 70, which implies Koli can't score 70 in the test => Amala can't score 50 in the test.

Students	Test scores	Project scores
Amala	40/60	80
Koli	60/80	40
Rini		60
Biman		
Mathew		
Shyamal	70	

It is given that Shyamal scored two more than Koli, but two less than Amala in the aggregate. Hence, the aggregate score of Amala is 4 more than Koli. It is also known that Amala had the highest aggregate score.

Case 1: The test score of Amala is 40

Students	Test scores	Project scores	Aggregate score
Amala	40	80	40(1-x)+80x
Koli	60	40	60(1-x)+40x
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, 40(1-x)+80x = 60(1-x)+40x+4

=> 60x = 24

=> x = 0.4

Hence, the aggregate score obtained by Amala is 40(1-0.4)+80*4 = 56

The minimum aggregate score of Shyamal is 70(1-0.4)+ 40*0.4 = 58, which is greater than Amala.

Hence, **Case 1 is not possible.**

Hence, the table is given below:

Students	Test scores	Project scores	Aggregate score
Amala	60	80	60(1-x)+80x
Koli	80	40	80(1-x)+40x
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, 60(1-x)+80x = 80(1-x)+40x+4

=> 60+20x = 84-40x

=> 60x = 24 => x = 0.4

Hence, the aggregate score of Amala is 60(1-0.4)+80*0.4 = 68, which implies the aggregate score of Shyamal is (68-2) = 66

Hence, the score obtained by Shyamal in Project is {66-70*(0.6)}/0.4 = 60.

It is also known that Biman scored second lowest in the test, which implies the score of Biman in the test is 50, and he scored the lowest in the aggregate. It is also known that Mathew scored more than Rini in the project, but less than her in the test. Hence, Mathew scored 80 in the project (since Rini scored 60 in the project), and Biman scored 40 in the project.

Similarly, Rini Scored more than Mathew on the test, which implies the score obtained by Rini is 60, and the score obtained by Mathew is 40 in the test.

Hence, the final table will look like this:

Students	Test scores (T)	Project scores(P)	Aggregate score (T*0.6+P*0.4)	Project Pair
Amala	60	80	68	Amala, Mathew
Koli	80	40	64	Koli, Biman
Rini	60	60	60	Rini, Shyamal
Biman	50	40	46	Biman, Koli
Mathew	40	80	56	Mathew, Amala
Shyamal	70	60	66	Shyamal, Rini

Hence, the score obtained by Rini in the project is 60

VIDEO SOLUTION

38. **A**

It is given that Indu and Jatin both scored 100% in exactly one section. We can say that Jatin scored 100% marks in DI. Therefore, Jatin's composite score = 2*20+16+14 = 70

It is given that Jatin's composite score was 10 more than Indu's. Therefore, we can say that Indu's composite score = 70 - 10 = 60.

Indu also scored 100% in exactly one section.

Case 1: Indu scored 100% marks in DI.

If Indu scored 100% marks in DI, then Indu's score in GA = 60 - 2*20 - 8 = 12 which is less than 70% of maximum possible marks. Indu already has less than 70% in WE, therefore we Indu can't be recruited . Hence, we can reject this case.

Consequently, we can say that Indu scored 100% marks in WE. Therefore, Indu's score in DI = $\frac{60-8-20}{2}$ = 16

It is also given that Danish, Harini, and Indu had scored the same marks the in GA.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8		16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14		6	
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

We are given that, among the four recruited, Geeta had the lowest composite score.

Maximum composite score that Geeta can get = 2*14 + 6 + 20 = 54 {Assuming 100% marks in WE}. Since, Geeta was recruitedat a composite score of 54 or less we can say that Ester was definitely recruited.

It is given that no two candidates had the same composite score. We can see that Chetna's composite score is 54. Hence, Geeta can't have a composite score of 54. Therefore, we can say that Geeta's composite score is 53 or less.

We already know the four people(Jatin, Indu, Geeta, Ester) which were recruited. Hence, we cab say that Danish was rejected at a composite score of 51. Hence, we can say that Geeta's composite score in 52 or more.

Consequently, we can say that Geeta's composite score if either 52 or 53. Therefore we can say that Geeta scored either 18 {52-(2*14+6)} or 19 {53-(2*14+6)} marks in WE.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8		16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14	18/19	6	52/53
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

Ajay was the unique highest scorer in WE.

Case 1: Geeta scored 19 marks in WE.

We can say that if Geeta scored 19 marks in WE, then Ajay scored 20 marks in DI. In that case Ajay's composite score = 2*8 + 20 + 16 = 52. Which is a possible case.

Case 1: Geeta scored 18 marks in WE.

We can say that if Geeta scored 18 marks in WE, then Ajay can score either 19 or 20 marks in DI.

If Ajay scored 20 marks in DI then in that case Ajay's composite score = 2*8 + 20 + 16 = 52 which will be same as Geeta's composite score. Hence, we can say that in this case Ajay can't score 20 marks.

If Ajay scored 19 marks in DI then in that case Ajay's composite score = 2*8 + 19 + 16 = 51 which will be same as Danish's composite score. Hence, we can say that in this case Ajay can't score 19 marks.

Therefore, we can say that case 2 is not possible at all.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8	20	16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14	19	6	53
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

Let us check all the statement one by one.

Option A: Bala scored same as Jatin in DI. We can say that Bala scored 20 marks in DI. In that Bala's composite score = 2*20 + 9 + 11 = 60 which is sam as Indu's composite score. Therefore, we can say that this is a false statement. Hence, option A is the correct answer.

VIDEO SOLUTION

39. **9**

It's given in the question that the first runner up polled 10,000 more votes than the second runner up in constituency A. Now the first runner up has got 95000 votes, hence the second runner up will get 85000 votes.

Now the remaining votes will be 500000-275000-95000-85000=45000

From 2, None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000 => the person who got 5th highest votes must have got > 600030/6 => $\geq$ 100006. Since it is also given that the difference of votes is $\geq$ 10000, the only possible case is winner, 1st runner up, 2nd runner up, 3rd runner up, 4th runner up must have got 140006,130006,120006,110006,100006 respectively which sums upto exactly 600030.

Let the number of votes polled in D be 100x.

From 3, The winning candidate in constituency D must have got 15x+37500

The table now looks like:

	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	100x
No. of votes polled by the winning candidate	2,75,000	48,750	140006	37500+5x
No. of votes polled by the first runner up	95,000		130006	37,500
No. of votes polled by the second runner up	85000		120006	30,000
% of valid votes polled by the third runner up			16.67%	10%

Total votes in A = 500000 => the candidates who got $\leq$ 83333 must have lost their security deposits => candidates till 2nd runner up didn't lose their deposit =>all the candidates who received 500000-275000-95000-85000=**45000** lost their deposits.

The percentage of votes polled in total by all the candidates who lost their security deposits while contesting for constituency A = 45000*100/500000 =9%

VIDEO SOLUTION

40. **13**

It is given that Indu and Jatin both scored 100% in exactly one section. We can say that Jatin scored 100% marks in DI. Therefore, Jatin's composite score = 2*20+16+14 = 70

It is given that Jatin's composite score was 10 more than Indu's. Therefore, we can say that Indu's composite score = 70 - 10 = 60.

Indu also scored 100% in exactly one section.

Case 1: Indu scored 100% marks in DI.

If Indu scored 100% marks in DI, then Indu's score in GA = 60 - 2*20 - 8 = 12 which is less than 70% of maximum possible marks. Indu already has less than 70% in WE, therefore we Indu can't be recruited . Hence, we can reject this case.

Consequently, we can say that Indu scored 100% marks in WE. Therefore, Indu's score in DI = $\dfrac{60-8-20}{2}$ = 16

It is also given that Danish, Harini, and Indu had scored the same marks the in GA.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8		16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14		6	
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

We are given that, among the four recruited, Geeta had the lowest composite score.

Maximum composite score that Geeta can get = 2*14 + 6 + 20 = 54 {Assuming 100% marks in WE}. Since, Geeta was recruitedat a composite score of 54 or less we can say that Ester was definitely recruited.

It is given that no two candidates had the same composite score. We can see that Chetna's composite score is 54. Hence, Geeta can't have a composite score of 54. Therefore, we can say that Geeta's composite score is 53 or less.

We already know the four people(Jatin, Indu, Geeta, Ester) which were recruited. Hence, we cab say that Danish was rejected at a composite score of 51. Hence, we can say that Geeta's composite score in 52 or more.

Consequently, we can say that Geeta's composite score if either 52 or 53. Therefore we can say that Geeta scored either 18 {52-(2*14+6)} or 19 {53-(2*14+6)} marks in WE.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8		16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14	18/19	6	52/53
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

Ajay was the unique highest scorer in WE.

Case 1: Geeta scored 19 marks in WE.

We can say that if Geeta scored 19 marks in WE, then Ajay scored 20 marks in DI. In that case Ajay's composite score = 2*8 + 20 + 16 = 52. Which is a possible case.

Case 1: Geeta scored 18 marks in WE.

We can say that if Geeta scored 18 marks in WE, then Ajay can score either 19 or 20 marks in DI.

If Ajay scored 20 marks in DI then in that case Ajay's composite score = 2*8 + 20 + 16 = 52 which will be same as Geeta's composite score. Hence, we can say that in this case Ajay can't score 20 marks.

If Ajay scored 19 marks in DI then in that case Ajay's composite score = 2*8 + 19 + 16 = 51 which will be same as Danish's composite score. Hence, we can say that in this case Ajay can't score 19 marks.

Therefore, we can say that case 2 is not possible at all.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8	20	16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14	19	6	53
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

It is given that all the candidates except Ajay and Danish had different marks in DI and Bala's composite score was less than Chetna's composite score.

Let us assume that Bala scored 'x' marks in DI.

$\Rightarrow$ 2*x + 9 + 11 < 54

$\Rightarrow$ x < 17

We can see that Bala's score will be less than 17. Bala's maximum score in DI will be the largest possibel number less than 17 which is not same as any other candidate's score in DI. From, the table we can see that 16, 15 and 14 are already taken by Indu, Falak and Geeta respectively.

Therefore, we can say that Bala can score a maximum of 13 marks in DI.

VIDEO SOLUTION

41. **C**
In this set, we are told that the amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down and no firm raised the same amount of money in two consecutive years.

The increase or decrease can be ± 1 or ± 2. => (1)

We are also told that each firm raised Rs. 1 crore in its first and last year of existence

Consider A:

It raised money for 8 years

=> The raising pattern looks like follows:

1, a, b, c, d, e, f, 1 => where a, b, c,..,, f are the unknown amounts raised.

Also a + b + c + d + e + f = 21 - 2 = 19.

We can observe that 19/6 is slightly greater than 3 => The average amount raised should be around 3.

If a = 3 and f = 3 => b + c + d + e = 13 (not possible) as the minimum case would be (4, 5, 6, 4) => Not possible.

If a = 3 and f = 2 => b + c + d + e = 14 (not possible) as the minimum case would be (4, 5, 4, 3) => Not possible.

=> a = 2 and f = 2 => b + c + d + e = 15 the minimum case is (3, 4, 5, 3) or (3, 5, 4, 3) which gives a sum of 15.

So, the possible cases for A are:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

Consider B:

The patterns looks as follows:

1, a, b, 1

If a = 2, b has to be equal to 3 to satisfy (1)

if a = 3, b has to be equal to 2 to satisfy (1)

=> The possible cases for B are:

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

Consider C:

The pattern looks as follows:

1, ..., 1

Let us assume there are 2 gaps between => a + b = 7 (Not possible) as maximum case would be 1, 3, 2, 1

Let us assume there are 3 gaps between => a + b + c = 7, the minimum case possible is 1, 2, 3, 2, 1 => Satisfies.

Now, if there are 4 gaps => a + b + c + d = 7 => The average value is 7/4 which is less than 2 => Not possible.

=> The possible cases for C are:

C	2013	2014	2015	2016	2017
	1	2	3	2	1

Consider D:

The pattern looks as follows:

1, a, b, c, 1

=> a + b + c = 8

When a = 2 and c = 2 => b = 4 => 2, 4, 2 => Satisfies.

When a = 2 and c = 3, b should be 3 (Not satisfying (1))

When a = 3 and c = 3, b should be 2 (Not satisfying (1))

=> The possible cases for D are:

D	2011	2012	2013	2014	2015
	1	2	4	2	1

Consider E:

The pattern looks as follows:

1,.....,1

For 1 or 2 gaps, we can't get a sum of 11.

Assume 3 gaps => a + b + c = 11, the maximum case is 3, 5, 3 => Satisfies.

Now, assume 4 gaps

=> a + b + c + d = 11, the minimum case is 2, 3, 4, 2 or 2, 4, 3, 2 which satisfies (1) and 2 + 3 + 4 + 2 = 11.

=> The possible cases for E are:

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

In summary, the possible cases for all 5 companies is:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

C	2013	2014	2015	2016	2017
	1	2	3	2	1

D	2011	2012	2013	2014	2015
	1	2	4	2	1

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

Money raised in 2015 is 2 + 1 + 3 + 1 + 0/1 = 7 or 8.

VIDEO SOLUTION

42. **C**

In this set, we are told that the amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down and no firm raised the same amount of money in two consecutive years.

The increase or decrease can be ± 1 or ± 2. => (1)

We are also told that each firm raised Rs. 1 crore in its first and last year of existence

Consider A:

It raised money for 8 years

=> The raising pattern looks like follows:

1, a, b, c, d, e, f, 1 => where a, b, c,..,, f are the unknown amounts raised.

Also a + b + c + d + e + f = 21 - 2 = 19.

We can observe that 19/6 is slightly greater than 3 => The average amount raised should be around 3.

If a = 3 and f = 3 => b + c + d + e = 13 (not possible) as the minimum case would be (4, 5, 6, 4) => Not possible.

If a = 3 and f = 2 => b + c + d + e = 14 (not possible) as the minimum case would be (4, 5, 4, 3) => Not possible.

=> a = 2 and f = 2 => b + c + d + e = 15 the minimum case is (3, 4, 5, 3) or (3, 5, 4, 3) which gives a sum of 15.

So, the possible cases for A are:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

Consider B:

The patterns looks as follows:

1, a, b, 1

If a = 2, b has to be equal to 3 to satisfy (1)

if a = 3, b has to be equal to 2 to satisfy (1)

=> The possible cases for B are:

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

Consider C:

The pattern looks as follows:

1, ..., 1

Let us assume there are 2 gaps between => a + b = 7 (Not possible) as maximum case would be 1, 3, 2, 1

Let us assume there are 3 gaps between => a + b + c = 7, the minimum case possible is 1, 2, 3, 2, 1 => Satisfies.

Now, if there are 4 gaps => a + b + c + d = 7 => The average value is 7/4 which is less than 2 => Not possible.

=> The possible cases for C are:

C	2013	2014	2015	2016	2017
	1	2	3	2	1

Consider D:

The pattern looks as follows:

1, a, b, c, 1

=> a + b + c = 8

When a = 2 and c = 2 => b = 4 => 2, 4, 2 => Satisfies.

When a = 2 and c = 3, b should be 3 (Not satisfying (1))

When a = 3 and c = 3, b should be 2 (Not satisfying (1))

=> The possible cases for D are:

D	2011	2012	2013	2014	2015
	1	2	4	2	1

Consider E:

The pattern looks as follows:

1,.....,1

For 1 or 2 gaps, we can't get a sum of 11.

Assume 3 gaps => a + b + c = 11, the maximum case is 3, 5, 3 => Satisfies.

Now, assume 4 gaps

=> a + b + c + d = 11, the minimum case is 2, 3, 4, 2 or 2, 4, 3, 2 which satisfies (1) and 2 + 3 + 4 + 2 = 11.

=> The possible cases for E are:

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

In summary, the possible cases for all 5 companies is:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

C	2013	2014	2015	2016	2017
	1	2	3	2	1

D	2011	2012	2013	2014	2015
	1	2	4	2	1

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

Given that E raised 3 in 2013 => in 2012 he could have raised a minimum of 4 crores.

=> Minimum amount is 4 + 1 + 0 + 2 + 4 = 11.

VIDEO SOLUTION

43. **D**

Given that in every month, both online and offline registration numbers were multiples of 10.

From (2), in Jan, the number of offline registrations was twice that of online registrations.

=> If x is number of online registrations => 2x is the number of offline registrations => 3x is the total number of registrations.

According to the data given in the table => 3x should lie between the minimum and maximum total number of registrations. => x = 40 (as x should also be a multiple of 10)

=> In Jan => (40,80) are the online and offline registrations respectively.

Similarly from (3) => In Apr (80,40) are the online and offline registrations respectively.

From-5, the number of online registrations is highest in may => In may there are 100 online registrations. The lowest possible number of offline registrations is 30 and maximum possible total registrations is 130 => In May (100,30) are the online and offline registrations respectively.

Let us assume, 'x' to be the number of offline registrations in May = number of online registrations in March.

Let us capture all this data in a table:

Month	Online	Offline	Total
Jan	40	80	120
Feb	y	x	
Mar	x	z	
Apr	80	40	120
May	100	30	130

From the table given in the question, 50 is the median for Offline data

=> x should lie between 50 and 80 (included)

For 80 to be the median for the online data => y lie between 80 and 100 (included).

Now, consider Feb => Minimum value of y + x = 80 + 50 = 130 (which is the maximum value possible of the total possible registrations)

=> x = 50 and y = 80

Since, 110 is the minimum number of total registrations, the only possibility is in March => 50 + z = 110 => z = 60.

Now, filling the complete table we get,

Month	Online	Offline	Total
Jan	40	80	120
Feb	80	50	130
Mar	50	60	110
Apr	80	40	120
May	100	30	130

1) In May, there are 30 offline registrations (smallest) => True

2) In Mar, we have smallest number of total registrations => False.

VIDEO SOLUTION

44. **5**
The set's starting point is that the sum of each row must be 100

Clue 3 tells us that all the missing elements in the Carbs column are multiple of 5. Similarly, clue 4 tells us that the missing elements of the other three columns are multiples of 4.

Note that we look at clue 1, which says that the carbs in C1 and C2 must be more than any carbs in any pseudo cereal.
We have the reference point of P1 at 66

Trying to fill in for C1:
The possible values are 75, 80, 85, 90, 95
Since 12 grams is of other nutrients, we can eliminate 90 and 95.
If we take 85, 85+12 = 97, which would leave 3 grams of protein, but from clue 4, we know this has to be a multiple of 4.

Taking 75, 75+12 = 87 would leave 13 grams of protein, which, too, is eliminated.
Leaving only 80 grams of carb in C1 and 8 grams of protein.

Similar logic is to be applied for C2; we have 13 grams already present, so the values of 90 and 95 are eliminated.
Taking 85 carbs would give 2 grams of protein, which can be eliminated.
80 and 70 would also not work for the same reason.
Leavin has only 75 grams of carbs, leaving 12 grams of protein.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2		14		8

Using clues 1 and 2 together, we can determine that the carbs in P2 must be less than 75 but at least greater than 62.
The only possible values are 65 and 70
Putting 65 grams of carbs in P2 gives us 100-65-14-8 = 13 grams of protein, which is invalid.

Putting 70 grams of carbs in P2 gives 100-70-14-8 = 8 grams of protein.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

Clue 1 gives us that the protein in M2 should be less than that present in either of the pseudo-cereals, which we right now have a baseline of 14.
So the protein in M2 (and M3) can be 12, 8, 4, 0

The protein and carbs in M2 should add up to 100-7-16 = 77
This is only possible if the protein count is 12, giving carbs as 65

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

The protein in M3 can be 0, 4, 8 or 12

We are given in clue five that the protein in P1 is double that in M3.
The protein in P1 thus can be 0, 8, 16 or 24
Since this P1 protein also has to be more than M1 and M2 protein, it can not be 0 or 8
Leaving only 16 or 24 as the valid values.

The protein and fats in P1 must add up to 100-66-10 = 24
If P1 had 24 grams of protein, then it would have 0 grams of fat, but in clue 4, we are given that all missing fats are **non-zero multiples of 4**
Hence, the only possible protein value in P1 is 16, with 8 grams of fats. Giving 8 grams of protein in M3 and 24 grams of others in M3.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56	8	12	24
Pseudo-cereals	P1	66	16	8	10
	P2	70	14	8	8

We can see that all other foodgrains except M3 had more carbohydrates than M1.
Therefore, 5 is the correct answer.

VIDEO SOLUTION

45. **D**

It's given in the question that the first runner up polled 10,000 more votes than the second runner up in constituency A. Now the first runner up has got 95000 votes, hence the second runner up will get 85000 votes.

Now the remaining votes will be 500000-275000-95000-85000=45000

From 2, None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000 => the person who got 5th highest votes must have got > 600030/6 => $\geq$ 100006. Since it is also given that the difference of votes is $\geq$ 10000, the only possible case is winner, 1st runner up, 2nd runner up, 3rd runner up, 4th runner up must have got 140006,130006,120006,110006,100006 respectively which sums upto exactly 600030.

Let the number of votes polled in D be 100x.

From 3, The winning candidate in constituency D must have got 37500+5x.

The table now looks like:

	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	100x
No. of votes polled by the winning candidate	2,75,000	48,750	140006	37500+5x
No. of votes polled by the first runner up	95,000		130006	37,500
No. of votes polled by the second runner up	85000		120006	30,000
% of valid votes polled by the third runner up			16.67%	10%

Number of votes polled to winning candidate must be 140006.

VIDEO SOLUTION

46. **C**

It's given in the question that the first runner up polled 10,000 more votes than the second runner up in constituency A. Now the first runner up has got 95000 votes, hence the second runner up will get 85000 votes.

Now the remaining votes will be 500000-275000-95000-85000=45000

From 2, None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000 => the person who got 5th highest votes must have got > 600030/6 => $\geq$ 100006. Since it is also given that the difference of votes is $\geq$ 10000, the only possible case is winner, 1st runner up, 2nd runner up, 3rd runner up, 4th runner up must have got 140006,130006,120006,110006,100006 respectively which sums upto exactly 600030.

From 3, The winning candidate in constituency D must have got 1.05*37500 = 39375.

Let the number of votes polled in D be 100x.

The table now looks like:

	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	100x
No. of votes polled by the winning candidate	2,75,000	48,750	140006	39,375
No. of votes polled by the first runner up	95,000		130006	37,500
No. of votes polled by the second runner up	85000		120006	30,000
% of valid votes polled by the third runner up			16.67%	10%

As calculated in the previous question, candidate D got 175000 votes. The winner got 5% more votes than first runner up, hence the winner got 0.05*175000 i.e 8750 more votes than first runner up . Thus 8750 is the winning margin for constituency D. Moreover margin in constituency is atleast 10000 . Hence in the increasing order C will always come after D which is not happening in the third option. Hence that is the correct answer.

VIDEO SOLUTION

47. **D**

It is given that: High Q > Best Ed > Cosmopolitan and Education Aid > A-one

We can say that High Q > Cosmopolitan

We can see that both High Q and Cosmopolitan got same points in reputation (R) and placement quality (P). High Q received more points in infrastructure (I) than Cosmopolitan whereas Cosmopolitan received more points in faculty Quality (F) than High Q.

Hence, we can say that Infrastructure's weight should be greater than Faculty quality. i.e. I > F

Similarly, We can see that both Best Ed and Cosmopolitan got same points in faculty Quality (F) and placement quality (P). Best Ed received more points in reputation (R) than Cosmopolitan whereas Cosmopolitan received more points in infrastructure (I) than Best Ed.

Hence, we can say that reputation's weight should be greater than infrastructure. i.e. R > I

Similarly, We can see that both Education Aid and A-one got same points in faculty Quality (F) and reputation (R). Education Aid received more points in infrastructure (I) than A-one whereas A-one received more points in placement quality (P) than Education Aid.

Hence, we can say that reputation's weight should be greater than infrastructure. i.e. I > P

So basically there are two possible cases: R > I > P > F or R > I > F > P

Case 1: Order of weights assigned = R > I > P > F

R = 0.4, I = 0.3. P = 0.2, F = 0.1

In this case overall score received by Best Ed = 0.1*40+0.4*30+0.2*20+0.3*20 = 26

In this case overall score received by High Q = 0.1*30+0.4*20+0.2*20+0.3*40 = 27

We can see that High Q's overall score is higher than Best Ed. Hence, this is a possible case.

Case 2: Order of weights assigned = R > I > F > P

R = 0.4, I = 0.3. P = 0.1, F = 0.2

In this case overall score received by Best Ed = 0.2*40+0.4*30+0.1*20+0.3*20 = 28

In this case overall score received by High Q = 0.2*30+0.4*20+0.1*20+0.3*40 = 28

We can see that High Q's overall score is not greater than the overall score received Best Ed. Hence, this case is not possible.

Now that we know the weight of each parameter, we can calculate the overall score and accreditation received by each college.

	F (0.1)	R (0.4)	P (0.2)	I (0.3)	Overall score	Accreditation
A-one	50	50	50	40	47	AAA
Best Ed	40	30	20	20	26	BBA
Cosmopolitan	40	20	20	30	25	BBA
Dominance	20	20	40	30	27	BBA
Education Aid	50	50	40	50	48	AAA
Fancy	50	50	40	40	45	AAA
Global	30	0	20	20	13	Junk
High Q	30	20	20	40	27	BBA

We can see that weight of the faculty quality parameter = 0.1. Hence,option D is the correct answer.

VIDEO SOLUTION

48. **A**

It is given that: High Q > Best Ed > Cosmopolitan and Education Aid > A-one

We can say that High Q > Cosmopolitan

We can see that both High Q and Cosmopolitan got same points in reputation (R) and placement quality (P). High Q received more points in infrastructure (I) than Cosmopolitan whereas Cosmopolitan received more points in faculty Quality (F) than High Q.

Hence, we can say that Infrastructure's weight should be greater than Faculty quality. i.e. I > F

Similarly, We can see that both Best Ed and Cosmopolitan got same points in faculty Quality (F) and placement quality (P). Best Ed received more points in reputation (R) than Cosmopolitan whereas Cosmopolitan received more points in infrastructure (I) than Best Ed.

Hence, we can say that reputation's weight should be greater than infrastructure. i.e. R > I

Similarly, We can see that both Education Aid and A-one got same points in faculty Quality (F) and reputation (R). Education Aid received more points in infrastructure (I) than A-one whereas A-one received more points in placement quality (P) than Education Aid.

Hence, we can say that reputation's weight should be greater than infrastructure. i.e. I > P

So basically there are two possible cases: R > I > P > F or R > I > F > P

Case 1: Order of weights assigned = R > I > P > F

R = 0.4, I = 0.3. P = 0.2, F = 0.1

In this case overall score received by Best Ed = 0.1*40+0.4*30+0.2*20+0.3*20 = 26

In this case overall score received by High Q = 0.1*30+0.4*20+0.2*20+0.3*40 = 27

We can see that High Q's overall score is higher than Best Ed. Hence, this is a possible case.

Case 2: Order of weights assigned = R > I > F > P

R = 0.4, I = 0.3. P = 0.1, F = 0.2

In this case overall score received by Best Ed = 0.2*40+0.4*30+0.1*20+0.3*20 = 28

In this case overall score received by High Q = 0.2*30+0.4*20+0.1*20+0.3*40 = 28

We can see that High Q's overall score is not greater than the overall score received Best Ed. Hence, this case is not possible.

Now that we know the weight of each parameter, we can calculate the overall score and accreditation received by each college.

	F (0.1)	R (0.4)	P (0.2)	I (0.3)	Overall score	Accreditation
A-one	50	50	50	40	47	AAA
Best Ed	40	30	20	20	26	BBA
Cosmopolitan	40	20	20	30	25	BBA
Dominance	20	20	40	30	27	BBA
Education Aid	50	50	40	50	48	AAA
Fancy	50	50	40	40	45	AAA
Global	30	0	20	20	13	Junk
High Q	30	20	20	40	27	BBA

From the table we can see that none of the mentioned college received an overall scores between 31 and 40, both inclusive. Hence, option A is the correct answer.

VIDEO SOLUTION

49. **D**

Since A, B, C are mutual strangers,(B, C)are strangers for A, (A, C) are strangers for B, (A, B) are strangers for C.

Since the total number of strangers+ acquaintances+ friends for any among the 7 is 6.

The number of friends for Amudha is 1, the number of friends for Gowtham is 1.

Using statement 3 Chandran and Gowtham are friends.

Using statement 2: Amudha, Dinesh, and Fani are Ezil's friends. Similarly, Ezil is a friend of Amudha, Dhinesh, and Fani.

Using statement 4 Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan, Hence the number of acquaintances of Bharatan is equal to the number of friends of Amudha.

Using statement 5 Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance Bharatan is a friend of Amudha, Hence the number of acquaintances of Amudha is equal to the number of friends of Bharatan.

Hence Bharatan has one friend, 1 Acquaintance, 4 strangers.

For Amudha we are yet to find a relationship with Dhinesh, Fani, and Gowtham. Any among the three can be the stranger for Amudha, considering the three different cases.

Case 1 :

Considering Fani as an acquaintance of Amudha, then Dhinesh and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Fani is an acquaintance of Amudha, hence Amudha is an Acquaintance of Fani, Fani is a friend of Bharathan and hence Bharathan is a friend of Fani. But Fani has only one friend and Ezhil is already a friend of Fani.

Hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1, (F)	4 (B, C, D, G)
Bharatan	1, (F)	1, (E)	4, (A, C, D, G)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 2 :

Considering Gowtham as an acquaintance of Amudha, then Dhinesh and Fani are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.Since Gowtham is a friend of Bharatan, Bharatan must be a friend of Gowtham. But Gowtham can only have one friend and it already mentioned that Chandran is a friend of Gowtham and hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (G)	4 (B, C, D, F)
Bharatan	1, (G)	1, (E)	4, (A, C, D, F)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 3 :

Considering Dhinesh as an acquaintance of Amudha, then Fani and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Since Fani, Gowtham are strangers to Amudha, Bharatan. Amudha, Bharatan are strangers to Fani, Gowtham.

The 2 strangers to Gowtham and his only friend are known. Hence his three acquaintances are (Dhinesh, Ezhil, Fani).

Hence Gowtham is an acquaintance of Dhinesh, Ezhil, and Fani.

Dhinesh is an acquaintance of Amudha and hence Amudha must be an acquaintance of Dhinesh.

Dhinesh is a friend of Bharatan and hence Bharatan is a friend of Dhinesh.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4 (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	G	1	(A, B)
Dhinesh	E, B	G, A	2
Ezhil	A, D, F	G	1
Fani	1 (E)	G	A, B
Gowtham	1 (C)	3 (D, E, F)	2 (A, B)

Ezhil is an acquaintance of Bharatan and hence Bharatan is an acquaintance of Ezhil. The only stranger to Ezhil who is left is Chandran. Hence Ezhil is a stranger to Chandran.

The two strangers to Dhinesh who are left are Chandran and Fani. Chandran is a stranger to Dhinesh and hence Dhinesh is a stranger to Chandran.

The only acquaintance of Chandran who is left is Fani.

The remaining relationships with Fani are Dhinesh and Chandran. Dhinesh is a stranger to Fani and Chandran is an acquaintance of Fani.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4, (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	1, (G)	1 (F)	4, (A, B, E, D)
Dhinesh	2, (E, B)	2, (G, A)	2, (C, F)
Ezhil	3, (A, D, F)	2, (G, B)	1, (C).
Fani	1 (E)	2, (G, C)	3, (A, B, D)
Gowtham	1 (C)	3 (D, E, F)	2, (A, B)

In the given options Bharatan and Ezhil are Acauaintances, Fani and Gowtham are acquaintances.

VIDEO SOLUTION

50. **B**

Since A, B, C are mutual strangers,(B, C)are strangers for A, (A, C) are strangers for B, (A, B) are strangers for C.

Since the total number of strangers+ acquaintances+ friends for any among the 7 is 6.

The number of friends for Amudha is 1, the number of friends for Gowtham is 1.

Using statement 3 Chandran and Gowtham are friends.

Using statement 2: Amudha, Dinesh, and Fani are Ezil's friends. Similarly, Ezil is a friend of Amudha, Dhinesh, and Fani.

Using statement 4 Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan, Hence the number of acquaintances of Bharatan is equal to the number of friends of Amudha.

Using statement 5 Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance Bharatan is a friend of Amudha, Hence the number of acquaintances of Amudha is equal to the number of friends of Bharatan.

Hence Bharatan has one friend, 1 Acquaintance, 4 strangers.

For Amudha we are yet to find a relationship with Dhinesh, Fani, and Gowtham. Any among the three can be the stranger for Amudha, considering the three different cases.

Case 1 :

Considering Fani as an acquaintance of Amudha, then Dhinesh and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Fani is an acquaintance of Amudha, hence Amudha is an Acquaintance of Fani, Fani is a friend of Bharathan and hence Bharathan is a friend of Fani. But Fani has only one friend and Ezhil is already a friend of Fani.

Hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1, (F)	4 (B, C, D, G)
Bharatan	1, (F)	1, (E)	4, (A, C, D, G)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 2 :

Considering Gowtham as an acquaintance of Amudha, then Dhinesh and Fani are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.Since Gowtham is a friend of Bharatan, Bharatan must be a friend of Gowtham. But Gowtham can only have one friend and it already mentioned that Chandran is a friend of Gowtham and hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (G)	4 (B, C, D, F)
Bharatan	1, (G)	1, (E)	4, (A, C, D, F)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 3 :

Considering Dhinesh as an acquaintance of Amudha, then Fani and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Since Fani, Gowtham are strangers to Amudha, Bharatan. Amudha, Bharatan are strangers to Fani, Gowtham.

The 2 strangers to Gowtham and his only friend are known. Hence his three acquaintances are (Dhinesh, Ezhil, Fani).

Hence Gowtham is an acquaintance of Dhinesh, Ezhil, and Fani.

Dhinesh is an acquaintance of Amudha and hence Amudha must be an acquaintance of Dhinesh.

Dhinesh is a friend of Bharatan and hence Bharatan is a friend of Dhinesh.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4 (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	G	1	(A, B)
Dhinesh	E, B	G, A	2
Ezhil	A, D, F	G	1
Fani	1 (E)	G	A, B
Gowtham	1 (C)	3 (D, E, F)	2 (A, B)

Ezhil is an acquaintance of Bharatan and hence Bharatan is an acquaintance of Ezhil. The only stranger to Ezhil who is left is Chandran. Hence Ezhil is a stranger to Chandran.

The two strangers to Dhinesh who are left are Chandran and Fani. Chandran is a stranger to Dhinesh and hence Dhinesh is a stranger to Chandran.

The only acquaintance of Chandran who is left is Fani.

The remaining relationships with Fani are Dhinesh and Chandran. Dhinesh is a stranger to Fani and Chandran is an acquaintance of Fani.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4, (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	1, (G)	1 (F)	4, (A, B, E, D)
Dhinesh	2, (E, B)	2, (G, A)	2, (C, F)
Ezhil	3, (A, D, F)	2, (G, B)	1, (C).
Fani	1 (E)	2, (G, C)	3, (A, B, D)
Gowtham	1 (C)	3 (D, E, F)	2, (A, B)

Fani is an acquaintance of Chandran

VIDEO SOLUTION

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51. **A**

In this set, we are told that the amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down and no firm raised the same amount of money in two consecutive years.

The increase or decrease can be ± 1 or ± 2. => (1)

We are also told that each firm raised Rs. 1 crore in its first and last year of existence

Consider A:

It raised money for 8 years

=> The raising pattern looks like follows:

1, a, b, c, d, e, f, 1 => where a, b, c,..,, f are the unknown amounts raised.

Also a + b + c + d + e + f = 21 - 2 = 19.

We can observe that 19/6 is slightly greater than 3 => The average amount raised should be around 3.

If a = 3 and f = 3 => b + c + d + e = 13 (not possible) as the minimum case would be (4, 5, 6, 4) => Not possible.

If a = 3 and f = 2 => b + c + d + e = 14 (not possible) as the minimum case would be (4, 5, 4, 3) => Not possible.

=> a = 2 and f = 2 => b + c + d + e = 15 the minimum case is (3, 4, 5, 3) or (3, 5, 4, 3) which gives a sum of 15.

So, the possible cases for A are:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

Consider B:

The patterns looks as follows:

1, a, b, 1

If a = 2, b has to be equal to 3 to satisfy (1)

if a = 3, b has to be equal to 2 to satisfy (1)

=> The possible cases for B are:

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

Consider C:

The pattern looks as follows:

1, ..., 1

Let us assume there are 2 gaps between => a + b = 7 (Not possible) as maximum case would be 1, 3, 2, 1

Let us assume there are 3 gaps between => a + b + c = 7, the minimum case possible is 1, 2, 3, 2, 1 => Satisfies.

Now, if there are 4 gaps => a + b + c + d = 7 => The average value is 7/4 which is less than 2 => Not possible.

=> The possible cases for C are:

C	2013	2014	2015	2016	2017
	1	2	3	2	1

Consider D:

The pattern looks as follows:

1, a, b, c, 1

=> a + b + c = 8

When a = 2 and c = 2 => b = 4 => 2, 4, 2 => Satisfies.

When a = 2 and c = 3, b should be 3 (Not satisfying (1))

When a = 3 and c = 3, b should be 2 (Not satisfying (1))

=> The possible cases for D are:

D	2011	2012	2013	2014	2015
	1	2	4	2	1

Consider E:

The pattern looks as follows:

1,.....,1

For 1 or 2 gaps, we can't get a sum of 11.

Assume 3 gaps => a + b + c = 11, the maximum case is 3, 5, 3 => Satisfies.

Now, assume 4 gaps

=> a + b + c + d = 11, the minimum case is 2, 3, 4, 2 or 2, 4, 3, 2 which satisfies (1) and 2 + 3 + 4 + 2 = 11.

=> The possible cases for E are:

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

In summary, the possible cases for all 5 companies is:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

C	2013	2014	2015	2016	2017
	1	2	3	2	1

D	2011	2012	2013	2014	2015
	1	2	4	2	1

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

Given that total amount raised in 2014 is 12

=> 3 + 3/2 + 2 + 2 + 1/2 = 12 =>

=> possible case is 3 + 3 + 2 + 2 + 2 = 12.

A) In 2013, B raised 2 crores and E also raised 3/4 crores => Not Possible.

B) In 2013, A could have raised 5/4 and D raised 4 => Possible.

C) In 2014, A raised 3 and B raised 3 => Possible.

D) In 2014, B raised 3 where as E raised 2 => 3 > 2 => Possible.

VIDEO SOLUTION

52. **A**

Since A, B, C are mutual strangers,(B, C)are strangers for A, (A, C) are strangers for B, (A, B) are strangers for C.

Since the total number of strangers+ acquaintances+ friends for any among the 7 is 6.

The number of friends for Amudha is 1, the number of friends for Gowtham is 1.

Using statement 3 Chandran and Gowtham are friends.

Using statement 2: Amudha, Dinesh, and Fani are Ezil's friends. Similarly, Ezil is a friend of Amudha, Dhinesh, and Fani.

Using statement 4 Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan, Hence the number of acquaintances of Bharatan is equal to the number of friends of Amudha.

Using statement 5 Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance Bharatan is a friend of Amudha, Hence the number of acquaintances of Amudha is equal to the number of friends of Bharatan.

Hence Bharatan has one friend, 1 Acquaintance, 4 strangers.

For Amudha we are yet to find a relationship with Dhinesh, Fani, and Gowtham. Any among the three can be the stranger for Amudha, considering the three different cases.

Case 1 :

Considering Fani as an acquaintance of Amudha, then Dhinesh and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Fani is an acquaintance of Amudha, hence Amudha is an Acquaintance of Fani, Fani is a friend of Bharathan and hence Bharathan is a friend of Fani. But Fani has only one friend and Ezhil is already a friend of Fani.

Hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1, (F)	4 (B, C, D, G)
Bharatan	1, (F)	1, (E)	4, (A, C, D, G)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 2 :

Considering Gowtham as an acquaintance of Amudha, then Dhinesh and Fani are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.Since Gowtham is a friend of Bharatan, Bharatan must be a friend of Gowtham. But Gowtham can only have one friend and it already mentioned that Chandran is a friend of Gowtham and hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (G)	4 (B, C, D, F)
Bharatan	1, (G)	1, (E)	4, (A, C, D, F)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 3 :

Considering Dhinesh as an acquaintance of Amudha, then Fani and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Since Fani, Gowtham are strangers to Amudha, Bharatan. Amudha, Bharatan are strangers to Fani, Gowtham.

The 2 strangers to Gowtham and his only friend are known. Hence his three acquaintances are (Dhinesh, Ezhil, Fani).

Hence Gowtham is an acquaintance of Dhinesh, Ezhil, and Fani.

Dhinesh is an acquaintance of Amudha and hence Amudha must be an acquaintance of Dhinesh.

Dhinesh is a friend of Bharatan and hence Bharatan is a friend of Dhinesh.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4 (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	G	1	(A, B)
Dhinesh	E, B	G, A	2
Ezhil	A, D, F	G	1
Fani	1 (E)	G	A, B
Gowtham	1 (C)	3 (D, E, F)	2 (A, B)

Ezhil is an acquaintance of Bharatan and hence Bharatan is an acquaintance of Ezhil. The only stranger to Ezhil who is left is Chandran. Hence Ezhil is a stranger to Chandran.

The two strangers to Dhinesh who are left are Chandran and Fani. Chandran is a stranger to Dhinesh and hence Dhinesh is a stranger to Chandran.

The only acquaintance of Chandran who is left is Fani.

The remaining relationships with Fani are Dhinesh and Chandran. Dhinesh is a stranger to Fani and Chandran is an acquaintance of Fani.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4, (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	1, (G)	1 (F)	4, (A, B, E, D)
Dhinesh	2, (E, B)	2, (G, A)	2, (C, F)
Ezhil	3, (A, D, F)	2, (G, B)	1, (C).
Fani	1 (E)	2, (G, C)	3, (A, B, D)
Gowtham	1 (C)	3 (D, E, F)	2, (A, B)

Gowtham's Acquaintances are Ghinesh, Ezhil, Fani.

VIDEO SOLUTION

53. **3**

Given that the means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

=> The sum of ratings given by R1, R2, R3 R4, R5 are 5*means = 17, 11, 19, 14, and 17 respectively.

Similarly the sum of ratings received by U, V, W, X and Y are 5*means = 11, 19, 17, 18, and 13 respectively.

Also capturing the absolute data given in the partial information (a) and (b) and representing as a table, we get:

	R1	R2	R3	R4	R5	Total
U	1					11
V					5	19
W	5	1	5			17
X		5	5			18
Y		1	1			13
Total	17	11	19	14	17	

Now,

Consider U

Given median = 2, mode = 2 and range = 3

=> His ratings should be of the form 1, a , 2, b, 4 => 1 + 2 + 4 + a + b = 11 => a + b = 4. For mode = 2 => a = b = 2

=> U's ratings are 1, 2, 2, 2, 4.

Consider V

Given median = 4, mode = 4 and range = 3

=> His ratings should be of the form 2, a, 4, b, 5 => 2 + 4 + 5 + a + b = 19 => a + b = 8 => For mode = 4 => a = b = 4

=> V's ratings are 2, 4, 4, 4, 5.

Consider W

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => 1 + a + 4 + 5 + 5 = 17 => a = 2

=> W's ratings are 1, 2, 4, 5, 5.

Consider X

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => a + 1 + 4 + 5 + 5 = 18 => a = 3

=> X's ratings are 1, 3, 4, 5, 5

Consider Y

Given median = 3, mode = 1 & 4, Range = 3

=> His ratings are 1, 1, 3, 4, 4.

Capturing this data in the table, we get:

	R1	R2	R3	R4	R5	Total	Entries
U	1					11	2,2,2,4
V					5	19	2,4,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R3 => The two missing entries should add up to 19 - 1 - 5 - 5 = 8, (only possibility is 4 + 4) => We can fill the row "U" and 4 in the row "V"

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V			4		5	19	2,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R2 => Missing entry should be 11 - 2 - 1 - 5 - 1 = 2

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Consider column R1, the missing elements should add up to 17 - 5 - 4 - 1 = 7 (3 + 4 or 4 + 3) ----(1)

Consider R5, the missing elements should add up to 10 => 2 + 4 + 4 or 4 + 3 + 3 (not possible) as (1) requires a 3.

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X		5	5		4	18	1,3
Y		1	1		4	13	3,4
Total	17	11	19	14	17		

Now, we can fill column R1 as 3 + 4 and the remaining in column R4 and we can get the complete table

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X	3	5	5	1	4	18	
Y	4	1	1	3	4	13	
Total	17	11	19	14	17		

=> From the table, we can see that R1 gave a rating of 3 to Xavier.

VIDEO SOLUTION

54. **C**

x is the fraction of male members in the secondary section.

0.6*80 + 0.55*80 + 640x = 800*0.475

=> 0.6 + 0.55 + 8x = 4.75

=> 8x = 3.6 => x = 0.45

=> 45% are males and 55% are females => 288 males and 352 females

=> There are 144 vegetarian males and 208 vegetarian females .

Also there are 144 non-vegetarian males and 144 non-vegetarian females.

So leaving vegetarian females all others are same in nos.

VIEW SOLUTION

55. **A**

There are 640 secondary section students, 80 11th class students and 80 12th class students.

If the fraction of vegetarians in 12th class is x, then

80x + 80*0.5 + 640*0.55 = 0.53*800

=> x + 0.5 + 4.4 = 5.3 => x = 0.4

=> 40% are vegetarians in class 12.

VIEW SOLUTION

56. **E**
There are 640 secondary section students, 80 11th class students and 80 12th class students.

If the fraction of vegetarians in 12th class is x, then

80x + 80*0.5 + 640*0.55 = 0.53*800

=> x + 0.5 + 4.4 = 5.3 => x = 0.4

=> There are 32 vegetarian students in class 12.

25% of class 12 i.e 8 are males.

So there are 24 vegetarian female students.

No. of non-veg males = 40 and veg females=24. Required answer 40 - 24 = 16.

VIEW SOLUTION

57. **B**
According to given condition No of male students in 11,12 and in total are 44 , 48 and 380.

So No of males in secondary are 380-44-48=288.

288 students => 45% of total secondary students.

VIEW SOLUTION

58. **12**
The set's starting point is that the sum of each row must be 100

Clue 3 tells us that all the missing elements in the Carbs column are multiple of 5. Similarly, clue 4 tells us that the missing elements of the other three columns are multiples of 4.

Note that we look at clue 1, which says that the carbs in C1 and C2 must be more than any carbs in any pseudo cereal.
We have the reference point of P1 at 66

Trying to fill in for C1:
The possible values are 75, 80, 85, 90, 95
Since 12 grams is of other nutrients, we can eliminate 90 and 95.
If we take 85, 85+12 = 97, which would leave 3 grams of protein, but from clue 4, we know this has to be a multiple of 4.

Taking 75, 75+12 = 87 would leave 13 grams of protein, which, too, is eliminated.
Leaving only 80 grams of carb in C1 and 8 grams of protein.

Similar logic is to be applied for C2; we have 13 grams already present, so the values of 90 and 95 are eliminated.
Taking 85 carbs would give 2 grams of protein, which can be eliminated.
80 and 70 would also not work for the same reason.
Leavin has only 75 grams of carbs, leaving 12 grams of protein.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2		14		8

Using clues 1 and 2 together, we can determine that the carbs in P2 must be less than 75 but at least greater than 62.
The only possible values are 65 and 70
Putting 65 grams of carbs in P2 gives us 100-65-14-8 = 13 grams of protein, which is invalid.

Putting 70 grams of carbs in P2 gives 100-70-14-8 = 8 grams of protein.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

Clue 1 gives us that the protein in M2 should be less than that present in either of the pseudo-cereals, which we right now have a baseline of 14.
So the protein in M2 (and M3) can be 12, 8, 4, 0

The protein and carbs in M2 should add up to 100-7-16 = 77
This is only possible if the protein count is 12, giving carbs as 65

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

The protein in M3 can be 0, 4, 8 or 12

We are given in clue five that the protein in P1 is double that in M3.
The protein in P1 thus can be 0, 8, 16 or 24
Since this P1 protein also has to be more than M1 and M2 protein, it can not be 0 or 8
Leaving only 16 or 24 as the valid values.

The protein and fats in P1 must add up to 100-66-10 = 24
If P1 had 24 grams of protein, then it would have 0 grams of fat, but in clue 4, we are given that all missing fats are **non-zero multiples of 4**
Hence, the only possible protein value in P1 is 16, with 8 grams of fats. Giving 8 grams of protein in M3 and 24 grams of others in M3.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56	8	12	24
Pseudo-cereals	P1	66	16	8	10
	P2	70	14	8	8

We can see that there were 12 grams of protein in M2.

VIDEO SOLUTION

59. **D**

From given conditions we find that :

	London(l)	Paris(p)	Beijing(b)	NY(n)	Total
Round1 (r1)	30	24	16	12	82
Round2(r2)	30	24(p r1) +3 (n r1) +4 (B r1) +1(n - representative)	12 (b r1) +9(n r1)	0	83
Round3(r3)	30+8 (Beijing r1 , r2)	24(p r1,r2) +4(b r1 ;p r2) +4(b r1,r2) +3 (n r1, p r2) +1 (n-representative) +1(b-representative)	0	0	75

Required value is 8*100/12 = 66.67 %. Hence option D.

VIDEO SOLUTION

60. **C**

It's given in the question that the first runner up polled 10,000 more votes than the second runner up in constituency A. Now the first runner up has got 95000 votes, hence the second runner up will get 85000 votes.

Now the remaining votes will be 500000-275000-95000-85000=45000

From 2, None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000 => the person who got 5th highest votes must have got > 600030/6 => $\geq$ 100006. Since it is also given that the difference of votes is $\geq$ 10000, the only possible case is winner, 1st runner up, 2nd runner up, 3rd runner up, 4th runner up must have got 140006,130006,120006,110006,100006 respectively which sums upto exactly 600030.

From 3, Let the total votes in D be 100x => The winning candidate in constituency D must have got 37500+5x.

The table now looks like:

	Constituency			
	A	**B**	**C**	**D**
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	100x
No. of votes polled by the winning candidate	2,75,000	48,750	140006	37500+5x
No. of votes polled by the first runner up	95,000		130006	37,500
No. of votes polled by the second runner up	85000		120006	30,000
% of valid votes polled by the third runner up			16.67%	10%

The candidates who didn't lose the deposit must have got <16.67% => 3rd runner up must surely didn't get the deposit.
Also, the candidates who got security deposit must have got 65% of votes.
Case I:
Let top three candidates got the security deposit => 37500+5x+37500+30000 =65x => x=1750 => 100x= **175000**

Case II:
Let top three candidates got the security deposit **=>** 37500+5x+37500 =65x => 60x = 1250 => x=125000 but 16.66% of 125000 =20834 => 2nd runner up must got security deposit. So, this case is not valid.

For all the constituencies lets look at the candidates who lost their security deposit.

A (500000-275000-95000-85000)=45000.

B (325000-48750)=276250

C (0) and D (61250)=175000-46250-30000-37500

Hence percentage will be 382500/1600000× 100=23.91%

VIDEO SOLUTION

CAT Percentile Predictor

61. **C**
It's given in the question that the first runner up polled 10,000 more votes than the second runner up in constituency A. Now the first runner up has got 95000 votes, hence the second runner up will get 85000 votes.

Now the remaining votes will be 500000-275000-95000-85000=45000

From 2, None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000 => the person who got 5th highest votes must have got > 600030/6 => $\geq$ 100006. Since it is also given that the difference of votes is $\geq$ 10000, the only possible case is winner, 1st runner up, 2nd runner up, 3rd runner up, 4th runner up must have got 140006,130006,120006,110006,100006 respectively which sums upto exactly 600030.

From 3, Let the total votes in D be 100x => The winning candidate in constituency D must have got 37500+5x.

The table now looks like:

	Constituency			
	A	**B**	**C**	**D**
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	100x
No. of votes polled by the winning candidate	2,75,000	48,750	140006	37500+5x
No. of votes polled by the first runner up	95,000		130006	37,500
No. of votes polled by the second runner up	85000		120006	30,000
% of valid votes polled by the third runner up			16.67%	10%

The candidates who didn't lose the deposit must have got <16.67% => 3rd runner up must surely didn't get the deposit.
Also, the candidates who got security deposit must have got 65% of votes.
Case I:
Let top three candidates got the security deposit => 37500+5x+37500+30000 =65x => x=1750 => 100x= **175000**

Case II:
Let top three candidates got the security deposit **=>** 37500+5x+37500 =65x => 60x = 1250 => x=125000 but 16.66% of 125000 =20834 => 2nd runner up must got security deposit. So, this case is not valid.

The increasing order C will always come after D which is not happening in the third option. Hence that is the correct answer.

VIDEO SOLUTION

62. **B**

From 1, Let sales amounts in the Apparel departments for Delhi and Kolkata in 2018 be **a.**

From 2, Let sales amounts in the Apparel departments for Mumbai and Bengaluru in 2018 and for Delhi be **b.**

From 3, let the sales amounts in the Home Décor departments for Mumbai and Kolkata in 2018 be **c** and that of Delhi and Bengaluru be **s** and **t** respectively.

Let the Let sales amounts in the Apparel departments for Mumbai and Bengaluru in 2019 be **m** and **n** respectively.

The table looks like:

Sales Amounts (Crore Rupees)								
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparel	a	b	b	m	b	n	a	54
Electronics	78	98	82	102	90	70	80	100
Home Décor	s	100	c	72	t	80	c	54

From 6, **s=80 and t=60.**

From 5, s+c+t+c+70=306 => 2c=96 => **c=48.**

From 4, (b-a)+(m-b)+(n-b)+(54-a)=(20)+20-20+20 => 54+m+n-2a-b=40=> 2a+b=14+m+n => 2b=28+2m+2n-4a...(i)

From 7, b-a=n-b => 2b=a+n...(ii) and m-b=54-a=> a+m=54+b=>2b=2a+2m-108...(iii)

From 8, 108=b+n=>2b=216-2n...(iv)

from (i) & (iii), 28+2m+2n-4a=2a+2m-108 => 6a=136+2n => 3a=68+n...(v)

from (ii) & (iv), 216-2n=a+n => a+3n=216=> 3a=648-9n...(vi)

from (v) & (vi), 648-9n=68+n => **n=58** => **a=42 => b=50 => m=62.**

The final table looks like this

Sales Amount (Crore Rupees)								
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparels	42	50	50	62	50	58	42	54
Electronics	78	98	82	102	90	70	80	100
Home Decor	80	100	48	72	60	80	48	54

∴ total sales amount, in Crore Rupees, in 2019 for the chain of departmental stores =900 cr.

VIDEO SOLUTION

63. **D**

From 1, Let sales amounts in the Apparel departments for Delhi and Kolkata in 2018 be **a.**

From 2, Let sales amounts in the Apparel departments for Mumbai and Bengaluru in 2018 and for Delhi be **b.**

From 3, let the sales amounts in the Home Décor departments for Mumbai and Kolkata in 2018 be **c** and that of Delhi and Bengaluru be **s** and **t** respectively.

Let the Let sales amounts in the Apparel departments for Mumbai and Bengaluru in 2019 be **m** and **n** respectuvely.

The table looks like:

Sales Amounts (Crore Rupees)								
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparel	a	b	b	m	b	n	a	54
Electronics	78	98	82	102	90	70	80	100
Home Décor	s	100	c	72	t	80	c	54

From 6, **s=80 and t=60.**
From 5, s+c+t+c+70=306 => 2c=96 => **c=48.**

From 4, (b-a)+(m-b)+(n-b)+(54-a)=(20)+20-20+20 => 54+m+n-2a-b=40=> 2a+b=14+m+n => 2b=28+2m+2n-4a...(i)

From 7, b-a=n-b => 2b=a+n...(ii) and m-b=54-a=> a+m=54+b=>2b=2a+2m-108...(iii)

From 8, 108=b+n=>2b=216-2n...(iv)

from (i) & (iii), 28+2m+2n-4a=2a+2m-108 => 6a=136+2n => 3a=68+n...(v)

from (ii) & (iv), 216-2n=a+n => a+3n=216=> 3a=648-9n...(vi)

from (v) & (vi), 648-9n=68+n => **n=58** => **a=42 => b=50 => m=62.**

The final table looks like this

Sales Amount (Crore Rupees)								
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparels	42	50	50	62	50	58	42	54
Electronics	78	98	82	102	90	70	80	100
Home Decor	80	100	48	72	60	80	48	54

Hence the percentage change is 50.

VIDEO SOLUTION

64. **A**

From 1, Let sales amounts in the Apparel departments for Delhi and Kolkata in 2018 be **a.**

From 2, Let sales amounts in the Apparel departments for Mumbai and Bengaluru in 2018 and for Delhi be **b.**

From 3, let the sales amounts in the Home Décor departments for Mumbai and Kolkata in 2018 be **c** and that of Delhi and Bengaluru be **s** and **t** respectively.

Let the Let sales amounts in the Apparel departments for Mumbai and Bengaluru in 2019 be **m** and **n** respectuvely.

The table looks like:

Sales Amounts (Crore Rupees)								
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparel	a	b	b	m	b	n	a	54
Electronics	78	98	82	102	90	70	80	100
Home Décor	s	100	c	72	t	80	c	54

From 6, **s=80 and t=60.**
From 5, s+c+t+c+70=306 => 2c=96 => **c=48.**

From 4, (b-a)+(m-b)+(n-b)+(54-a)=(20)+20-20+20 => 54+m+n-2a-b=40=> 2a+b=14+m+n => 2b=28+2m+2n-4a...(i)

From 7, b-a=n-b => 2b=a+n...(ii) and m-b=54-a=> a+m=54+b=>2b=2a+2m-108...(iii)

From 8, 108=b+n=>2b=216-2n...(iv)

from (i) & (iii), 28+2m+2n-4a=2a+2m-108 => 6a=136+2n => 3a=68+n...(v)

from (ii) & (iv), 216-2n=a+n => a+3n=216=> 3a=648-9n...(vi)

from (v) & (vi), 648-9n=68+n => **n=58** => **a=42 => b=50 => m=62.**

The final table looks like this

Sales Amount (Crore Rupees)								
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparels	42	50	50	62	50	58	42	54
Electronics	78	98	82	102	90	70	80	100
Home Decor	80	100	48	72	60	80	48	54

Increase in sales =12

VIDEO SOLUTION

65. **D**

From 1, Let sales amounts in the Apparel departments for Delhi and Kolkata in 2018 be **a.**

From 2, Let sales amounts in the Apparel departments for Mumbai and Bengaluru in 2018 and for Delhi be **b.**

From 3, let the sales amounts in the Home Décor departments for Mumbai and Kolkata in 2018 be **c** and that of Delhi and Bengaluru be **s** and **t** respectively.

Let the Let sales amounts in the Apparel departments for Mumbai and Bengaluru in 2019 be **m** and **n** respectuvely.

The table looks like:

Sales Amounts (Crore Rupees)								
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparel	a	b	b	m	b	n	a	54
Electronics	78	98	82	102	90	70	80	100
Home Décor	s	100	c	72	t	80	c	54

From 6, **s=80 and t=60.**
From 5, s+c+t+c+70=306 => 2c=96 => **c=48.**

From 4, (b-a)+(m-b)+(n-b)+(54-a)=(20)+20-20+20 => 54+m+n-2a-b=40=> 2a+b=14+m+n => 2b=28+2m+2n-4a...(i)

From 7, b-a=n-b => 2b=a+n...(ii) and m-b=54-a=> a+m=54+b=>2b=2a+2m-108...(iii)

From 8, 108=b+n=>2b=216-2n...(iv)

from (i) & (iii), 28+2m+2n-4a=2a+2m-108 => 6a=136+2n => 3a=68+n...(v)

from (ii) & (iv), 216-2n=a+n => a+3n=216=> 3a=648-9n...(vi)

from (v) & (vi), 648-9n=68+n => **n=58** => **a=42 => b=50 => m=62.**

The final table looks like this

Sales Amount (Crore Rupees)								
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparels	42	50	50	62	50	58	42	54
Electronics	78	98	82	102	90	70	80	100
Home Decor	80	100	48	72	60	80	48	54

VIDEO SOLUTION

66. **24**
The set's starting point is that the sum of each row must be 100

Clue 3 tells us that all the missing elements in the Carbs column are multiple of 5. Similarly, clue 4 tells us that the missing elements of the other three columns are multiples of 4.

Note that we look at clue 1, which says that the carbs in C1 and C2 must be more than any carbs in any pseudo cereal.
We have the reference point of P1 at 66

Trying to fill in for C1:
The possible values are 75, 80, 85, 90, 95
Since 12 grams is of other nutrients, we can eliminate 90 and 95.
If we take 85, 85+12 = 97, which would leave 3 grams of protein, but from clue 4, we know this has to be a multiple of 4.

Taking 75, 75+12 = 87 would leave 13 grams of protein, which, too, is eliminated.
Leaving only 80 grams of carb in C1 and 8 grams of protein.

Similar logic is to be applied for C2; we have 13 grams already present, so the values of 90 and 95 are eliminated.
Taking 85 carbs would give 2 grams of protein, which can be eliminated.
80 and 70 would also not work for the same reason.
Leavin has only 75 grams of carbs, leaving 12 grams of protein.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2		14		8

Using clues 1 and 2 together, we can determine that the carbs in P2 must be less than 75 but at least greater than 62.
The only possible values are 65 and 70
Putting 65 grams of carbs in P2 gives us 100-65-14-8 = 13 grams of protein, which is invalid.

Putting 70 grams of carbs in P2 gives 100-70-14-8 = 8 grams of protein.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

Clue 1 gives us that the protein in M2 should be less than that present in either of the pseudo-cereals, which we right now have a baseline of 14.
So the protein in M2 (and M3) can be 12, 8, 4, 0

The protein and carbs in M2 should add up to 100-7-16 = 77
This is only possible if the protein count is 12, giving carbs as 65

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

The protein in M3 can be 0, 4, 8 or 12

We are given in clue five that the protein in P1 is double that in M3.
The protein in P1 thus can be 0, 8, 16 or 24
Since this P1 protein also has to be more than M1 and M2 protein, it can not be 0 or 8
Leaving only 16 or 24 as the valid values.

The protein and fats in P1 must add up to 100-66-10 = 24
If P1 had 24 grams of protein, then it would have 0 grams of fat, but in clue 4, we are given that all missing fats are **non-zero multiples of 4**
Hence, the only possible protein value in P1 is 16, with 8 grams of fats. Giving 8 grams of protein in M3 and 24 grams of others in M3.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56	8	12	24
Pseudo-cereals	P1	66	16	8	10
	P2	70	14	8	8

There were 24 grams of other nutrients in M3.

VIDEO SOLUTION

67. **A**

It is given that there are only three female students - Amala, Koli, and Rini - and only three male students - Biman, Mathew, and Shyamal - in a course.

It is also known that the aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

Let the project score component be x, which implies the test score component will be (1-x). The projects are done in groups of two, with each group consisting of a female and a male student, which implies there are three groups for the project. It is also known that both the group members obtain the same score in the project. The score obtained in the project is 40, 60, and 80, respectively.

Therefore, we can say that each female student will consist of a different group, and no two male students or female students will be in the same group.

For the test scores, there are six scores given for six students among which four are distinct and the remaining two are average scores, which is 60. It is also known that the maximum score possible is 80, and the minimum score is 40.

Hence, the distinct scores are 80, 70, 50, and 40 (since all the test scores are multiple of 10), and the remaining two scores are 60, and 60, respectively.

From point 3, we know that Amala's score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Hence, we can say the score obtained by Amala in the project is 80, and the score obtained by Koli is 40, which implies the score obtained by Rini in the project is 60. Now, Koli scored 20 more than Amala in the test, which implies the score obtained by Koli can be either 80, 70, or 60.

The score obtained by them is given below:

Students	Test scores	Project scores
Amala	40/50/60	80
Koli	60/70/80	40
Rini		60
Biman		
Mathew		
Shyamal		

It is known that Amala had the highest aggregate score, and Shyamal scored the second highest on the test. He scored two more than Koli, but two less than Amala in the aggregate.

Hence, the score obtained by Shyamal in the test is 70, which implies Koli can't score 70 in the test => Amala can't score 50 in the test.

Students	Test scores	Project scores
Amala	40/60	80
Koli	60/80	40
Rini		60
Biman		
Mathew		
Shyamal	70	

It is given that Shyamal scored two more than Koli, but two less than Amala in the aggregate. Hence, the aggregate score of Amala is 4 more than Koli. It is also known that Amala had the highest aggregate score.

Case 1: The test score of Amala is 40

Students	Test scores	Project scores	Aggregate score
Amala	40	80	40(1-x)+80x
Koli	60	40	60(1-x)+40x
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, $40(1-x)+80x = 60(1-x)+40x+4$

$\Rightarrow 60x = 24$

$\Rightarrow x = 0.4$

Hence, the aggregate score obtained by Amala is 40(1-0.4)+80*4 = 56

The minimum aggregate score of Shyamal is 70(1-0.4)+ 40*0.4 = 58, which is greater than Amala.

Hence, **Case 1 is not possible.**

Hence, the table is given below:

Students	Test scores	Project scores	Aggregate score
Amala	60	80	60(1-x)+80x
Koli	80	40	80(1-x)+40x
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, 60(1-x)+80x = 80(1-x)+40x+4

=> 60+20x = 84-40x

=> 60x = 24 => x = 0.4

Hence, the aggregate score of Amala is 60(1-0.4)+80*0.4 = 68, which implies the aggregate score of Shyamal is (68-2) = 66

Hence, the score obtained by Shyamal in Project is {66-70*(0.6)}/0.4 = 60.

It is also known that Biman scored second lowest in the test, which implies the score of Biman in the test is 50, and he scored the lowest in the aggregate. It is also known that Mathew scored more than Rini in the project, but less than her in the test. Hence, Mathew scored 80 in the project (since Rini scored 60 in the project), and Biman scored 40 in the project.

Similarly, Rini Scored more than Mathew on the test, which implies the score obtained by Rini is 60, and the score obtained by Mathew is 40 in the test.

Hence, the final table will look like this:

Students	Test scores (T)	Project scores(P)	Aggregate score (T*0.6+P*0.4)	Project Pair
Amala	60	80	68	Amala, Mathew
Koli	80	40	64	Koli, Biman
Rini	60	60	60	Rini, Shyamal
Biman	50	40	46	Biman, Koli
Mathew	40	80	56	Mathew, Amala
Shyamal	70	60	66	Shyamal, Rini

Hence, the weight of the test component is 0.6

The correct option is A

VIDEO SOLUTION

68. **A**

Since A, B, C are mutual strangers,(B, C)are strangers for A, (A, C) are strangers for B, (A, B) are strangers for C.

Since the total number of strangers+ acquaintances+ friends for any among the 7 is 6.

The number of friends for Amudha is 1, the number of friends for Gowtham is 1.

Using statement 3 Chandran and Gowtham are friends.

Using statement 2: Amudha, Dinesh, and Fani are Ezil's friends. Similarly, Ezil is a friend of Amudha, Dhinesh, and Fani.

Using statement 4 Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan, Hence the number of acquaintances of Bharatan is equal to the number of friends of Amudha.

Using statement 5 Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance Bharatan is a friend of Amudha, Hence the number of acquaintances of Amudha is equal to the number of friends of Bharatan.

Hence Bharatan has one friend, 1 Acquaintance, 4 strangers.

For Amudha we are yet to find a relationship with Dhinesh, Fani, and Gowtham. Any among the three can be the stranger for Amudha, considering the three different cases.

Case 1 :

Considering Fani as an acquaintance of Amudha, then Dhinesh and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Fani is an acquaintance of Amudha, hence Amudha is an Acquaintance of Fani, Fani is a friend of Bharathan and hence Bharathan is a friend of Fani. But Fani has only one friend and Ezhil is already a friend of Fani.

Hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1, (F)	4 (B, C, D, G)
Bharatan	1, (F)	1, (E)	4, (A, C, D, G)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 2 :

Considering Gowtham as an acquaintance of Amudha, then Dhinesh and Fani are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.Since Gowtham is a friend of Bharatan, Bharatan must be a friend of Gowtham. But Gowtham can only have one friend and it already mentioned that Chandran is a friend of Gowtham and hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (G)	4 (B, C, D, F)
Bharatan	1, (G)	1, (E)	4, (A, C, D, F)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 3 :

Considering Dhinesh as an acquaintance of Amudha, then Fani and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Since Fani, Gowtham are strangers to Amudha, Bharatan. Amudha, Bharatan are strangers to Fani, Gowtham.

The 2 strangers to Gowtham and his only friend are known. Hence his three acquaintances are (Dhinesh, Ezhil, Fani).

Hence Gowtham is an acquaintance of Dhinesh, Ezhil, and Fani.

Dhinesh is an acquaintance of Amudha and hence Amudha must be an acquaintance of Dhinesh.

Dhinesh is a friend of Bharatan and hence Bharatan is a friend of Dhinesh.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4 (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	G	1	(A, B)
Dhinesh	E, B	G, A	2
Ezhil	A, D, F	G	1
Fani	1 (E)	G	A, B
Gowtham	1 (C)	3 (D, E, F)	2 (A, B)

Ezhil is an acquaintance of Bharatan and hence Bharatan is an acquaintance of Ezhil. The only stranger to Ezhil who is left is Chandran. Hence Ezhil is a stranger to Chandran.

The two strangers to Dhinesh who are left are Chandran and Fani. Chandran is a stranger to Dhinesh and hence Dhinesh is a stranger to Chandran.

The only acquaintance of Chandran who is left is Fani.

The remaining relationships with Fani are Dhinesh and Chandran. Dhinesh is a stranger to Fani and Chandran is an acquaintance of Fani.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4, (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	1, (G)	1 (F)	4, (A, B, E, D)
Dhinesh	2, (E, B)	2, (G, A)	2, (C, F)
Ezhil	3, (A, D, F)	2, (G, B)	1, (C).
Fani	1 (E)	2, (G, C)	3, (A, B, D)
Gowtham	1 (C)	3 (D, E, F)	2, (A, B)

Dhinesh is an acquaintance of Amudha

VIDEO SOLUTION

69. **D**

Since tara received the same grade in exactly 3 subjects , only one possibility is getting B in 3 subjects and F in other 2. So in operation her grade would be either B or F.Among options only manab has B in operations . hence option D.

VIDEO SOLUTION

70. **120**

Given that in every month, both online and offline registration numbers were multiples of 10.

From (2), in Jan, the number of offline registrations was twice that of online registrations.

=> If x is number of online registrations => 2x is the number of offline registrations => 3x is the total number of registrations.

According to the data given in the table => 3x should lie between the minimum and maximum total number of registrations. => x = 40 (as x should also be a multiple of 10)

=> In Jan => (40,80) are the online and offline registrations respectively.

Similarly from (3) => In Apr (80,40) are the online and offline registrations respectively.

From-5, the number of online registrations is highest in may => In may there are 100 online registrations. The lowest possible number of offline registrations is 30 and maximum possible total registrations is 130 => In May (100,30) are the online and offline registrations respectively.

Let us assume, 'x' to be the number of offline registrations in May = number of online registrations in March.

Let us capture all this data in a table:

Month	Online	Offline	Total
Jan	40	80	120
Feb	y	x	
Mar	x	z	
Apr	80	40	120
May	100	30	130

From the table given in the question, 50 is the median for Offline data

=> x should lie between 50 and 80 (included)

For 80 to be the median for the online data => y lie between 80 and 100 (included).

Now, consider Feb => Minimum value of y + x = 80 + 50 = 130 (which is the maximum value possible of the total possible registrations)

=> x = 50 and y = 80

Since, 110 is the minimum number of total registrations, the only possibility is in March => 50 + z = 110 => z = 60.

Now, filling the complete table we get,

Month	Online	Offline	Total
Jan	40	80	120
Feb	80	50	130
Mar	50	60	110
Apr	80	40	120
May	100	30	130

The total number of registrations in April is 120.

VIDEO SOLUTION

71. **A**

Given that in every month, both online and offline registration numbers were multiples of 10.

From (2), in Jan, the number of offline registrations was twice that of online registrations.

=> If x is number of online registrations => 2x is the number of offline registrations => 3x is the total number of registrations.

According to the data given in the table => 3x should lie between the minimum and maximum total number of registrations. => x = 40 (as x should also be a multiple of 10)

=> In Jan => (40,80) are the online and offline registrations respectively.

Similarly from (3) => In Apr (80,40) are the online and offline registrations respectively.

From-5, the number of online registrations is highest in may => In may there are 100 online registrations. The lowest possible number of offline registrations is 30 and maximum possible total registrations is 130 => In May (100,30) are the online and offline registrations respectively.

Let us assume, 'x' to be the number of offline registrations in May = number of online registrations in March.

Let us capture all this data in a table:

Month	Online	Offline	Total
Jan	40	80	120
Feb	y	x	
Mar	x	z	
Apr	80	40	120
May	100	30	130

From the table given in the question, 50 is the median for Offline data

=> x should lie between 50 and 80 (included)

For 80 to be the median for the online data => y lie between 80 and 100 (included).

Now, consider Feb => Minimum value of y + x = 80 + 50 = 130 (which is the maximum value possible of the total possible registrations)

=> x = 50 and y = 80

Since, 110 is the minimum number of total registrations, the only possibility is in March => 50 + z = 110 => z = 60.

Now, filling the complete table we get,

Month	Online	Offline	Total
Jan	40	80	120
Feb	80	50	130
Mar	50	60	110
Apr	80	40	120
May	100	30	130

Total registrations in Jan = Apr = 120 and Feb = May = 130.

VIDEO SOLUTION

72. **40**

Given that in every month, both online and offline registration numbers were multiples of 10.

From (2), in Jan, the number of offline registrations was twice that of online registrations.

=> If x is number of online registrations => 2x is the number of offline registrations => 3x is the total number of registrations.

According to the data given in the table => 3x should lie between the minimum and maximum total number of registrations. => x = 40 (as x should also be a multiple of 10)

=> In Jan => (40,80) are the online and offline registrations respectively.

Similarly from (3) => In Apr (80,40) are the online and offline registrations respectively.

From-5, the number of online registrations is highest in may => In may there are 100 online registrations. The lowest possible number of offline registrations is 30 and maximum possible total registrations is 130 => In May (100,30) are the online and offline registrations respectively.

Let us assume, 'x' to be the number of offline registrations in May = number of online registrations in March.

Let us capture all this data in a table:

Month	Online	Offline	Total
Jan	40	80	120
Feb	y	x	
Mar	x	z	
Apr	80	40	120
May	100	30	130

From the table given in the question, 50 is the median for Offline data

=> x should lie between 50 and 80 (included)

For 80 to be the median for the online data => y lie between 80 and 100 (included).

Now, consider Feb => Minimum value of y + x = 80 + 50 = 130 (which is the maximum value possible of the total possible registrations)

=> x = 50 and y = 80

Since, 110 is the minimum number of total registrations, the only possibility is in March => 50 + z = 110 => z = 60.

Now, filling the complete table we get,

Month	Online	Offline	Total
Jan	40	80	120
Feb	80	50	130
Mar	50	60	110
Apr	80	40	120
May	100	30	130

The number of online registrations in Jan is 40.

VIDEO SOLUTION

73. **A**

It is given that there are only three female students - Amala, Koli, and Rini - and only three male students - Biman, Mathew, and Shyamal - in a course.

It is also known that the aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

Let the project score component be x, which implies the test score component will be (1-x). The projects are done in groups of two, with each group consisting of a female and a male student, which implies there are three groups for the project. It is also known that both the group members obtain the same score in the project. The score obtained in the project is 40, 60, and 80, respectively.

Therefore, we can say that each female student will consist of a different group, and no two male students or female students will be in the same group.

For the test scores, there are six scores given for six students among which four are distinct and the remaining two are average scores, which is 60. It is also known that the maximum score possible is 80, and the minimum score is 40.

Hence, the distinct scores are 80, 70, 50, and 40 (since all the test scores are multiple of 10), and the remaining two scores are 60, and 60, respectively.

From point 3, we know that Amala's score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Hence, we can say the score obtained by Amala in the project is 80, and the score obtained by Koli is 40, which implies the score obtained by Rini in the project is 60. Now, Koli scored 20 more than Amala in the test, which implies the score obtained by Koli can be either 80, 70, or 60.

The score obtained by them is given below:

Students	Test scores	Project scores
Amala	40/50/60	80
Koli	60/70/80	40
Rini		60
Biman		
Mathew		
Shyamal		

It is known that Amala had the highest aggregate score, and Shyamal scored the second highest on the test. He scored two more than Koli, but two less than Amala in the aggregate.

Hence, the score obtained by Shyamal in the test is 70, which implies Koli can't score 70 in the test => Amala can't score 50 in the test.

Students	Test scores	Project scores
Amala	40/60	80
Koli	60/80	40
Rini		60
Biman		
Mathew		
Shyamal	70	

It is given that Shyamal scored two more than Koli, but two less than Amala in the aggregate. Hence, the aggregate score of Amala is 4 more than Koli. It is also known that Amala had the highest aggregate score.

Case 1: The test score of Amala is 40

Students	Test scores	Project scores	Aggregate score
Amala	40	80	40(1-x)+80x
Koli	60	40	60(1-x)+40x
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, 40(1-x)+80x = 60(1-x)+40x+4

=> 60x = 24

=> x = 0.4

Hence, the aggregate score obtained by Amala is 40(1-0.4)+80*4 = 56

The minimum aggregate score of Shyamal is 70(1-0.4)+ 40*0.4 = 58, which is greater than Amala.

Hence, **Case 1 is not possible.**

Hence, the table is given below:

Students	Test scores	Project scores	Aggregate score
Amala	60	80	60(1-x)+80x
Koli	80	40	80(1-x)+40x
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, 60(1-x)+80x = 80(1-x)+40x+4

=> 60+20x = 84-40x

=> 60x = 24 => x = 0.4

Hence, the aggregate score of Amala is 60(1-0.4)+80*0.4 = 68, which implies the aggregate score of Shyamal is (68-2) = 66

Hence, the score obtained by Shyamal in Project is {66-70*(0.6)}/0.4 = 60.

It is also known that Biman scored second lowest in the test, which implies the score of Biman in the test is 50, and he scored the lowest in the aggregate. It is also known that Mathew scored more than Rini in the project, but less than her in the test. Hence, Mathew scored 80 in the project (since Rini scored 60 in the project), and Biman scored 40 in the project.

Similarly, Rini Scored more than Mathew on the test, which implies the score obtained by Rini is 60, and the score obtained by Mathew is 40 in the test.

Hence, the final table will look like this:

Students	Test scores (T)	Project scores(P)	Aggregate score (T*0.6+P*0.4)	Project Pair
Amala	60	80	68	Amala, Mathew
Koli	80	40	64	Koli, Biman
Rini	60	60	60	Rini, Shyamal
Biman	50	40	46	Biman, Koli
Mathew	40	80	56	Mathew, Amala
Shyamal	70	60	66	Shyamal, Rini

Hence, the maximum aggregate score obtained is 68. The correct option is A

VIDEO SOLUTION

74. **40**

It is given that there are only three female students - Amala, Koli, and Rini - and only three male students - Biman, Mathew, and Shyamal - in a course.

It is also known that the aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

Let the project score component be x, which implies the test score component will be (1-x). The projects are done in groups of two, with each group consisting of a female and a male student, which implies there are three groups for the project. It is also known that both the group members obtain the same score in the project. The score obtained in the project is 40, 60, and 80, respectively.

Therefore, we can say that each female student will consist of a different group, and no two male students or female students will be in the same group.

For the test scores, there are six scores given for six students among which four are distinct and the remaining two are average scores, which is 60. It is also known that the maximum score possible is 80, and the minimum score is 40.

Hence, the distinct scores are 80, 70, 50, and 40 (since all the test scores are multiple of 10), and the remaining two scores are 60, and 60, respectively.

From point 3, we know that Amala's score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Hence, we can say the score obtained by Amala in the project is 80, and the score obtained by Koli is 40, which implies the score obtained by Rini in the project is 60. Now, Koli scored 20 more than Amala in the test, which implies the score obtained by Koli can be either 80, 70, or 60.

The score obtained by them is given below:

Students	Test scores	Project scores
Amala	40/50/60	80
Koli	60/70/80	40
Rini		60
Biman		
Mathew		
Shyamal		

It is known that Amala had the highest aggregate score, and Shyamal scored the second highest on the test. He scored two more than Koli, but two less than Amala in the aggregate.

Hence, the score obtained by Shyamal in the test is 70, which implies Koli can't score 70 in the test => Amala can't score 50 in the test.

Students	Test scores	Project scores
Amala	40/60	80
Koli	60/80	40
Rini		60
Biman		
Mathew		
Shyamal	70	

It is given that Shyamal scored two more than Koli, but two less than Amala in the aggregate. Hence, the aggregate score of Amala is 4 more than Koli. It is also known that Amala had the highest aggregate score.

Case 1: The test score of Amala is 40

Students	Test scores	Project scores	Aggregate score
Amala	40	80	40(1-x)+80x
Koli	60	40	60(1-x)+40x
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, 40(1-x)+80x = 60(1-x)+40x+4

=> 60x = 24

=> x = 0.4

Hence, the aggregate score obtained by Amala is 40(1-0.4)+80*4 = 56

The minimum aggregate score of Shyamal is 70(1-0.4)+ 40*0.4 = 58, which is greater than Amala.

Hence, **Case 1 is not possible.**

Hence, the table is given below:

Students	Test scores	Project scores	Aggregate score
Amala	60	80	60(1-x)+80x
Koli	80	40	80(1-x)+40x
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, 60(1-x)+80x = 80(1-x)+40x+4

=> 60+20x = 84-40x

=> 60x = 24 => x = 0.4

Hence, the aggregate score of Amala is 60(1-0.4)+80*0.4 = 68, which implies the aggregate score of Shyamal is (68-2) = 66

Hence, the score obtained by Shyamal in Project is {66-70*(0.6)}/0.4 = 60.

It is also known that Biman scored second lowest in the test, which implies the score of Biman in the test is 50, and he scored the lowest in the aggregate. It is also known that Mathew scored more than Rini in the project, but less than her in the test. Hence, Mathew scored 80 in the project (since Rini scored 60 in the project), and Biman scored 40 in the project.

Similarly, Rini Scored more than Mathew on the test, which implies the score obtained by Rini is 60, and the score obtained by Mathew is 40 in the test.

Hence, the final table will look like this:

Students	Test scores (T)	Project scores(P)	Aggregate score (T*0.6+P*0.4)	Project Pair
Amala	60	80	68	Amala, Mathew
Koli	80	40	64	Koli, Biman
Rini	60	60	60	Rini, Shyamal
Biman	50	40	46	Biman, Koli
Mathew	40	80	56	Mathew, Amala
Shyamal	70	60	66	Shyamal, Rini

Hence, the score obtained by Mathew in the test is 40

VIDEO SOLUTION

75. **C**

It is given that there are only three female students - Amala, Koli, and Rini - and only three male students - Biman, Mathew, and Shyamal - in a course.

It is also known that the aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

Let the project score component be x, which implies the test score component will be (1-x). The projects are done in groups of two, with each group consisting of a female and a male student, which implies there are three groups for the project. It is also known that both the group members obtain the same score in the project. The score obtained in the project is 40, 60, and 80, respectively.

Therefore, we can say that each female student will consist of a different group, and no two male students or female students will be in the same group.

For the test scores, there are six scores given for six students among which four are distinct and the remaining two are average scores, which is 60. It is also known that the maximum score possible is 80, and the minimum score is 40.

Hence, the distinct scores are 80, 70, 50, and 40 (since all the test scores are multiple of 10), and the remaining two scores are 60, and 60, respectively.

From point 3, we know that Amala's score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Hence, we can say the score obtained by Amala in the project is 80, and the score obtained by Koli is 40, which implies the score obtained by Rini in the project is 60. Now, Koli scored 20 more than Amala in the test, which implies the score obtained by Koli can be either 80, 70, or 60.

The score obtained by them is given below:

Students	Test scores	Project scores
Amala	40/50/60	80
Koli	60/70/80	40
Rini		60
Biman		
Mathew		
Shyamal		

It is known that Amala had the highest aggregate score, and Shyamal scored the second highest on the test. He scored two more than Koli, but two less than Amala in the aggregate.

Hence, the score obtained by Shyamal in the test is 70, which implies Koli can't score 70 in the test => Amala can't score 50 in the test.

Students	Test scores	Project scores
Amala	40/60	80
Koli	60/80	40
Rini		60
Biman		
Mathew		
Shyamal	70	

It is given that Shyamal scored two more than Koli, but two less than Amala in the aggregate. Hence, the aggregate score of Amala is 4 more than Koli. It is also known that Amala had the highest aggregate score.

Case 1: The test score of Amala is 40

Students	Test scores	Project scores	Aggregate score
Amala	40	80	40(1-x)+80x
Koli	60	40	60(1-x)+40x
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, $40(1-x)+80x = 60(1-x)+40x+4$

=> $60x = 24$

=> $x = 0.4$

Hence, the aggregate score obtained by Amala is $40(1-0.4)+80*4 = 56$

The minimum aggregate score of Shyamal is $70(1-0.4)+ 40*0.4 = 58$, which is greater than Amala.

Hence, **Case 1 is not possible.**

Hence, the table is given below:

Students	Test scores	Project scores	Aggregate score
Amala	60	80	60(1-x)+80x
Koli	80	40	80(1-x)+40x
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, 60(1-x)+80x = 80(1-x)+40x+4

=> 60+20x = 84-40x

=> 60x = 24 => x = 0.4

Hence, the aggregate score of Amala is 60(1-0.4)+80*0.4 = 68, which implies the aggregate score of Shyamal is (68-2) = 66

Hence, the score obtained by Shyamal in Project is {66-70*(0.6)}/0.4 = 60.

It is also known that Biman scored second lowest in the test, which implies the score of Biman in the test is 50, and he scored the lowest in the aggregate. It is also known that Mathew scored more than Rini in the project, but less than her in the test. Hence, Mathew scored 80 in the project (since Rini scored 60 in the project), and Biman scored 40 in the project.

Similarly, Rini Scored more than Mathew on the test, which implies the score obtained by Rini is 60, and the score obtained by Mathew is 40 in the test.

Hence, the final table will look like this:

Students	Test scores (T)	Project scores(P)	Aggregate score (T*0.6+P*0.4)	Project Pair
Amala	60	80	68	Amala, Mathew
Koli	80	40	64	Koli, Biman
Rini	60	60	60	Rini, Shyamal
Biman	50	40	46	Biman, Koli
Mathew	40	80	56	Mathew, Amala
Shyamal	70	60	66	Shyamal, Rini

From the table, we can see that (Amala, Mathew), (Koli, Biman), and (Shyama, Rini) are the three groups for the project.

Hence, the correct option is C

VIDEO SOLUTION

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